ENT ATLAS

CLINICAL

Approach to the patient

From the presenting complaint to the diagnosis

24 articles · November 2026

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CLINICAL · APPROACH TO

Approach to hearing loss

Almost every patient with hearing loss can be placed with three questions: is the loss conductive or sensorineural, is it in one ear or both, and how quickly did it come on? The answers point to the cause and to what must not be missed. A one-sided or sudden sensorineural loss always needs more than a hearing aid.

AT A GLANCE

- Sort by type, side and speed.
- Forks at the bedside; audiogram to quantify.
- Asymmetric or sudden sensorineural loss needs urgent work-up.
- Treat reversible causes, then rehabilitate.

1 CHAPTER ONE Overview

A method, not a list.

1.1 Why a method

The causes of hearing loss run into hundreds. A structured approach, starting with tuning forks and an audiogram, narrows them quickly and catches the few that are urgent or dangerous.

1.2 The three questions

THREE QUESTIONS

- TYPE **Conductive or sensorineural?** or mixed, or non-organic
- SIDE **One ear or both?** asymmetry needs imaging
- SPEED **Sudden, progressive or fluctuating?** sudden loss is an emergency

2 CHAPTER TWO Classification

Where the lesion is, and how bad.

2.1 By site

2.1.1 Conductive

Any lesion from the canal to the stapediovestibular joint that stops sound reaching the cochlea.

2.1.2 Sensorineural

A lesion of the cochlea (*sensory*), the cochlear nerve (*neural*), or the central pathways.

2.1.3 Mixed

Both together, as in otosclerosis with cochlear involvement or chronic otitis media with labyrinthine damage.

2.1.4 Non-organic

Claimed loss without an organic basis, deliberate or unconscious.

2.2 By degree

WHO GRADES OF HEARING LOSS (2021)

GRADE	BETTER-EAR AVERAGE AT 500, 1000, 2000 AND 4000 HZ
Normal	Below 20 dB
Mild	20 to under 35 dB
Moderate	35 to under 50 dB
Moderately severe	50 to under 65 dB
Severe	65 to under 80 dB
Profound	80 to under 95 dB
Complete or total	95 dB or more

WHERE THE BOOKS DIFFER · GRADING SYSTEMS

Dhingra

Uses the older WHO scale averaged over 500, 1000 and 2000 Hz, with 0 to 25 dB counted as normal and 91 dB or more as profound.

The 2021 World Report on Hearing lowered the threshold for normal hearing to under 20 dB, added 4000 Hz to the average and added a complete hearing loss grade, because even mild losses affect communication. Many British services use the British Society of Audiology bands, in which normal hearing is up to 20 dB.

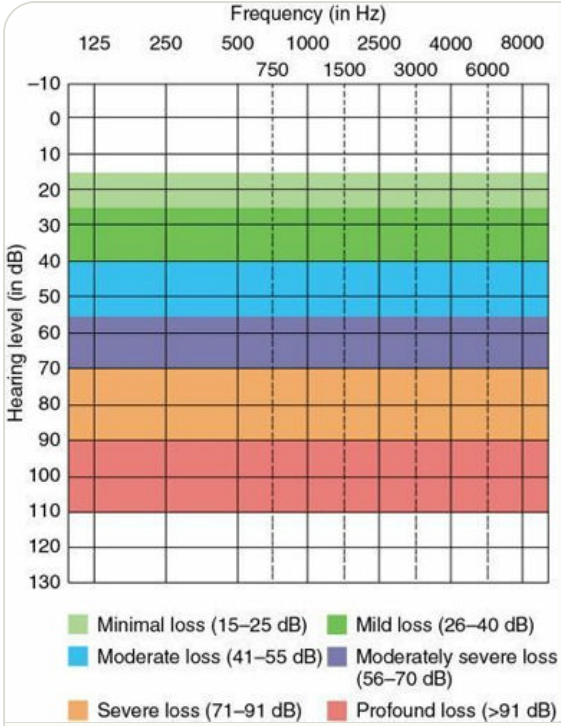


FIGURE 1 Classification of hearing loss by threshold. Most people with normal hearing have thresholds between −10 and +10 dB.

Dhingra, fig. 5.8

3 CHAPTER THREE Conductive loss

Sound that cannot get in.

3.1 Causes

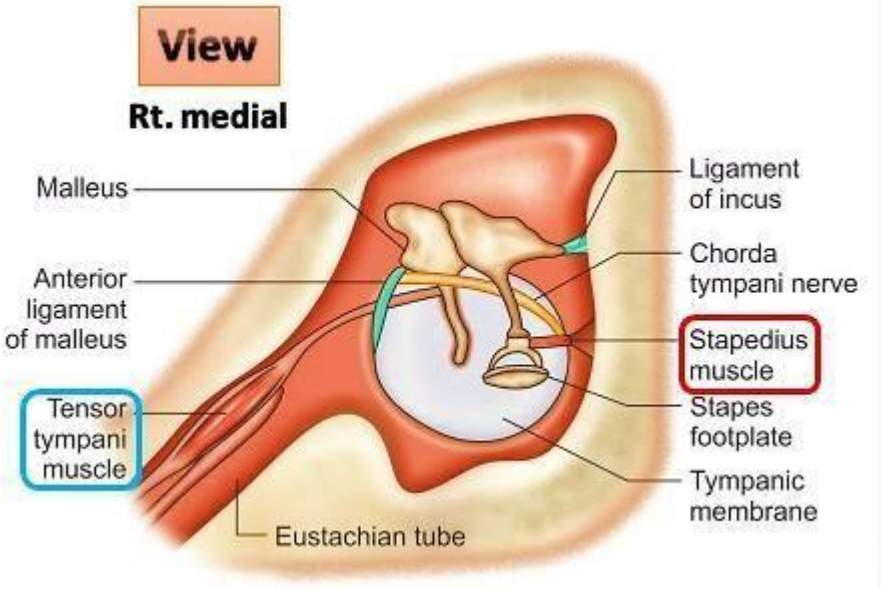


FIGURE 2 The drum, ossicular chain and intratympanic muscles: every conductive loss interrupts this chain somewhere.

Anatomy atlas · ENT Steps p. 51

3.1.1 Congenital

Canal atresia, fixation of the stapes or malleus head, ossicular discontinuity, and congenital cholesteatoma.

3.1.2 Acquired

- **Canal:** wax, foreign body, otitis externa, stenosis, tumours.
- **Drum:** perforation, scarring.
- **Middle ear:** fluid, blood, pus, mass, cholesteatoma, tympanosclerosis.
- **Ossicles:** disruption (trauma, erosion) or fixation (otosclerosis, tympanosclerosis).
- **Eustachian tube** failure.

3.2 Features

Low tones are affected more; speech discrimination is good; the patient may hear better in noise; voice is soft. A conductive loss alone never exceeds about 60 dB.

HOW BIG IS THE GAP?

- Complete canal block: about 30 dB.
- Perforation: 10 to 40 dB, more for larger and posterior holes.
- Malleus fixation: 10 to 25 dB.
- Ossicular discontinuity behind an intact drum: about 54 dB; with a perforation, about 38 dB, because an intact drum blocks sound reaching the round window.
- Closed oval window: about 60 dB.

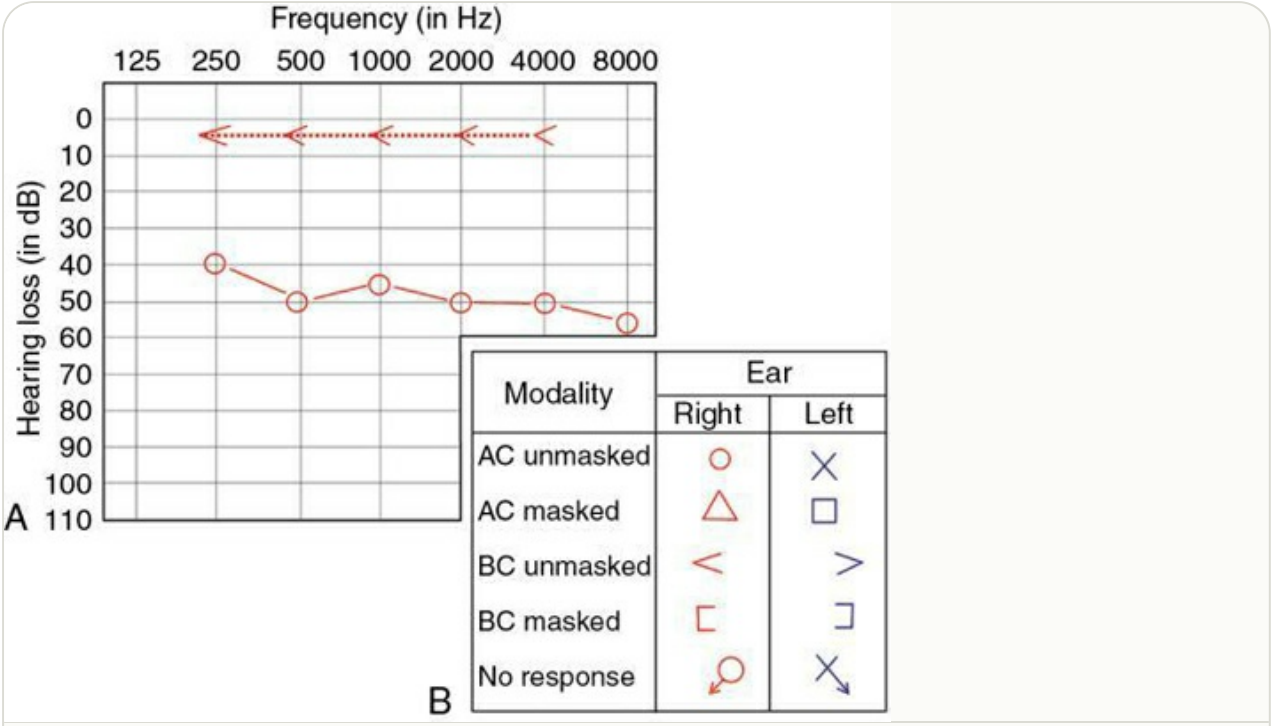


FIGURE 3 Audiogram of a right ear with a conductive loss and an air-bone gap, with the standard audiogram symbols.

Dhingra, fig. 5.1

3.3 Conductive loss behind an intact drum

When the drum looks normal, the list is **otosclerosis** (the commonest, with a family history, onset in early adult life, worsening in pregnancy, a Carhart notch at 2 kHz and absent stapedial reflexes), **ossicular discontinuity** after trauma, **malleus or**

incus fixation, **tympanosclerosis**, **congenital cholesteatoma** (a white mass behind an intact drum in a child), a small **effusion**, and **congenital ossicular anomalies**. A **third-window lesion** such as superior canal dehiscence produces an air–bone gap at low frequencies that is not truly conductive: bone conduction is better than normal, and the stapedial reflexes are present. Present reflexes with an apparent conductive loss should therefore stop a surgeon from operating on the stapes until a CT has excluded dehiscence.

3.4 Management

Clear the canal, drain fluid, remove a mass, repair the drum and chain (tympanoplasty), stapes surgery, or a hearing aid; bone-conduction devices suit patients who cannot wear an air-conduction aid, such as those with chronic discharge or canal atresia.

OPERATIVE GUIDES IN THEATRE 6 operations

Myringoplasty and Tympanoplasty
Tympanoplasty
Ossicular Chain Reconstruction
Ossicular reconstruction
Otosclerosis
Stapes surgery
Bone-Anchored Hearing Devices
Bone-anchored hearing
Implantable Middle Ear Devices
Implantable middle ear devices
Cochlear Implantation in Adults
Cochlear implant

4 CHAPTER FOUR Sensorineural loss

Damage at or beyond the cochlea.

4.1 Causes

CAUSES OF SENSORINEURAL LOSS

- CONGENITAL **Genetic, syndromic** intrauterine infection, prematurity, kernicterus
- INFECTION **Viral, meningitis** labyrinthitis, syphilis
- TRAUMA **Head injury, surgery** noise, barotrauma
- TOXIC **Ototoxic drugs**
- DEGENERATIVE **Presbycusis**
- INNER EAR **Ménière's disease** sudden hearing loss, autoimmune disease
- NEURAL **Vestibular schwannoma** multiple sclerosis, CP angle tumours
- SYSTEMIC **Diabetes, thyroid, renal disease** blood disorders, autoimmune disease

4.2 Features

High tones are usually affected more; discrimination is reduced, especially in noise; the loss can exceed 60 dB; there may be recruitment or tinnitus.

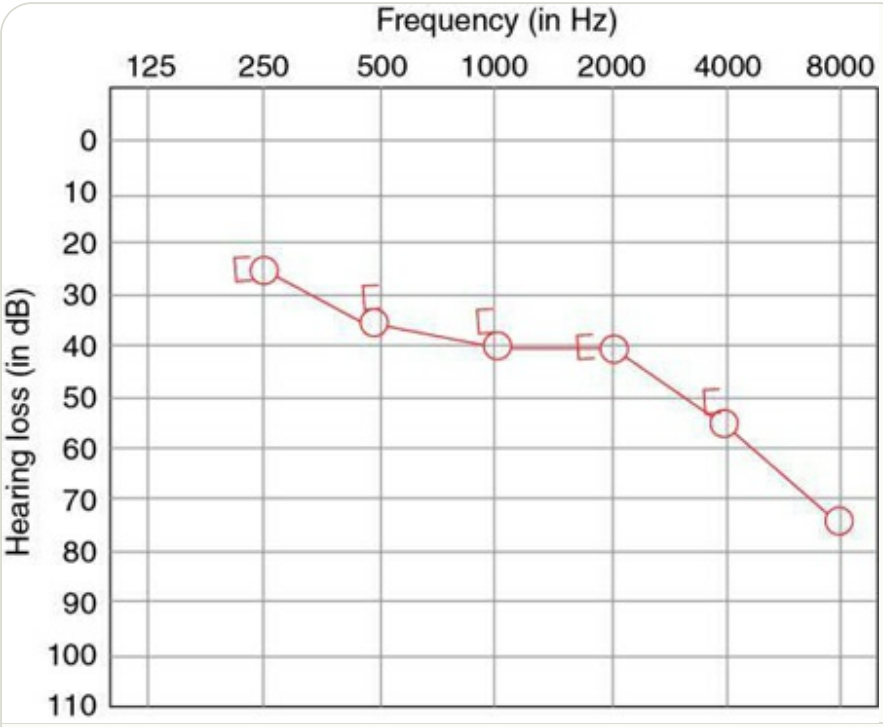


FIGURE 4 Audiogram of a right ear with sensorineural loss: air and bone conduction fall together, with no air-bone gap.

Dhingra, fig. 5.6

4.3 Cochlear or retrocochlear?

COCHLEAR OR RETROCOCHLEAR

TEST	COCHLEAR	RETROCOCHLEAR
Speech discrimination	Reduced, below 90%	Very poor, out of proportion
Roll-over	Absent	Present
Recruitment	Present	Absent
SISI	Over 70%	0–20%
Tone decay	Under 25 dB	Over 25 dB
Stapedial reflex	Present	Absent
Reflex decay	Absent	Present
ABR	Normal I–V	Wave V delayed or absent

Contrast MRI has largely replaced these special audiometric tests for excluding a retrocochlear lesion.

4.4 Management

Treat the cause where possible; protect what remains from noise and ototoxic drugs; then rehabilitate early with hearing aids or implants.

5 CHAPTER FIVE Assessment

Bedside to scanner.

5.1 History

Onset and speed, one or both ears, fluctuation, tinnitus, vertigo, otalgia and discharge, noise and drug exposure, head injury, family history, and general health.

5.2 Examination

Otoscopy under the microscope, tuning forks, and the cranial nerves.

TUNING FORK PATTERNS

TEST	CONDUCTIVE LOSS	SENSORINEURAL LOSS
Rinne	Negative (BC better than AC)	Positive (AC better than BC)
Weber	To the poorer ear	To the better ear
Absolute bone conduction	Normal	Reduced

5.3 Audiometry

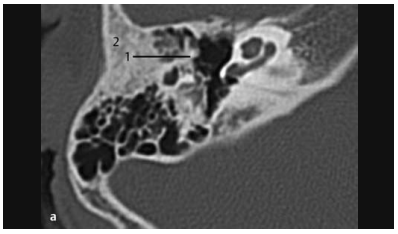
Pure-tone audiogram with masking, speech audiometry, tympanometry and stapedial reflexes; otoacoustic emissions and ABR when needed.

5.4 Further tests by suspicion

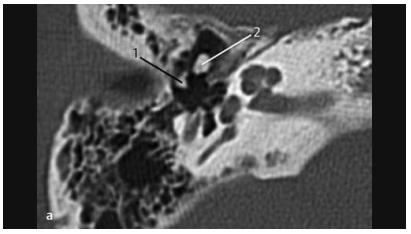
MRI with contrast for asymmetric or retrocochlear loss; CT for conductive loss with a normal drum, trauma or malformation; blood glucose, thyroid and renal function, full blood count and syphilis serology.

IMAGING IN THE APP 3 cases

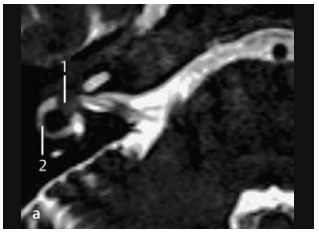
Ossicular fixation and luxation, and vestibular schwannoma.



Fixation of the Ossicular Chain
Bone disease · 3 images



Luxation of the Ossicular Chain
Trauma · 10 images



Vestibular Schwannoma
Tumours · 13 images

5.5 Non-organic hearing loss

CLUES TO NON-ORGANIC LOSS

- Inconsistent repeat audiograms (normal repeats agree within about 5 dB).
- No shadow curve with a claimed one-sided loss.
- Speech reception threshold more than 10 dB better than the pure-tone average.
- A positive Stenger test.
- Stapedial reflexes present despite claimed severe deafness.
- ABR or ASSR near the true threshold.

6 CHAPTER SIX Special situations

When to act fast.

6.1 Sudden sensorineural loss

An emergency: steroids the same day and MRI (see the sudden hearing loss topic).

6.2 Asymmetric sensorineural loss

MRI to exclude a vestibular schwannoma.

6.3 Treatable causes

Syphilis, hypothyroidism, autoimmune inner ear disease, serous labyrinthitis, perilymph fistula and ototoxic drugs.

6.4 Hearing loss in children

Universal **newborn hearing screening** uses automated otoacoustic emissions, followed by automated auditory brainstem responses for babies who do not pass, and all babies who spend time in neonatal intensive care have both. Permanent childhood hearing loss affects about 1 to 2 per 1000 births; about half is genetic

(most often *GJB2*, connexin 26), and congenital cytomegalovirus is the commonest non-genetic cause, tested within the first three weeks of life when antiviral treatment may help. Every child with permanent loss needs an aetiological work-up: imaging, genetics, CMV, ophthalmology, ECG for Jervell and Lange-Nielsen syndrome, and renal assessment when indicated. **Otitis media with effusion** is the commonest cause of acquired hearing loss in children and is managed by watchful waiting for three months, then hearing aids or ventilation tubes if a bilateral loss persists.

6.5 Age-related hearing loss

Presbycusis is a symmetrical high-frequency sensorineural loss. Untreated hearing loss in mid-life is the largest single modifiable risk factor for dementia identified by the Lancet Commission, and hearing aids are associated with reduced cognitive decline in those at risk; this is a strong reason to offer amplification early.

6.6 Implantable hearing

Cochlear implants are offered in the UK to adults and children whose hearing thresholds are 80 dB HL or worse at two or more frequencies between 500 and 4000 Hz in both ears and who gain inadequate benefit from hearing aids, measured by a sentence test score of 50% or less. **Bone-conduction implants**, **middle ear implants** and implants for single-sided deafness extend the options.

6.7 Single-sided deafness

Poor sound localisation and difficulty in noise, even with normal speech elsewhere. Bone-anchored devices, CROS aids or a cochlear implant help; the only hearing ear must be protected.

IN CLINIC

1. Do the tuning forks first: they take seconds.
2. Sudden sensorineural loss goes on the sudden hearing loss pathway today.
3. Asymmetric sensorineural loss needs MRI.
4. Look for treatable causes before fitting an aid.

5 Warn anyone with one hearing ear to protect it.

CHAPTER SEVEN

7 Pathway

From complaint to plan.

HEARING LOSS

1 Otoscopy and tuning forks

- → **wax, effusion, perforation or cholesteatoma**
  treat the ear
- → **conductive with a normal drum**
  audiogram, reflexes, CT if surgery considered
- → **sensorineural**
  step 2

2 Speed

- → **sudden, within 72 hours**
  sudden sensorineural loss pathway today
- → **fluctuating with vertigo**
  Ménière's pathway
- → **gradual**
  step 3

3 Symmetry

- → **asymmetric or unilateral**
  MRI of the internal auditory meatus
- → **symmetrical**
  age, noise, drugs, family history, systemic causes

4 Rehabilitation

- → hearing aids
- → bone-conduction device for discharging ears or atresia
- → cochlear implant if thresholds 80 dB or worse with poor aided speech

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Conductive versus sensorineural:** causes, features and fork tests.
- **WHO grades,** old and new.
- **Conductive loss behind an intact drum,** including third-window lesions.
- **Newborn screening** and the work-up of a deaf child.
- **Cochlear implant criteria.**
- **Cochlear versus retrocochlear** tests.
- **Non-organic hearing loss** and how to detect it.
- **Sizes of conductive loss** for common lesions.

8.2 Common pitfalls

COMMON PITFALLS

- Fitting a hearing aid to an asymmetric loss without MRI.
- Missing a sudden loss labelled as a blocked ear.
- Forgetting the treatable causes.

Written for this app from Oxford Textbook of Otolaryngology ch. 5, 11, 12, 83, Oxford Specialist Handbook ch. 3, 14, Oxford Handbook of ENT ch. 3, 4, Dhingra ch. 5. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for

patients.

CLINICAL · APPROACH TO

Approach to the discharging ear

A discharging ear is a sign as well as a symptom, and the discharge hides the diagnosis beneath it. The method is the same every time. Clean the ear under the microscope, look at the drum, and decide whether the discharge comes from the canal, through a perforation, or out of a retraction pocket. Then ask whether anything about the patient or the ear makes this more than a simple infection: diabetes, pain out of proportion, granulation tissue, blood, a facial palsy, vertigo, or a watery leak after a head injury.

AT A GLANCE

- Microsuction first: you cannot diagnose through pus.
- Mucoid = middle ear; scanty foul = cholesteatoma; watery after trauma = CSF.
- Diabetic, severe pain, granulation on the canal floor = necrotising otitis externa.
- Facial palsy, vertigo or headache with discharge = a complication.

1 CHAPTER ONE The first look

What the discharge tells you.

1.1 Character of the discharge

The discharge is not diagnostic, but it is a strong first clue.

WHAT THE DISCHARGE SUGGESTS

DISCHARGE	USUAL SOURCE	THINK OF
Mucopurulent, often profuse	Middle ear mucosa through a perforation or grommet	Acute otitis media with perforation; mucosal chronic otitis media; grommet otorrhoea
Scanty, foul-smelling, sometimes with white flakes	Keratin and anaerobes in a pocket	Cholesteatoma
Thin, serous or watery	Inflamed canal skin or drum surface	Chronic otitis externa, eczema, granular myringitis

DISCHARGE	USUAL SOURCE	THINK OF
Creamy debris with spores or hyphae	Canal	Otomycosis (white in *Candida*, black in *Aspergillus niger*, yellow in *A. flavus*)
Bloody	Trauma, granulation, polyp, tumour	Acute perforation, granulation tissue, aural polyp, carcinoma of the canal, glomus tumour
Clear and watery, profuse, worse on straining or bending forwards	Subarachnoid space	CSF otorrhoea after temporal bone fracture, surgery, or a spontaneous tegmen defect

Mucus is the useful discriminator: the canal has no mucous glands, so a mucoid discharge means the middle ear is open.

1.2 Red flags

WHEN A DISCHARGING EAR IS MORE THAN AN INFECTION

- **Diabetes, immunosuppression or old age with pain out of proportion and granulation on the canal floor:** necrotising otitis externa.
- **Facial weakness, vertigo, sensorineural loss, severe headache, fever, neck stiffness or drowsiness:** an intratemporal or intracranial complication.
- **Blood-stained discharge, a polyp or granulation that persists, or pain at night:** carcinoma of the canal or middle ear until biopsy says otherwise.
- **Clear watery discharge after head injury or surgery:** CSF leak.
- **Postauricular swelling pushing the pinna forwards:** mastoiditis with a subperiosteal abscess.

2 CHAPTER TWO History

Short questions with long answers.

2.1 The ear

Ask how long the ear has been discharging and whether this is a first episode or the latest of many. A short, painful history after swimming, scratching or a cold points to otitis externa or acute otitis media; a long history of intermittent painless discharge with hearing loss points to chronic otitis media.

Ask whether pain came before the discharge and was then relieved by it, which is the story of an acute otitis media that has perforated, or whether pain persists despite the discharge, which is otitis externa, furunculosis or something deeper. Ask about hearing loss, tinnitus, vertigo and facial weakness, about previous ear surgery or grommets, and about what the patient puts in the ear: cotton buds, hearing aids, ear plugs, water.

2.2 The patient

Diabetes, immunosuppression, chemotherapy and steroids all change the list. Ask about headache, fever and drowsiness. In an adult with a long-standing discharging ear, ask also about smoking and other head and neck symptoms, because pain and bleeding can be a sign of malignancy.

3 CHAPTER THREE Examination

Clean, then look.

3.1 The ear

Look at the pinna, the postauricular sulcus and the mastoid before touching the canal. Tenderness when the pinna is pulled or the tragus pressed is the sign of otitis

externa or a furuncle; a red, tender, oedematous pinna is cellulitis or, if the lobule is spared, perichondritis.

Then *clean the canal under the microscope or endoscope*. Meticulous microsuction is the key to diagnosis and to treatment, because drops cannot reach a canal full of debris and the drum cannot be judged through pus. Assess the canal skin for oedema, fungal debris, granulations on the floor and exposed bone, then the drum: intact, perforated centrally, perforated at the margin or in the attic, retracted, or obscured by a polyp.

Test the facial nerve every time. Perform the fistula test when there is an attic or posterosuperior lesion. Use tuning forks, and arrange audiometry for any ear that is not settling or where the inner ear may be involved.

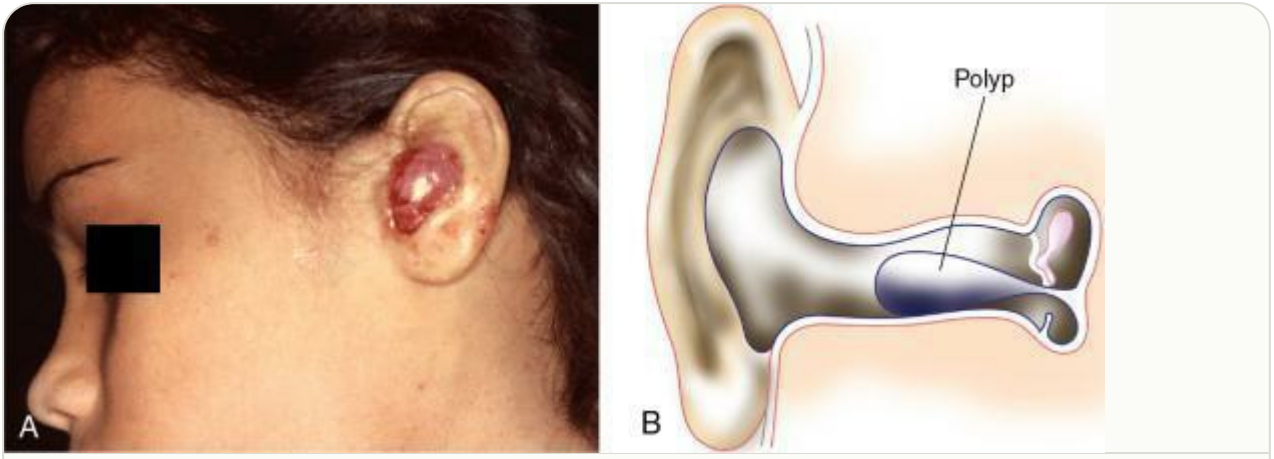


FIGURE 1 An aural polyp in the canal, and how a polyp from the promontory can present through a perforation.
Dhingra, fig. 11.9

3.2 Beyond the ear

If malignancy is possible, examine the neck, oral cavity and pharynx and pass a flexible endoscope. The nose and nasopharynx are worth examining in recurrent middle ear infection.

WHEN THE DRUM CANNOT BE SEEN

If the canal is too swollen or tender to clean fully, start antibiotic and steroid drops, with a wick if the canal is closed, and *bring the patient back* to finish the microsuction and see the drum. Too many cholesteatomas are treated for months as "otitis externa".

4 CHAPTER FOUR Causes

By site.

4.1 The canal

Acute otitis externa is bacterial in almost all cases, mostly *Pseudomonas aeruginosa* and *Staphylococcus aureus*. It is driven by water, cotton buds, narrow canals, hearing aids and dermatitis. It begins with itch and fullness, then pain, a swollen canal and discharge.

Otomycosis follows antibiotic drops or arises in humid climates, and its debris is often visibly fungal. **Furunculosis** is a staphylococcal abscess of a hair follicle in the outer canal and causes pain out of proportion to the findings. **Chronic otitis externa** means inflammation lasting more than three months or four acute episodes in a year; it is itchy more than painful and is often eczematous or due to contact allergy to drops. **Necrotising otitis externa** is osteitis of the skull base, most often in elderly diabetics. **Keratosis obturans**, **canal cholesteatoma** and **carcinoma** complete the canal list.

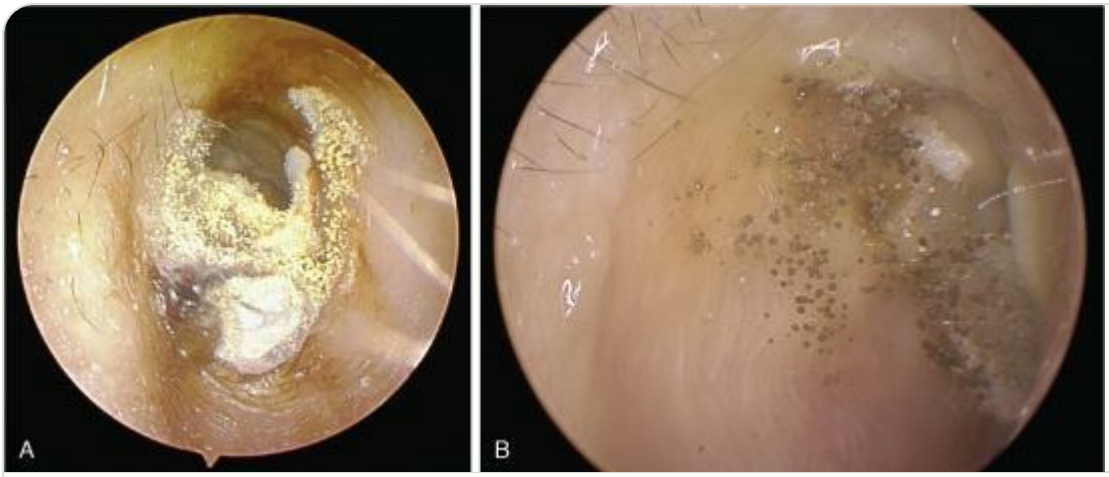


FIGURE 2 Otomycosis: (A) yellow and (B) black fungal debris in the canal.
Dhingra, fig. 8.11

4.2 The drum and middle ear

Acute otitis media perforates with sudden relief of pain and a mucopurulent discharge. **Grommet otorrhoea** is a perforation with a tube in it and is treated the same way. **Mucosal chronic otitis media** gives intermittent profuse mucoid discharge through a central perforation, usually after colds or water. **Cholesteatoma** gives scanty, foul, persistent discharge from an attic or marginal defect. **Granular myringitis** produces granulation on the outer surface of an intact drum, often after surgery, with watery or bloody discharge. **Tuberculous otitis media**, now rare, gives painless discharge through multiple perforations and fails to respond to the usual drops.

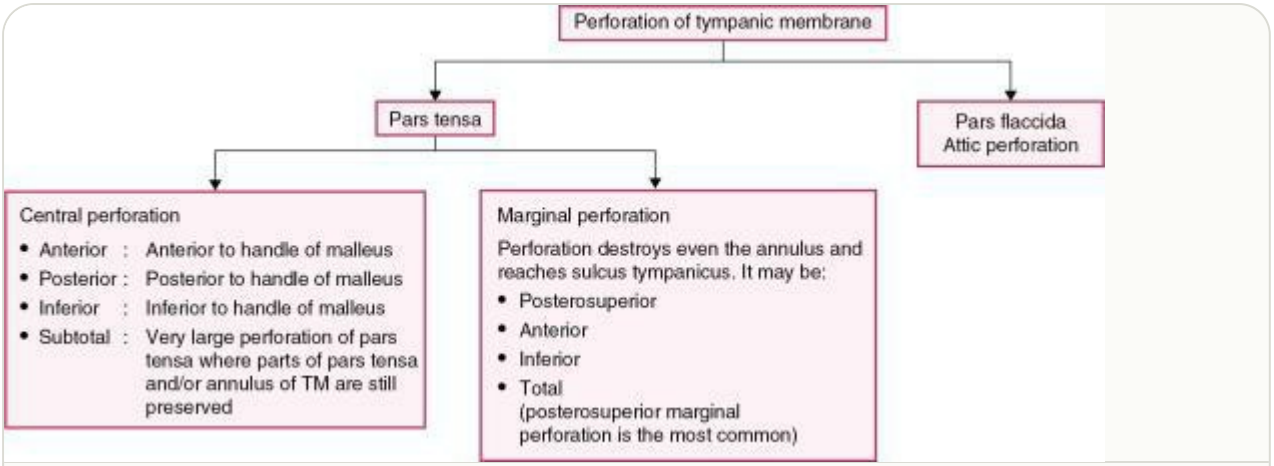


FIGURE 3 Sites of perforation: central perforations belong to the mucosal type; attic and posterosuperior marginal defects suggest cholesteatoma.

Dhingra, fig. 11.8

4.3 Beyond the ear

CSF otorrhoea follows temporal bone fracture, surgery, erosive disease and spontaneous tegmen defects linked to raised intracranial pressure; it is confirmed by beta-2 transferrin in the fluid. A **mastoid abscess** or a **parotid or skin process** can discharge into the canal.

WHERE THE BOOKS DIFFER · EAR DROPS WHEN THE DRUM IS OPEN

Dhingra

Topical antibiotic drops with steroid for both otitis externa and active chronic otitis media, with aural toilet.

Lee

Prefers non-ototoxic quinolone drops whenever the middle ear is open.

The Oxford Textbook takes the ENT-UK consensus view. The risk of an untreated infection to the inner ear outweighs the small risk of aminoglycoside drops through a perforation, provided the course is short and the patient is told to stop if dizziness or hearing loss appears. Quinolones are preferred where available, but quinolone-resistant *Pseudomonas* is a growing problem.

5 CHAPTER FIVE Pathway

From discharge to diagnosis.

THE DISCHARGING EAR

1 Clean the canal under the microscope

- → **canal cannot be cleaned**
  drops with a wick, review within days
- → **cleaned**
  look at the drum

2 Any red flag?

- → **diabetic with pain and granulation**
  CT, ESR or CRP, biopsy — treat as necrotising otitis externa
- → **palsy, vertigo or neurological signs**
  urgent CT, admit
- → **watery after trauma**
  beta-2 transferrin

3 Drum intact?

- → **yes, canal inflamed**
  otitis externa or otomycosis
- → **yes, granulation on the drum**
  granular myringitis
- → **yes, polyp or granulation in the canal**
  biopsy

4 Drum perforated?

- → **central, mucoid**
  mucosal chronic otitis media or perforated acute otitis media

→ **attic or marginal, foul**
cholesteatoma — CT

→ **grommet**
grommet otorrhoea

5 Not settling after two courses of appropriate drops?

→ swab for culture and fungi

→ biopsy granulation

→ CT temporal bone

→ consider tuberculosis

6 CHAPTER SIX Management

Scenario by scenario.

6.1 Otitis externa and otomycosis

Treatment is aural toilet, topical drops, analgesia and keeping the ear dry. Combination antibiotic and steroid drops for 7 to 10 days are the most effective treatment. When the canal is closed, an expanding wick is inserted with *aqueous* drops, which make it swell, and changed after two to seven days. Acidifying drops such as acetic acid suit mild cases and prevention.

Fungal otitis externa needs thorough removal of debris and two to three weeks of antifungal drops (clotrimazole) or acidifying drops. Systemic antibiotics are reserved for infection spreading beyond the canal: furunculosis, cellulitis, perichondritis (with antipseudomonal cover) and necrotising otitis externa. Water precautions follow, with ear plugs or cotton wool coated in petroleum jelly, and no cotton buds.

6.2 Active middle ear disease

A perforated acute otitis media in an adult needs analgesia and often oral amoxicillin. Grommet otorrhoea and active mucosal chronic otitis media need microsuction and topical antibiotic drops, with a swab only when the ear does not settle. Once the ear is dry, a perforation that keeps discharging is repaired by tympanoplasty. Cholesteatoma needs imaging and surgery, not repeated courses of drops.

6.3 Necrotising otitis externa and complications

Suspected necrotising otitis externa means admission, glycaemic control, culture before antibiotics, biopsy to exclude cancer, CT and MRI, and prolonged antipseudomonal treatment. Any complication of otitis media, such as mastoiditis, facial palsy, labyrinthitis or intracranial infection, needs intravenous antibiotics, imaging and usually drainage.

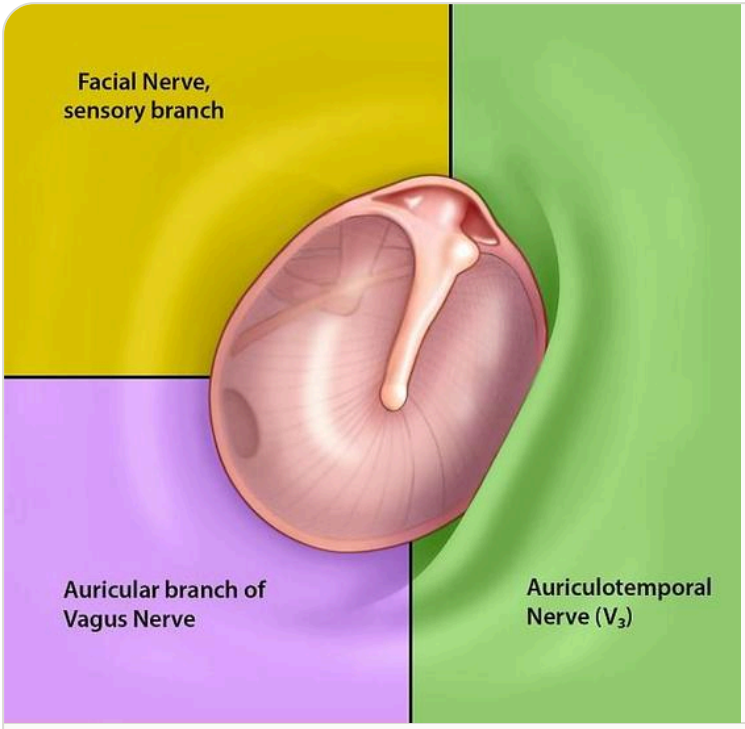


FIGURE 4 Sensory supply of the canal and drum: why canal disease is so painful.
Anatomy atlas · ENT Steps p. 35

6.4 Children

In children the discharging ear is usually acute otitis media with perforation or grommet otorrhoea. Grommet otorrhoea is treated with non-ototoxic quinolone drops after cleaning; recurrent episodes need a look at the adenoids and a swab. A child with a red, swollen, tender mastoid and an ear pushed forward has acute mastoiditis and needs admission, intravenous antibiotics and often myringotomy with or without cortical mastoidectomy.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Clean before you diagnose:** microsuction and a view of the drum.
- **Reading the discharge:** mucoid means the middle ear; foul and scanty means cholesteatoma; watery after trauma means CSF.
- **The red flags:** necrotising otitis externa, complications, malignancy.
- **Ototopical drops through a perforation:** the consensus position.
- **When to swab, scan and biopsy.**

7.2 Common pitfalls

COMMON PITFALLS

- Treating a canal full of pus with drops alone and never seeing the drum.
- Repeating antibiotic drops for months in what turns out to be cholesteatoma or carcinoma.
- Missing necrotising otitis externa in an elderly diabetic with "severe otitis externa".

- Using oil-based drops with an expanding wick.

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CLINICAL · APPROACH TO

Approach to earache

Earache has two sources. In *primary* otalgia the ear itself is diseased, and otoscopy shows it. In *referred* otalgia the ear is normal and the pain travels along one of the five nerves the ear shares with the teeth, jaw, throat, larynx and neck. About half of adult earache is referred, most often from the teeth and the temporomandibular joint, but in an adult smoker a painful ear that looks normal can be the first sign of a hidden cancer of the tongue base, tonsil, hypopharynx or larynx.

AT A GLANCE

- Normal ear, painful ear: look elsewhere.
- Routes are V, IX, X and C2–3.
- Exclude throat and larynx cancer in smokers.
- Treat the cause.

1 CHAPTER ONE Overview

Pain in the ear, cause in or beyond it.

1.1 Definitions

Otalgia is ear pain. It is **primary** (otogenic) when the cause lies in the external, middle or inner ear or the mastoid, and **referred** (secondary) when it arises in another structure that shares the ear's sensory supply.

1.2 The rule

If the canal, drum and hearing are normal, examine everything that shares the ear's nerves before calling the pain idiopathic or psychogenic, and keep the patient under review.

RED FLAGS IN EARACHE

- **An elderly or diabetic patient with severe pain, discharge and granulation on the canal floor:** necrotising otitis externa.

- **A normal ear with dysphagia, odynophagia, hoarseness, weight loss or a neck lump, especially in a smoker or drinker:** referred pain from upper aerodigestive tract cancer.
- **Facial weakness, vertigo or hearing loss with ear pain:** Ramsay Hunt syndrome, a complication of otitis media, or a tumour.
- **Pain with jaw claudication or scalp tenderness in an older patient:** giant cell arteritis.
- **Swelling behind the ear pushing the pinna forward:** mastoiditis.

CHAPTER TWO

2 Nerve supply

Why pain travels to the ear.

2.1 Five nerves

NERVES OF THE EAR AND WHERE THEIR PAIN COMES FROM

NERVE	SUPPLIES IN THE EAR	REFERRED FROM
Trigeminal (auriculotemporal)	Tragus, anterior canal and drum	Teeth, jaw joint, mouth, tongue, sinuses, parotid
Facial	Concha and posterior canal (sensory branch)	Geniculate ganglion, herpes zoster
Glossopharyngeal (tympanic)	Middle ear	Tonsil, base of tongue, pharynx, styloid, after tonsillectomy
Vagus (auricular, Arnold's)	Concha, posterior canal	Larynx, hypopharynx, vallecula, oesophagus
C2 and C3	Back of the pinna and mastoid skin	Cervical spine, neck injury

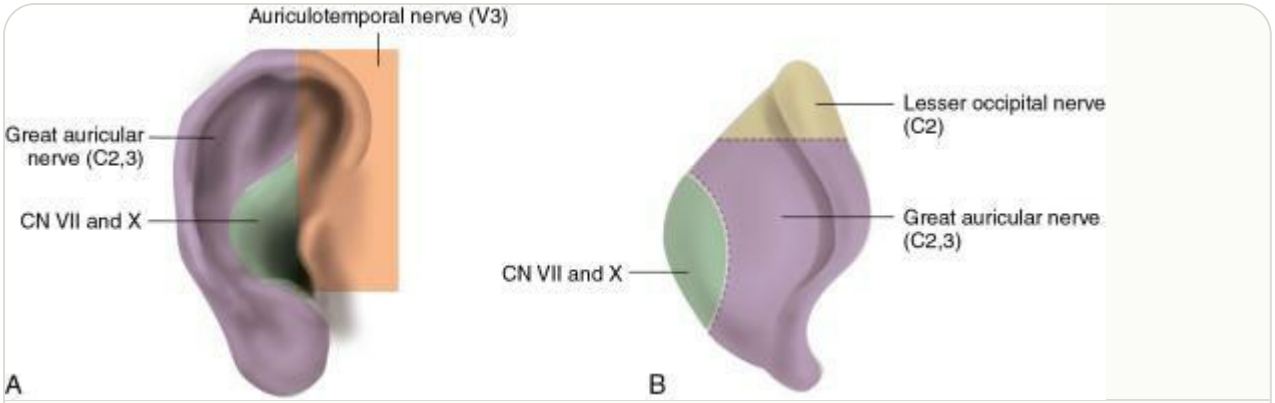


FIGURE 1 Sensory nerve supply of the pinna.
Dhingra, fig. 1.6

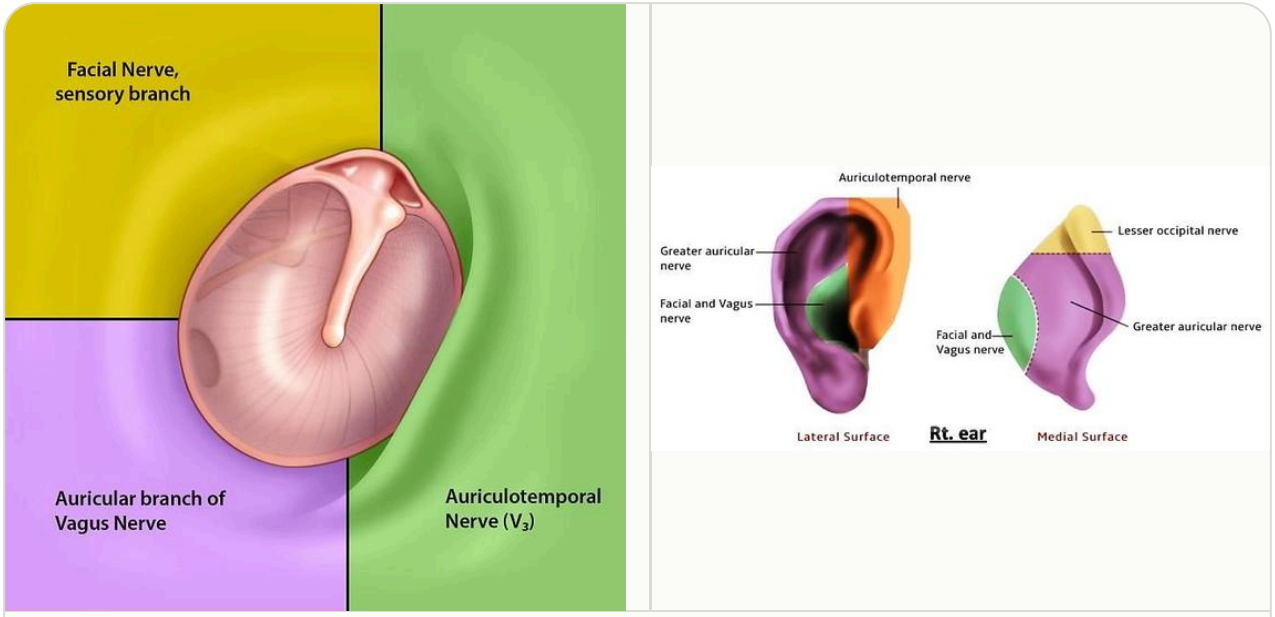


FIGURE 2 Sensory supply of the canal and drum, and of the auricle.
Anatomy atlas · ENT Steps p. 35, ENT Steps p. 30

3 CHAPTER THREE Primary otalgia

The ear is abnormal.

3.1 Causes in the ear

The **pinna** gives perichondritis, cellulitis, trauma, frostbite, chondrodermatitis nodularis helicis (a tender nodule on the helix that wakes the patient when lying on that side) and skin cancer. The **canal** gives acute otitis externa, furunculosis

(exquisite pain, a swelling in the outer canal), otomycosis, impacted wax or foreign body, necrotising otitis externa, keratosis obturans and carcinoma. The **middle ear** gives acute otitis media, which in adults is less common and should prompt a look at the nasopharynx, bullous myringitis, barotrauma, Eustachian tube dysfunction, mastoiditis and its complications, and carcinoma.

3.2 Herpes zoster oticus

Ramsay Hunt syndrome begins with deep, burning ear pain, often days before the vesicles in the concha, canal or palate, and is accompanied by facial palsy and sometimes hearing loss and vertigo. Early pain without a rash is easily mistaken for otitis externa or Bell's palsy.

4 CHAPTER FOUR Referred otalgia

The ear is normal.

4.1 Through the trigeminal nerve

Dental disease (caries, apical abscess, impacted third molars, cracked teeth) and **temporomandibular disorders** are the commonest causes of referred earache. Temporomandibular disorders cause preauricular pain worse with chewing, clicking, limited opening, bruxism and tender masticatory muscles; *Costen syndrome* links a faulty bite with earache, a blocked feeling, tinnitus and headache. Sinusitis, parotitis, oral ulcers and cancers of the tongue and floor of mouth also refer pain this way.

4.2 Through the glossopharyngeal nerve

Tonsillitis, quinsy, the days after tonsillectomy, cancer of the tonsil and tongue base, and an elongated styloid process or calcified stylohyoid ligament (**Eagle syndrome**), which causes throat and ear pain on swallowing and turning the head,

with a palpable styloid in the tonsillar fossa. **Glossopharyngeal neuralgia** causes paroxysmal stabbing pain in the throat and ear triggered by swallowing or talking.

4.3 Through the vagus

Cancer or inflammation of the vallecula, epiglottis, larynx, piriform fossa and oesophagus, and reflux. Pain from the hypopharynx is classically felt in the ear, sometimes before any other symptom.

4.4 Through C2 and C3

Cervical spondylosis, whiplash injury and other neck disease refer pain behind the ear.

4.5 Other causes

Giant cell arteritis, migraine, cardiac ischaemia (rarely), and **nervus intermedius (geniculate) neuralgia**, a paroxysmal deep ear pain. Psychogenic pain is diagnosed only after a full search.

5 CHAPTER FIVE Assessment

Look everywhere.

5.1 History

Ask about the character and timing of the pain, its relation to chewing, swallowing, talking and neck movement, discharge, hearing loss, itching and the water history. Ask about voice change, dysphagia, weight loss, smoking and alcohol, dental problems, bruxism and jaw clicking, neck pain and headaches.

5.2 Examination

After the pinna, otoscopy and tuning forks, examine the teeth and bite, palpate the temporomandibular joints and masticatory muscles while the patient opens and closes, and examine the mouth and tongue. Palpate the tongue base and tonsils with a gloved finger, examine the nasopharynx, and perform **flexible laryngoscopy** of the larynx and hypopharynx. Examine the neck for nodes, the parotid, the cervical spine and the temporal arteries, and test the cranial nerves.

THE SMOKER WITH A NORMAL EAR

An adult smoker or drinker with one-sided ear pain, a normal ear, and any throat symptom (dysphagia, hoarseness, a lump in the neck) needs endoscopy of the upper aerodigestive tract. Referred otalgia is cancer until the base of tongue, piriform fossa and larynx have been seen; if the examination is normal and the pain persists, image the neck and consider examination under anaesthesia.

5.3 Investigations

Investigations follow the findings: an orthopantomogram and dental review, imaging of the temporomandibular joint or styloid, CT or MRI of the neck and skull base, ESR and CRP for suspected arteritis, and biopsy of any lesion. In suspected necrotising otitis externa, CT and MRI with inflammatory markers and biopsy.

6 CHAPTER SIX Pathway

From pain to source.

EARACHE

1 **Otoscopy and hearing**

→ **abnormal ear**
primary otalgia — treat the ear disease

→ **diabetic or elderly with granulation**
necrotising otitis externa pathway

→ **vesicles or facial weakness**
Ramsay Hunt — steroid and antiviral

2 Normal ear

→ examine teeth, jaw joints, mouth, tongue base, tonsils, nasopharynx, larynx, neck, spine

3 Findings

→ **dental disease**
dentist

→ **joint tenderness and clicking**
temporomandibular disorder

→ **throat lesion or neck node**
biopsy and imaging

→ **triggered stabbing pain**
neuralgia

4 Nothing found

→ **smoker or drinker with persistent pain**
imaging and examination under anaesthesia

→ **others**
review, consider neuralgia or neck causes

7 CHAPTER SEVEN Management

Treat the source.

Treat the cause found. **Temporomandibular disorders** respond in most patients to explanation, a soft diet, warmth, simple analgesia, jaw exercises, avoidance of wide opening and gum chewing, and an occlusal splint from a dentist; a short course of low-

dose amitriptyline helps chronic muscle pain. **Eagle syndrome** is treated with analgesia and, if symptoms persist, styloidectomy through the tonsillar fossa or the neck. **Glossopharyngeal and nervus intermedius neuralgias** are treated like trigeminal neuralgia, with carbamazepine first and microvascular decompression or nerve section for refractory cases. Ear pain after tonsillectomy needs analgesia and reassurance.

OPERATIVE GUIDES IN THEATRE 2 operations

Transoral Removal of Elongated Styloid Process
Elongated styloid

Tonsillectomy
Tonsillectomy

PEARL

Ear pain after tonsillectomy is almost always referred pain through the glossopharyngeal nerve, not an ear infection.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The five nerves** and the sites that refer pain through each.
- **Referred otalgia in a smoker:** exclude cancer, including examination under anaesthesia.
- **Temporomandibular disorders** and Costen syndrome.
- **Eagle syndrome** and the neuralgias.
- **Primary otalgia red flags:** necrotising otitis externa, Ramsay Hunt syndrome.

8.2 Common pitfalls

COMMON PITFALLS

- Prescribing ear drops for a normal ear.
- Labelling pain psychogenic before examining the throat and larynx.
- Missing early Ramsay Hunt syndrome before the vesicles appear.

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CLINICAL · APPROACH TO

Approach to a blocked or full ear

"My ear feels blocked" is one of the commonest things patients say, and the cause can lie anywhere from the canal to the cochlea. Most blocked ears are wax, a middle ear effusion or a sluggish Eustachian tube. The ones not to miss are the sudden sensorineural hearing loss, which feels like a blocked ear and needs steroids within days, the unilateral effusion in an adult that hides a nasopharyngeal tumour, and the early Ménière's disease that announces itself as fullness with low-frequency loss.

AT A GLANCE

- Otoscopy and tympanometry before anything else.
- Sudden sensorineural loss presents as a blocked ear: test the hearing that day.
- Unilateral effusion in an adult: examine the nasopharynx.
- Autophony: patulous tube or superior canal dehiscence.

1 CHAPTER ONE Where is the block?

Canal, middle ear, tube or cochlea.

1.1 Four places

The sensation of fullness has four main sources. **The canal** contributes wax, debris, a foreign body, otitis externa or exostoses. **The middle ear** contributes an effusion, a retracted or perforated drum, haemotympanum or, rarely, a mass. **The Eustachian tube** fails to open (obstructive dysfunction) or stays open (patulous tube). **The inner ear** produces fullness in Ménière's disease, sudden sensorineural loss, superior canal dehiscence and other third-window lesions, and with raised intracranial pressure. Temporomandibular joint disorders and anxiety can also make an ear feel full.

DO NOT MISS

- **Sudden hearing loss with fullness:** test the hearing that day. A sensorineural loss of 30 dB or more over three frequencies within 72 hours needs steroids now and MRI later.

- **A unilateral middle ear effusion in an adult:** examine and, if in doubt, biopsy the nasopharynx.
- **Fullness with low-frequency loss and episodic vertigo:** Ménière's disease.
- **Fullness with pulsatile tinnitus, headache and visual blurring:** idiopathic intracranial hypertension.

2 CHAPTER TWO History

Small differences matter.

2.1 The questions

Ask whether the fullness came on suddenly or gradually, and whether it is constant or comes and goes. A *sudden* onset, especially on waking, with muffled hearing and tinnitus, is a sudden sensorineural loss until tested. Fullness after a cold, a flight or a dive suggests Eustachian tube dysfunction or barotrauma.

Ask whether the patient can "pop" the ear and whether that relieves it. Obstructive tube dysfunction improves when the ear pops; a patulous tube makes the patient hear their own voice and breathing loudly (*autophony*), improves when lying down or with a cold, and is common after weight loss and in pregnancy. Autophony with hearing one's own eye movements or footsteps, or vertigo with loud noise or straining, points to superior canal dehiscence.

Ask about episodic vertigo, fluctuating hearing and roaring tinnitus (Ménière's disease), nasal blockage, rhinitis and reflux (tube dysfunction), jaw pain and clicking (temporomandibular disorder), and in adults about epistaxis, nasal blockage, neck lumps and ethnicity (nasopharyngeal carcinoma).

3 CHAPTER THREE Examination and tests

Look, then measure.

3.1 Examination

Otoscopy shows wax, otitis externa, a perforation, a dull retracted drum with an air–fluid level or bubbles, or a blue drum of haemotympanum. Watch the drum during deep breathing: in a patulous tube it moves in and out with respiration. Tuning forks are essential, because they separate a conductive from a sensorineural loss at the bedside within minutes. With Weber to the blocked ear and Rinne negative, the loss is conductive; with Weber away from the blocked ear and Rinne positive, it is sensorineural.

Examine the nose and nasopharynx with an endoscope, and palpate the temporomandibular joints and the neck.

3.2 Tests

Pure tone audiometry is the key test. A conductive loss points to the canal or middle ear, a low-frequency sensorineural loss to hydrops, and a sudden flat or high-frequency sensorineural loss to sudden sensorineural hearing loss. **Tympanometry** shows a flat type B trace with normal canal volume in an effusion, a negative-pressure type C trace in tube dysfunction, and a normal type A trace in inner ear causes, with a large volume if the drum is perforated. Respiratory variation on tympanometry supports a patulous tube.

The **seven-item Eustachian Tube Dysfunction Questionnaire (ETDQ-7)** scores symptoms such as pressure, pain, a clogged feeling, symptoms with colds, popping, tinnitus and muffled hearing, and is used to select patients for treatment. **Tubomanometry** measures how the tube opens under nasopharyngeal pressure in specialist centres. **MRI** is used for an asymmetric sensorineural loss and to examine the nasopharynx, and **CT** for suspected dehiscence.

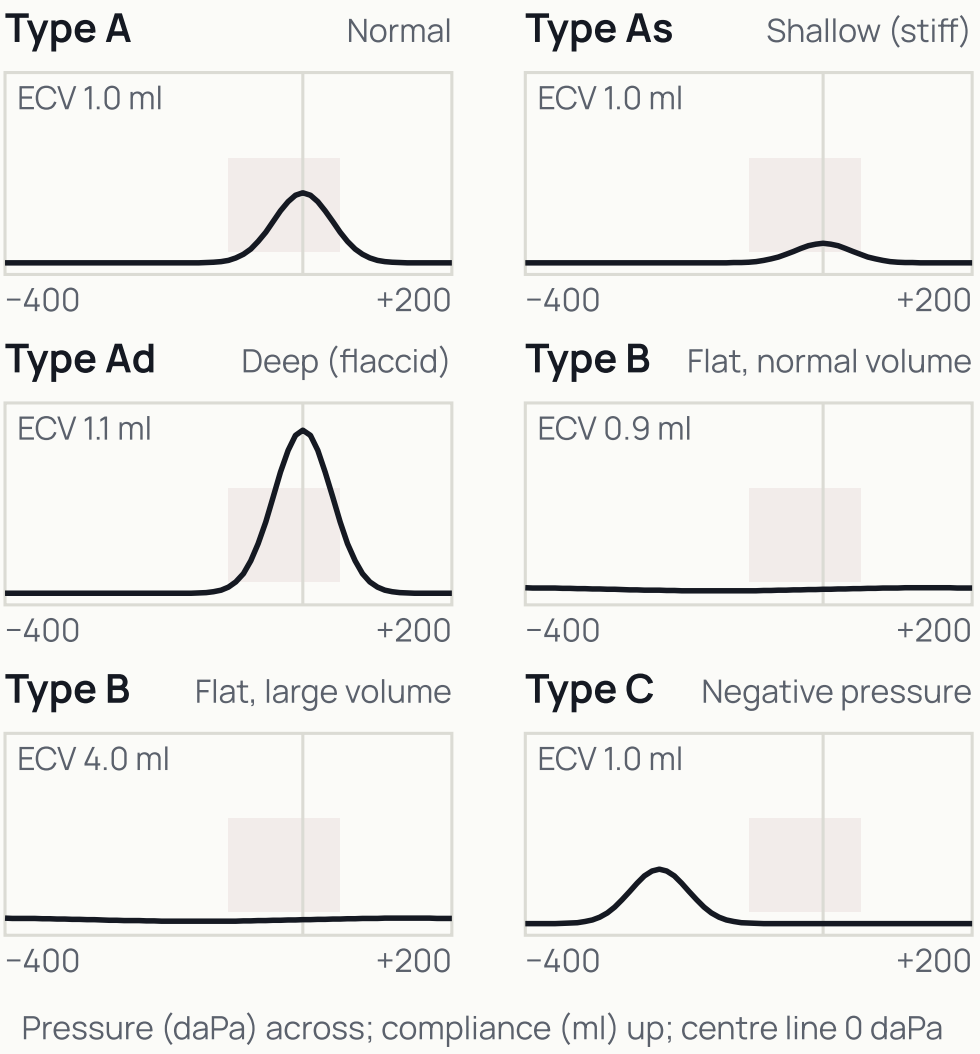


FIGURE 1 Tympanograms in the blocked ear: B with normal volume in an effusion, C in tube dysfunction, A in inner ear causes.

Drawn for ENT Atlas

4 CHAPTER FOUR Causes in turn

What to do for each.

4.1 Wax and the canal

Wax is the commonest cause. It is removed by microsuction, softening drops or irrigation, provided there is no perforation, previous surgery or infection. Otitis externa is treated as such.

4.2 Eustachian tube dysfunction

Obstructive dysfunction is usually *mucosal inflammation* in the cartilaginous tube, from allergic rhinitis, chronic rhinosinusitis, laryngopharyngeal reflux and smoke; in children it is adenoidal hypertrophy and infection, with cleft palate as an important cause. It affects about 1% of adults, and around 40% of children have a temporary episode before age ten.

Management starts with autoinflation (Valsalva or a nasal balloon) and treatment of the nose with saline and a steroid spray. Ventilation tubes treat persistent effusion and retraction. *Balloon dilation Eustachian tuboplasty* is an option for chronic obstructive and baro-challenged dysfunction: a balloon passed through the nose under endoscopic vision is inflated in the cartilaginous tube to about 10 bar for two minutes. Patients are selected by symptoms (ETDQ-7) and testing.

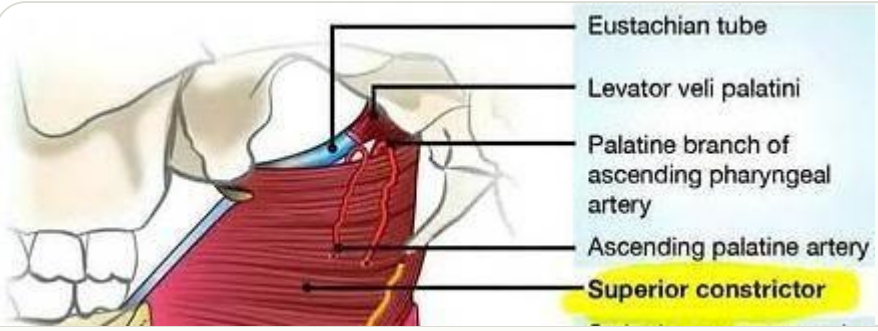


FIGURE 2 The Eustachian tube with the levator and tensor veli palatini, which open it on swallowing.

Anatomy atlas · ENT Steps p. 204

4.3 Patulous Eustachian tube

A patulous tube causes autophony, hearing one's own breathing, and fullness. It is linked to weight loss, pregnancy, hormones, radiotherapy and neuromuscular disease, and some patients "sniff" to close the tube, which can lead to retraction and cholesteatoma. Treatment is explanation, hydration, saline nasal drops and lying down, with augmentation or reconstruction of the tube for disabling cases.

4.4 Middle ear effusion in adults

An adult with a persistent unilateral middle ear effusion needs the nasopharynx examined with an endoscope, and an MRI or biopsy if anything looks abnormal, to

exclude *nasopharyngeal carcinoma*, especially in patients from endemic areas. Other causes are tube dysfunction after a cold, barotrauma, radiotherapy and sinonasal disease. Ventilation tubes relieve persistent effusions once the nasopharynx is clear.



FIGURE 3 Middle ear effusion: bubbles behind the drum on Valsalva.
Dhingra, fig. 10.2

4.5 The inner ear

Sudden sensorineural hearing loss often presents as a blocked ear with tinnitus; an audiogram confirms it and treatment with oral or intratympanic steroids begins at once, followed by MRI. **Ménière's disease** produces fluctuating fullness, roaring tinnitus and low-frequency loss before or during vertigo attacks. **Superior canal dehiscence** produces fullness, autophony, sound- or pressure-induced vertigo and an apparent conductive loss with normal tympanometry and preserved reflexes. **Raised intracranial pressure**, typically idiopathic intracranial hypertension, produces fullness with pulsatile tinnitus and headache.

THE BLOCKED OR FULL EAR

1 Otoscopy

→ wax or debris

remove and re-test

→ **effusion, retraction or haemotympanum**
go to step 3

→ **normal drum**
go to step 2

2 Tuning forks and audiogram today

→ **sensorineural loss, sudden**
steroids now, MRI

→ **low-frequency sensorineural, fluctuating, with vertigo**
Ménière's pathway

→ **normal hearing**
step 4

3 Effusion or negative pressure

→ **adult, unilateral, persistent**
endoscope the nasopharynx, biopsy if abnormal

→ **after a cold or flight**
autoinflation, nasal treatment, review

→ **persistent with hearing loss**
ventilation tube

4 Normal hearing and drum

→ **autophony, better lying down**
patulous tube

→ **popping and pressure with colds**
obstructive tube dysfunction — ETDQ-7, consider balloon dilation

→ **jaw clicking and tenderness**
temporomandibular disorder

→ **sound or pressure vertigo**
CT for dehiscence

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The four sources of fullness;** tuning forks and tympanometry to separate them.
- **Sudden sensorineural loss** presenting as a blocked ear.
- **The adult unilateral effusion** and the nasopharynx.
- **Obstructive versus patulous tube dysfunction:** symptoms, tests, treatment, balloon dilation.
- **Third-window disorders** that mimic conductive loss.

5.2 Common pitfalls

COMMON PITFALLS

- Telling a patient with sudden sensorineural loss that "it's just wax" or "Eustachian tube".
- Inserting a grommet in an adult unilateral effusion without examining the nasopharynx.
- Treating a patulous tube with decongestants, which make it worse.

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CLINICAL · APPROACH TO

Approach to tinnitus

Tinnitus is a sound with no outside source, and it is a symptom rather than a disease. Two kinds need imaging: tinnitus in one ear, and tinnitus that pulses with the heartbeat, because a tumour or a vascular lesion may lie behind them. Most other tinnitus needs good hearing care, explanation and sound therapy, not drugs.

AT A GLANCE

- Find the cause first.
- Pulsatile or one-sided tinnitus needs imaging.
- Hearing aids, sound therapy and counselling help most.
- Ask about mood.

1 CHAPTER ONE Overview

A sound no one else can hear.

1.1 Definition

The perception of sound, ringing, hissing, buzzing or roaring, in the absence of an external source.

1.2 Epidemiology

Between 10 and 15% of adults have tinnitus, but only about one in thirty find it at least moderately annoying and about one in two hundred are severely affected. Most have some hearing loss, often age-related or noise-induced. About half of those with troublesome tinnitus also have reduced tolerance of sound.

1.3 Why it matters

It can be the first symptom of a vestibular schwannoma, a glomus tumour or a dural arteriovenous fistula. And severe tinnitus disrupts sleep, concentration and mood.

2 CHAPTER TWO Classification

Who can hear it?

2.1 Subjective tinnitus

Heard only by the patient: by far the commonest.

2.2 Objective tinnitus

Can be heard by the examiner with a stethoscope or a Toynbee tube.

- **Vascular:** pulsatile, from arterial or venous flow.
- **Muscular:** clicking from palatal myoclonus, or from spasm of the stapedius or tensor tympani.
- **Patulous Eustachian tube:** a blowing sound with breathing.
- **Temporomandibular joint:** clicks with jaw movement.

2.3 A normal phenomenon

Sudden brief unilateral tapering tinnitus, a few seconds of ringing in one ear with fullness, happens to most people and needs no investigation.

3 CHAPTER THREE Causes

Ear, body and blood vessels.

3.1 Otological

Wax, middle ear fluid, otitis media, otosclerosis, Ménière's disease, presbycusis, noise-induced hearing loss, sudden hearing loss, ototoxicity and vestibular schwannoma.

3.2 General

Anaemia, hypertension, thyroid disease, hyperlipidaemia, vitamin B12 deficiency, head injury and multiple sclerosis.

3.3 Drugs

Aspirin and other salicylates, quinine, aminoglycosides, loop diuretics and some chemotherapy. Most drug-induced tinnitus reverses when the drug is stopped.

3.4 Pulsatile tinnitus

CAUSES OF PULSATILE TINNITUS

- ARTERIAL **Carotid stenosis, dissection, aneurysm** aberrant internal carotid artery, persistent stapedial artery
- VENOUS **High or dehiscent jugular bulb** sigmoid sinus diverticulum or dehiscence, idiopathic intracranial hypertension
- TUMOUR **Glomus tympanicum or jugulare** other vascular tumours
- SHUNT **Dural arteriovenous fistula** arteriovenous malformation
- HIGH OUTPUT **Anaemia, thyrotoxicosis, pregnancy** conductive hearing loss makes body sounds louder

3.5 How it arises

Reduced input from a damaged cochlea leads to reorganisation and increased spontaneous activity in the central auditory pathways. Everyone has some random activity in these pathways: in a silent room most people soon hear a faint hum or ringing. Normally, subconscious filters ignore it. When the brain labels the sound as a threat, the limbic and autonomic systems respond, attention is drawn to it, and a vicious circle of awareness, anxiety and amplification follows. This model explains why the loudness of tinnitus predicts distress so poorly, and why treatment aims at the reaction rather than the sound.

3.6 Disorders of sound tolerance

Hyperacusis is intolerance of everyday sounds that others find comfortable, often with tinnitus. **Misophonia** is a strong emotional reaction to specific sounds, often mouth sounds such as chewing, regardless of loudness. **Phonophobia** is fear of particular sounds, and **recruitment** is the abnormally rapid growth of loudness in a cochlear hearing loss. Facial palsy, by removing the stapedial reflex, can cause hyperacusis. Sound tolerance is treated with explanation, graded sound exposure and cognitive behavioural therapy; ear plugs worn constantly make it worse.

---

4 CHAPTER FOUR Assessment

Character, side, and mood.

4.1 History

Is it in one ear or both? Is it pulsatile? Is there hearing loss, vertigo or fullness? Noise exposure and drugs? How much does it disturb sleep, work and mood?

> **ASK ABOUT MOOD**
> Severe tinnitus is linked with anxiety and depression. Ask directly how the patient is coping, including any thoughts of self-harm, and arrange support where needed.

4.2 Examination

Otoscopy under the microscope (a red mass behind the drum, a blue drum, wax), tuning forks, cranial nerves, and auscultation over the ear, orbit and neck. Pressing on the carotid stops arterial tinnitus; pressing on the jugular vein or turning the head stops a venous hum. A rhythm with breathing points to a patulous tube; clicks with palatal movement mean myoclonus.

4.3 Investigations

4.3.1 Audiometry

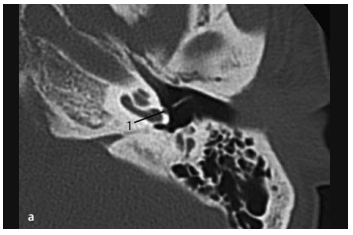
Pure-tone and speech audiometry, tympanometry, and tinnitus matching for pitch and loudness. Loudness and pitch do not predict how distressing it is.

4.3.2 Imaging

- **One-sided tinnitus or asymmetric hearing:** MRI of the internal auditory canals.
- **Pulsatile tinnitus:** CT angiography, MR angiography and venography; formal angiography if a dural fistula is suspected.
- **Palatal myoclonus:** MRI of the posterior fossa.

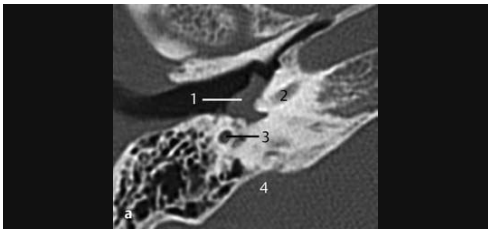
IMAGING IN THE APP 3 cases

A persistent stapedial artery, glomus tympanicum and vestibular schwannoma.



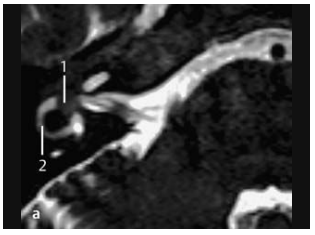
Persistent Stapedial Artery

Variants · 4 images



Glomus Tympanicum (Paraganglioma)

Tumours · 11 images



Vestibular Schwannoma

Tumours · 13 images

4.3.3 Blood tests

As suggested by the history: full blood count, thyroid function, lipids, glucose, B12.

4.3.4 Long-duration tympanometry

Shows rhythmic compliance changes in myoclonus or a patulous tube.

5 CHAPTER FIVE Management

Treat the cause, then the reaction.

5.1 Treat the cause

Remove wax, treat otitis media, operate on otosclerosis, stop ototoxic drugs where possible, and treat anaemia or thyroid disease.

5.2 Hearing aids

The first step whenever there is hearing loss: restoring external sound reduces the contrast with the tinnitus.

OPERATIVE GUIDES IN THEATRE 3 operations

Bone-Anchored Hearing Devices
Bone-anchored hearing

Implantable Middle Ear Devices
Implantable devices

Cochlear Implantation in Adults
Cochlear implant

5.3 Explanation and counselling

Explaining that tinnitus is common, real and not a sign of a brain tumour (once excluded) relieves much of the distress.

5.4 Sound therapy

5.4.1 Masking

Background sound, a bedside sound generator or a wearable masker, alone or combined with a hearing aid.

5.4.2 Tinnitus retraining therapy

Directive counselling plus low-level broadband sound, aiming at habituation: the tinnitus is still there but no longer noticed or disturbing.

WHERE THE BOOKS DIFFER · RETRAINING SOUND EXPOSURE

Dhingra	Lee
At least 8 hours a day for 18 to 24 months.	Up to 16 hours a day for about a year.

5.5 Psychological therapy

Cognitive behavioural therapy has the strongest evidence for reducing distress; relaxation and biofeedback also help. Treat anxiety, depression and insomnia.

5.6 Specific objective causes

- **Dural arteriovenous fistula:** embolisation, surgery or radiosurgery; fistulas with cortical venous drainage can bleed.
- **Palatal myoclonus:** muscle relaxants or botulinum toxin.
- **Stapedial or tensor spasm:** muscle relaxants, or division of the tendon.
- **Patulous tube:** see the Eustachian tube topic.

5.7 Treatments without proven benefit

No drug reliably suppresses tinnitus, and betahistine is not recommended for it. Transcranial magnetic stimulation and implanted stimulators remain investigational. A cochlear implant often reduces tinnitus in patients who qualify for it because of their hearing loss.

5.8 Stepped care

Severity is measured with a validated questionnaire such as the **Tinnitus Handicap Inventory** or the Tinnitus Functional Index. Most patients need only explanation, hearing care and self-help; those with greater distress move to audiologist-led tinnitus support and sound therapy, and then to cognitive behavioural therapy delivered by a psychologist, with access to mental health services for those at risk.

6 CHAPTER SIX Pathway

From symptom to plan.

TINNITUS

1 Pulsatile?

- → **yes, audible to examiner or not**
  otoscopy, auscultation, CT angiography or MR angiography and venography
- → **young woman with obesity and headache**
  fundoscopy — idiopathic intracranial hypertension
- → **red mass behind the drum**
  glomus — imaging

2 Unilateral or asymmetric hearing?

- → **yes**
  MRI of the internal auditory meatus
- → **sudden hearing loss with it**
  treat as sudden sensorineural loss

3 Bilateral, non-pulsatile

- → audiogram
- → treat wax, effusion, otosclerosis
- → hearing aids for hearing loss

4 Distress

- → explanation and self-help
- → sound therapy and tinnitus counselling
- → cognitive behavioural therapy
- → urgent help if risk of self-harm

7 CHAPTER SEVEN Prognosis

Most get better with time.

About a quarter of patients find their tinnitus nearly resolves, half improve, and a quarter stay the same. Habituation is the realistic goal.

PEARL

Most people adapt to tinnitus. What a clinician says in the first consultation strongly shapes how well they do.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Subjective versus objective** tinnitus and examples of each.
- **Causes of pulsatile tinnitus,** and how to investigate them.
- **Unilateral tinnitus:** MRI for schwannoma.
- **Tinnitus retraining therapy** and the aim of habituation.
- **Hyperacusis and misophonia.**

- **Stepped care** and cognitive behavioural therapy.

8.2 Common pitfalls

COMMON PITFALLS

- Telling a patient that nothing can be done.
- Missing pulsatile tinnitus because nobody asked.
- Forgetting to image one-sided tinnitus.
- Ignoring the patient's mood.

Written for this app from Oxford Textbook of Otolaryngology ch. 6, Oxford Specialist Handbook ch. 3, 18, Oxford Handbook of ENT ch. 3, 7, Dhingra ch. 5, 22 and K. J. Lee ch. 18. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · APPROACH TO

Approach to dizziness and vertigo

"Dizzy" means different things to different patients: spinning, swaying, light-headedness, or a vague sense that the world is not right. The diagnosis rarely comes from the word they use. It comes from the timing and the triggers. Seconds of vertigo on rolling over in bed is BPPV. Hours of vertigo with a full, roaring ear is Ménière's disease, and similar attacks with headache or light sensitivity are vestibular migraine. Days of continuous vertigo is vestibular neuritis, or a stroke. Chronic daily unsteadiness after an earlier vestibular upset is persistent postural-perceptual dizziness. Most dizzy patients can be diagnosed from the history and the eyes, and the one you must not miss is the cerebellar stroke.

AT A GLANCE

- Classify by timing and trigger: episodic triggered, episodic spontaneous, acute continuous, chronic.
- Dix–Hallpike for positional vertigo; HINTS for acute continuous vertigo.
- A normal head impulse in acute vertigo is a red flag for stroke.
- Vestibular migraine is the commonest cause of spontaneous episodic vertigo.

1 CHAPTER ONE The history

Timing and triggers.

1.1 What kind of dizziness?

Ask the patient to describe the sensation without using the word "dizzy". **Vertigo** is an illusion of movement, usually spinning, and points to the vestibular system. **Disequilibrium** is unsteadiness on the feet without a feeling in the head, typical of bilateral vestibular loss, neuropathy and cerebellar disease. **Presyncope** is light-headedness and a sense of impending faint, typical of orthostatic and cardiac causes. **Visual or motion sensitivity**, feeling worse in supermarkets, crowds or scrolling screens, suggests migraine, persistent postural-perceptual dizziness or an uncompensated vestibular loss. Vestibular migraine can produce any of these.

1.2 Timing

TIMING POINTS TO THE CAUSE

PATTERN	TYPICAL DURATION	MOST LIKELY
Episodic, triggered by head position	Seconds (under a minute)	BPPV
Episodic, triggered by pressure, straining or loud sound	Seconds	Superior canal dehiscence, perilymph fistula
Episodic, spontaneous, many brief spells a day	Seconds to a minute	Vestibular paroxysmia
Episodic, spontaneous, with fluctuating hearing, tinnitus and fullness	20 minutes to 12 hours	Ménière's disease
Episodic, spontaneous, with headache, photophobia or visual aura (though not always)	5 minutes to 72 hours	Vestibular migraine
Single acute episode, continuous, with nausea and imbalance	Days, then slowly recovers	Vestibular neuritis (with hearing loss: labyrinthitis); posterior circulation stroke
Transient attacks with other brainstem symptoms	Minutes	Vertebrobasilar TIA
Chronic, most days for more than three months	Persistent	Persistent postural-perceptual dizziness, uncompensated or bilateral vestibular loss, multisensory imbalance of the elderly

1.3 Triggers and associations

Ask what brings the dizziness on. Rolling over in bed, looking up or bending down suggests BPPV. Standing up suggests orthostatic hypotension. Straining, lifting, coughing or loud noise suggests a third window. Sleep loss, stress, caffeine, menstruation and bright light suggest migraine.

Then ask about the ear and the brain. Hearing loss, tinnitus and fullness put the labyrinth at the centre, and fluctuating hearing suggests Ménière's disease. But

acute vertigo with sudden hearing loss can also be an anterior inferior cerebellar artery stroke, because that artery supplies both the labyrinth and the lateral pons. Headache, double vision, dysarthria, dysphagia, numbness, weakness and falls point to a central cause. Ask about migraine past and present, motion sickness as a child, anxiety, ototoxic drugs (gentamicin), head injury, recent ear surgery and barotrauma.

CENTRAL RED FLAGS IN THE DIZZY PATIENT

- Cannot stand or walk unaided during an attack.
- New headache, especially occipital, or neck pain (vertebral dissection).
- Diplopia, dysarthria, dysphagia, facial numbness, limb weakness or ataxia.
- Direction-changing, vertical or purely torsional nystagmus, or nystagmus not suppressed by fixation.
- A *normal* head impulse test in a patient with acute continuous vertigo and nystagmus.
- Vascular risk factors with acute vertigo and sudden hearing loss.

2 CHAPTER TWO Examination

The eyes tell you where.

2.1 Eyes and head

Look for spontaneous nystagmus with the eyes in the primary position and on gaze to each side, with and without fixation (Frenzel lenses remove fixation). **Peripheral** nystagmus is horizontal-torsional, *direction-fixed* with the fast phase away from the affected ear, increases on gaze towards the fast phase (Alexander's law) and is suppressed by fixation. **Central** nystagmus may change direction with gaze, be purely vertical or torsional, and is not suppressed by fixation.

The **head impulse test** looks at the vestibulo-ocular reflex. The patient fixes on your nose while you turn the head briskly through about 15° to each side. A corrective saccade back to the target after a turn towards one side means that canal is weak. The **test of skew**, alternate covering of each eye, reveals a vertical

refixation in skew deviation, a central sign. Check smooth pursuit and saccades, which are normal in peripheral disease.

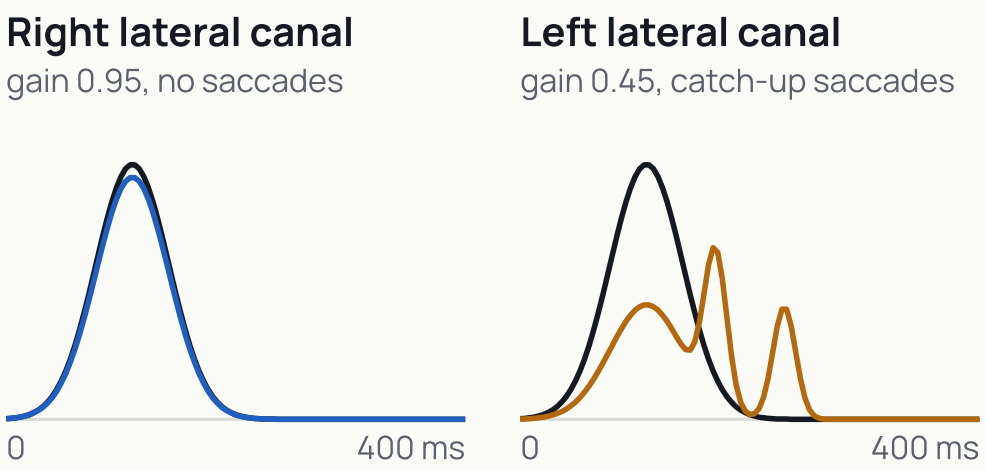


FIGURE 1 The head impulse at the bedside and on video: a weak canal shows low gain and catch-up saccades.

Drawn for ENT Atlas

2.2 Positional testing

The **Dix–Hallpike test** looks for posterior canal BPPV. Turn the head 45° towards the test ear, lie the patient back quickly with the head hanging about 20° below horizontal, and watch the eyes for 30 seconds. A positive test gives upbeating, torsional nystagmus with the upper pole of the eye beating towards the lower ear. It starts after a latency of a few seconds, lasts under a minute, reverses on sitting up and fatigues with repetition. The **supine roll test**, turning the head to each side while lying flat, looks for horizontal canal BPPV, which gives horizontal geotropic or apogeotropic nystagmus.

Positional nystagmus that has no latency, does not fatigue, is purely vertical (especially downbeating), or comes without vertigo suggests a central lesion.

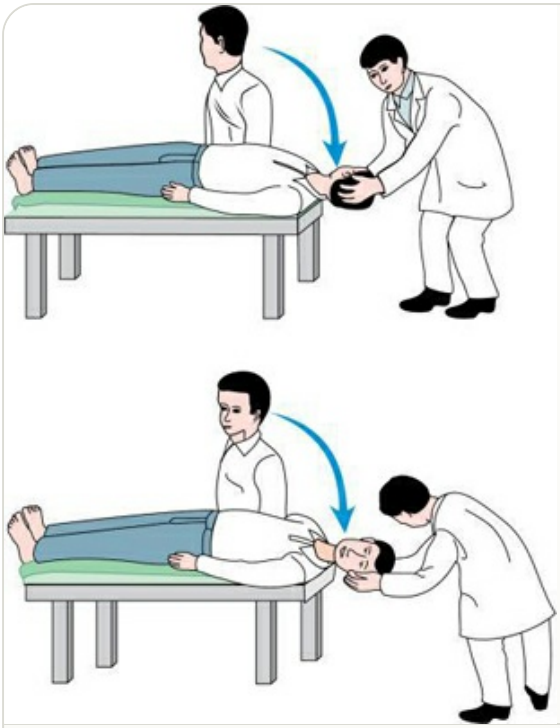


FIGURE 2 The Dix–Hallpike manoeuvre.
Dhingra, fig. 6.1

2.3 The rest of the examination

Otoscopy, tuning forks and a fistula test (pressure on the tragus or with a pneumatic speculum) cover the ear. A cranial nerve and cerebellar examination covers the brain: finger–nose, heel–shin, rapid alternating movements, Romberg and tandem gait. Lying and standing blood pressure looks for orthostatic hypotension, defined as a fall of at least 20 mmHg systolic or 10 mmHg diastolic within three minutes of standing (some definitions allow up to ten); a heart rate rising by 30 or more without hypotension suggests postural tachycardia syndrome. Finally, ask the patient to walk.

3 CHAPTER THREE Acute continuous vertigo

Ear or brain?

3.1 HINTS

In a patient who has been continuously vertiginous for hours or days, with nausea and spontaneous nystagmus, the question is whether this is vestibular neuritis or a stroke in the posterior circulation, which accounts for roughly one in ten such patients. **HINTS** (Head Impulse, Nystagmus, Test of Skew), done by a trained examiner, is more sensitive for stroke in the first 48 hours than early MRI, which misses small infarcts.

A *peripheral* pattern is reassuring only when all three agree: an *abnormal* head impulse test (a corrective saccade towards the affected side), direction-fixed horizontal nystagmus, and no skew. *Any* central feature (a normal head impulse, direction-changing nystagmus or skew) means stroke until proved otherwise. **HINTS-plus** adds new hearing loss as a central warning, because of anterior inferior cerebellar artery infarcts. Inability to stand unaided is an additional central sign.

HINTS IN ACUTE CONTINUOUS VERTIGO

TEST	PERIPHERAL (NEURITIS)	CENTRAL (STROKE)
Head impulse	Abnormal: catch-up saccade	Normal
Nystagmus	Direction-fixed, horizontal-torsional, suppressed by fixation	Direction-changing on gaze, vertical or purely torsional
Test of skew	No vertical refixation	Vertical refixation (skew)
Plus: hearing	Usually normal (labyrinthitis if lost)	New loss suggests AICA infarct
Gait	Leans but can stand	Cannot stand unaided

3.2 Management

A central pattern needs urgent MRI with diffusion-weighted imaging and the stroke team. A peripheral pattern is managed with short-term vestibular sedatives and antiemetics (prochlorperazine, cyclizine) for no more than a few days, fluids, and early mobilisation with vestibular rehabilitation exercises. A short course of oral steroids may be given early for vestibular neuritis. Sedatives slow compensation if they are continued.

CHAPTER FOUR

4 Recurrent episodic vertigo

The big three.

4.1 BPPV

BPPV is the commonest cause of vertigo: brief spinning on head movement, often first noticed on getting out of bed, typically in older adults and after head injury or a period in bed. The posterior canal is affected in about 85 to 90%. The diagnosis is made by the Dix–Hallpike test, and the cure is a repositioning manoeuvre: the *Epley* (or Semont) manoeuvre for the posterior canal, and the *Lempert (barbecue) roll* or Gufoni manoeuvre for the horizontal canal. Recurrence is common, so patients can be taught self-treatment. Medication has no role, and atypical or refractory cases need MRI.

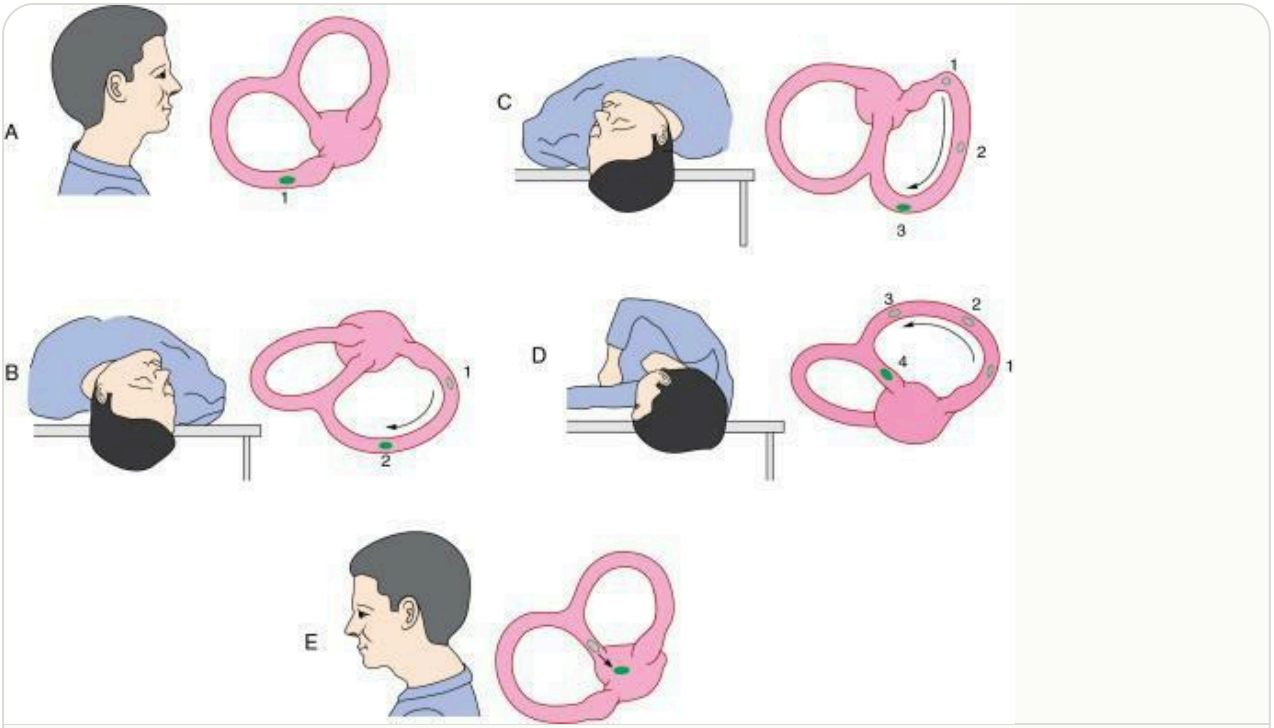


FIGURE 3 The Epley manoeuvre for posterior canal BPPV, with the position of the debris in the canal at each step.

Dhingra, fig. 7.1

4.2 Ménière's disease

Definite Ménière's disease (Bárány Society) requires two or more spontaneous episodes of vertigo, each lasting 20 minutes to 12 hours, an audiometrically documented low- to mid-frequency sensorineural loss in the affected ear on at least one occasion, and fluctuating aural symptoms (hearing, tinnitus or fullness) in that ear, not better explained by another diagnosis. An audiogram is essential, and asymmetric sensorineural loss means MRI.

Treatment aims to reduce the number and severity of attacks. It runs from acute vestibular sedatives through lifestyle measures (moderation of salt, caffeine and alcohol, though the evidence is weak), betahistine (in widespread use despite a large trial showing no benefit over placebo) and diuretics, to intratympanic steroid, intratympanic gentamicin and, rarely, surgery.

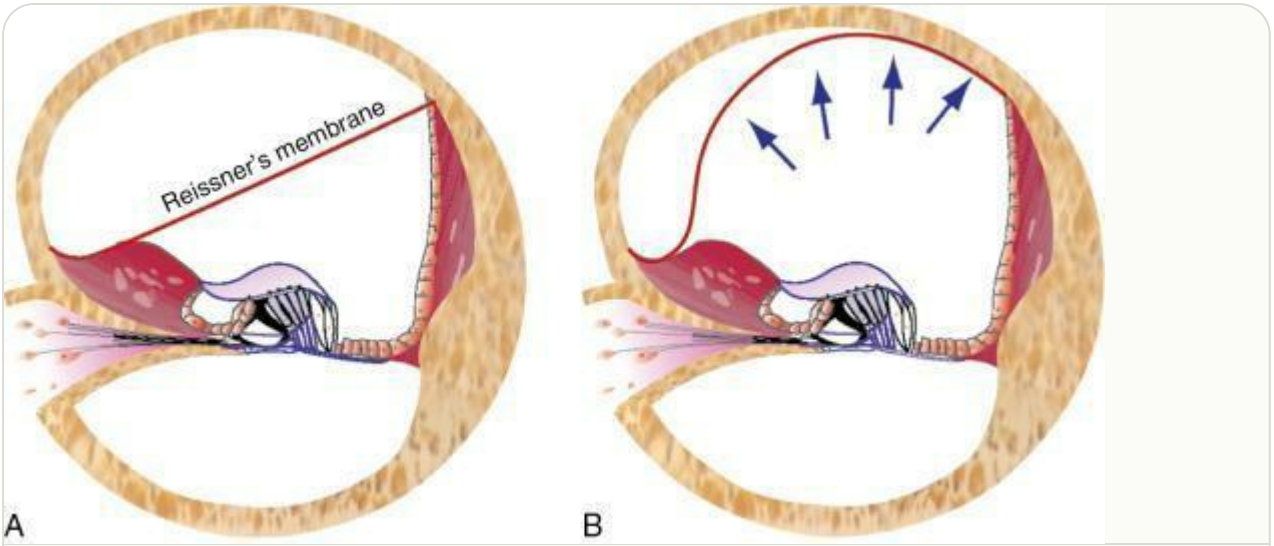


FIGURE 4 (A) The normal cochlear duct. (B) Endolymphatic hydrops: the duct distended and Reissner's membrane bulging.

Dhingra, fig. 15.1

4.3 Vestibular migraine

Vestibular migraine is the commonest cause of spontaneous recurrent vertigo, and it is underdiagnosed because many attacks come without headache. *Definite* vestibular migraine (Bárány Society and International Headache Society) requires at least five episodes of moderate or severe vestibular symptoms lasting 5 minutes to 72 hours. The patient must have current or previous migraine, and at least half the episodes must carry at least one migrainous feature: a migraine-type headache, photophobia with phonophobia, or a visual aura.

There is no confirmatory test, so the diagnosis is clinical. Treatment begins with sleep, regular meals, hydration and the identification of triggers such as caffeine and alcohol. Acute attacks are treated with prochlorperazine and, for headache, a triptan with an NSAID and an antiemetic. Frequent attacks need prophylaxis: propranolol, amitriptyline or nortriptyline, or candesartan. The trial evidence comes from migraine headache rather than vestibular migraine itself, and CGRP antibodies are emerging.

MÉNIÈRE'S OR VESTIBULAR MIGRAINE?

The two overlap more than the textbooks admit, and some patients have both. Documented, progressive low-frequency hearing loss favours Ménière's disease. Headache, photophobia, visual aura, motion sensitivity, a long migraine history and

attacks lasting more than 12 hours or only minutes favour migraine. When in doubt, a trial of migraine prophylaxis is safe and often diagnostic.

4.4 Less common episodic causes

Superior canal dehiscence causes vertigo with straining or loud sound, autophony and pulsatile tinnitus, with low bone conduction thresholds; it is confirmed by VEMPs and CT. **Perilymph fistula** follows stapes surgery, head injury or barotrauma. **Vestibular paroxysmia**, from a vascular loop compressing the eighth nerve, causes many brief spells a day and responds to carbamazepine. **Episodic ataxia type 2** (CACNA1A) is rare and responds to acetazolamide. **Vertebrobasilar TIAs** cause brief vertigo with other brainstem symptoms in patients with vascular risk.

5 CHAPTER FIVE Chronic dizziness

Months of unsteadiness.

5.1 The causes

Chronic dizziness usually has one of five causes, alone or in combination. An *uncompensated unilateral vestibular loss* after neuritis or surgery leaves symptoms on quick head turns. *Bilateral vestibular loss*, from gentamicin, meningitis, bilateral Ménière's disease or idiopathic causes, gives oscillopsia and unsteadiness in the dark. *Persistent postural-perceptual dizziness* is non-spinning dizziness and unsteadiness present on most days for more than three months, worse when upright, moving or in busy visual surroundings; it usually follows a vestibular event, migraine or anxiety, examination and tests are normal, and it is treated with vestibular rehabilitation, cognitive behavioural therapy and SSRIs. The *multisensory imbalance of the elderly* combines vestibular, visual, proprioceptive and musculoskeletal decline with drugs. *Central causes*, cerebellar degeneration, multiple sclerosis and posterior fossa tumours, are rare but must be sought.

Vestibular rehabilitation is the treatment for most chronic vestibular dizziness, and vestibular sedatives should be stopped.

5.2 Medical causes

Many dizzy patients have no ear disease at all. Orthostatic hypotension, arrhythmias and aortic stenosis cause presyncope. Hypoglycaemia, anaemia and thyroid disease, drugs (antihypertensives, anticonvulsants, sedatives), anxiety and panic, and hyperventilation complete the list. Hyperventilation causes dizziness without nystagmus, unless the eighth nerve is demyelinated or compressed.

6 CHAPTER SIX Investigations

When and which.

6.1 Tests that help

A **pure tone audiogram** is needed for every patient with ear symptoms and is required to diagnose Ménière's disease. **MRI of the internal auditory meati and brain** is needed for asymmetric hearing loss or tinnitus, central signs, atypical positional nystagmus and suspected stroke. **Vestibular tests** (videonystagmography, calorics, video head impulse test, VEMPs, rotational chair) quantify and localise vestibular loss; they are useful when the history is unclear, before ablative treatment, and in suspected bilateral loss or dehiscence. **CT temporal bones** is for suspected dehiscence or fistula. An **ECG**, lying and standing blood pressure, glucose and blood count are for presyncope.

THE DIZZY PATIENT

1 Describe the dizziness and its timing

→ **single acute continuous attack**
go to step 2

→ **recurrent episodes**
go to step 3

→ **chronic, most days**
go to step 4

→ **presyncope**
lying and standing BP, ECG, medical review

2 Acute continuous vertigo: do HINTS-plus

→ **any central sign, cannot stand, new headache or neurology**
urgent MRI-DWI and stroke team

→ **all peripheral**
vestibular neuritis — short sedation, early mobilisation, consider steroids

3 Recurrent episodes: what triggers them?

→ **head position, seconds**
Dix–Hallpike, then Epley

→ **spontaneous, 20 minutes to 12 hours, fluctuating hearing**
audiogram, MRI — Ménière's

→ **spontaneous with migrainous features**
vestibular migraine

→ **straining or sound**
CT, VEMP — dehiscence or fistula

4 Chronic dizziness

→ **head-movement related**
uncompensated loss — vestibular rehabilitation

→ **oscillopsia, worse in the dark**
bilateral loss — vestibular tests

→ **visual and motion sensitivity, normal tests**
persistent postural-perceptual dizziness

→ **cerebellar signs**
MRI

6.2 Children

Dizziness in children is uncommon and is usually benign. Vestibular migraine and *benign paroxysmal vertigo of childhood* (brief attacks in toddlers, a migraine precursor) are the commonest causes. Otitis media with effusion, vestibular neuritis, head injury and BPPV after trauma follow, along with the third-window effect of an enlarged vestibular aqueduct. Posterior fossa tumours, seizures and cardiac causes are the rare serious ones to exclude.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Classification by timing and trigger,** not by the word "dizzy".
- **Peripheral versus central nystagmus;** Alexander's law.
- **HINTS and HINTS-plus:** why a normal head impulse is alarming.
- **Dix–Hallpike technique and the Epley manoeuvre;** horizontal canal BPPV.
- **Diagnostic criteria** for Ménière's disease and vestibular migraine.
- **Persistent postural-perceptual dizziness.**
- **When to scan;** the role of vestibular rehabilitation.

7.2 Common pitfalls

COMMON PITFALLS

- Labelling acute vertigo "labyrinthitis" without examining the eyes.
- Prescribing long-term vestibular sedatives, which delay compensation.
- Missing vestibular migraine because the patient has no headache during attacks.

- Giving betahistine for BPPV instead of doing an Epley manoeuvre.

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CLINICAL · APPROACH TO

Approach to facial weakness

A face that will not move is frightening for the patient and often treated too casually by the doctor. Most acute lower motor neuron palsies are idiopathic and recover, but Bell's palsy is a diagnosis of exclusion, and the approach is designed to find the patient who does not fit: the stroke that spares the forehead, the cholesteatoma behind the drum, the zoster vesicles in the concha, the parotid cancer under the angle of the jaw, and the slow, twitching palsy that is a tumour until imaging says otherwise. Whatever the cause, the eye comes first.

AT A GLANCE

- Forehead spared = upper motor neuron.
- Examine ear, parotid, other cranial nerves and skin for vesicles.
- Prednisolone within 72 hours; eye protection always.
- Slowly progressive or recurrent same-side palsy = tumour until imaged.

1 CHAPTER ONE Upper or lower?

The forehead decides.

1.1 The first distinction

The upper face receives fibres from both cerebral hemispheres, while the lower face receives fibres only from the opposite hemisphere. A **supranuclear (upper motor neuron)** lesion therefore weakens the lower face on the opposite side and spares the forehead, and emotional smiling may still move the mouth. A **lower motor neuron** lesion, at or beyond the facial nucleus in the pons, paralyses the whole of one side of the face, forehead included.

A forehead-sparing palsy of sudden onset is a stroke until proven otherwise and goes down the stroke pathway, not to the ENT clinic. A pontine lesion can weaken the face together with the sixth nerve, because the motor fibres loop around the abducens nucleus.

RED FLAGS IN FACIAL WEAKNESS

- **Forehead sparing, limb weakness, dysarthria or other brainstem signs:** stroke pathway now.
- **Slow onset over weeks, twitching or spasm before the weakness, or a palsy that recurs on the same side:** tumour until imaging says otherwise.
- **No recovery at all by three to four months, or an outcome worse than House–Brackmann III:** look again for a cause.
- **A discharging ear, cholesteatoma or acute otitis media:** an otological palsy that needs surgery or drainage.
- **A parotid mass or a facial skin cancer:** perineural malignancy.
- **Bilateral palsy:** Lyme disease, Guillain–Barré syndrome, sarcoidosis, leukaemia, HIV.

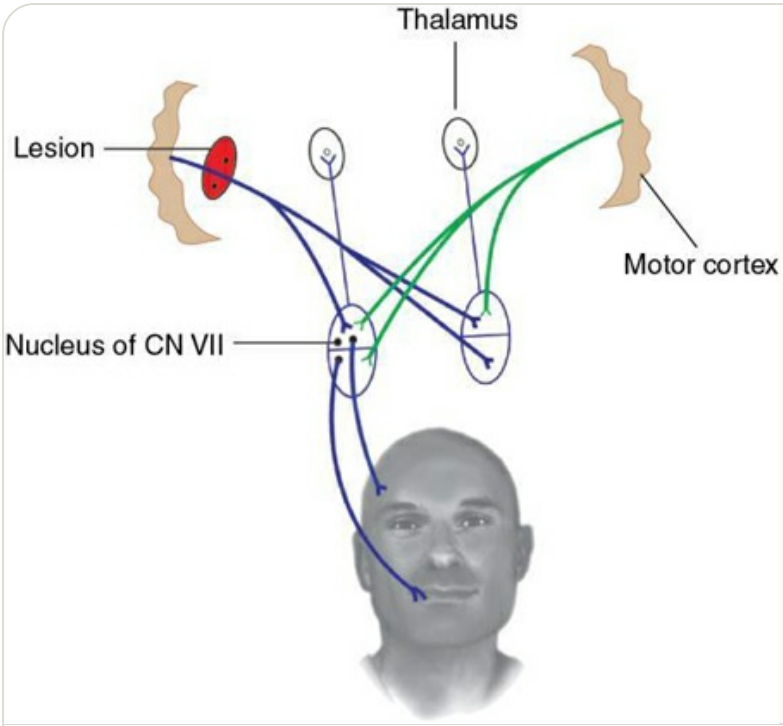


FIGURE 1 The forehead receives bilateral cortical innervation and is spared in supranuclear palsy.
Dhingra, fig. 14.1

2 CHAPTER TWO History

Speed, sequence and company.

2.1 Onset and course

The single most useful question is how quickly the weakness developed. Bell's palsy reaches its worst within 72 hours, often within 24 to 48 hours, and is frequently preceded by pain behind the ear. A palsy that progresses over weeks, starts with twitching, or never begins to improve suggests a tumour of the nerve, the temporal bone or the parotid.

2.2 Associated symptoms

Ask about taste disturbance and hyperacusis, which place the lesion proximal to the chorda tympani and the nerve to stapedius but are common in Bell's palsy; about ear pain, a rash or blisters in the ear or mouth (Ramsay Hunt syndrome); about hearing loss, tinnitus and vertigo (zoster, cholesteatoma, a temporal bone fracture or a tumour of the internal auditory meatus or cerebellopontine angle); about discharge, previous ear disease and ear surgery; and about head injury and recent parotid or ear operations.

Ask also about tick bites, travel and rashes (Lyme disease), ascending weakness and numbness (Guillain–Barré syndrome), diabetes and hypertension, pregnancy, and previous episodes. Recurrent alternating palsies with facial swelling and a fissured tongue suggest Melkersson–Rosenthal syndrome.

3 CHAPTER THREE Examination

Grade, locate, find the cause.

3.1 The face and the eye

Look at the face at rest and with each movement: raising the brows, closing the eyes gently and tightly, smiling, puffing the cheeks and showing the teeth. Grade the weakness with the **House–Brackmann scale**, which runs from I (normal) to VI (total paralysis). The key break lies between III and IV: in grade III the eye still closes completely with effort, and from grade IV it no longer closes. Note **Bell's phenomenon** (the eye rolls up on attempted closure) and any exposure of the cornea, because an eye that does not close and does not roll up is at high risk.

HOUSE-BRACKMANN GRADING

GRADE	DESCRIPTION	EYE CLOSURE	FOREHEAD
I	Normal	Normal	Normal
II	Slight weakness	Complete, minimal effort	Moderate to good
III	Obvious, not disfiguring	Complete with full effort	Slight to moderate
IV	Disfiguring weakness	Incomplete	None
V	Barely perceptible movement	Incomplete	None
VI	Total paralysis	None	None

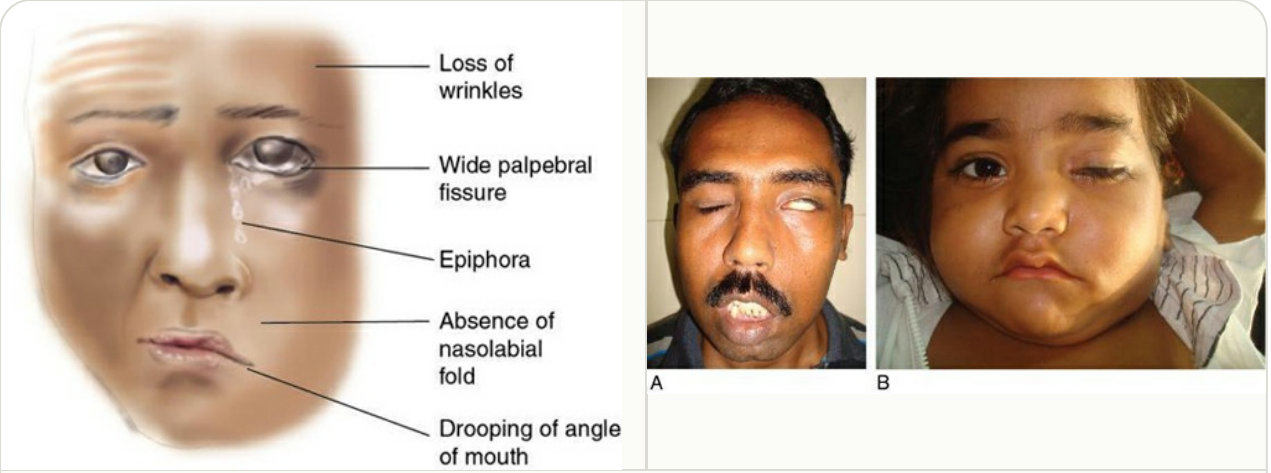


FIGURE 2 First, the features of a left lower motor neuron facial palsy. Second, Bell's palsy in an adult and a child.

Dhingra, fig. 14.7

3.2 Looking for a cause

Examine the ear under the microscope for cholesteatoma, otitis media, a mass behind the drum or blood, and look for **vesicles** in the concha, the canal, the soft palate and the anterior tongue. Test hearing with tuning forks. Palpate the parotid and the neck, and look at the facial skin and scalp for a cancer that may spread along the nerve. Examine the other cranial nerves, the limbs and the gait.

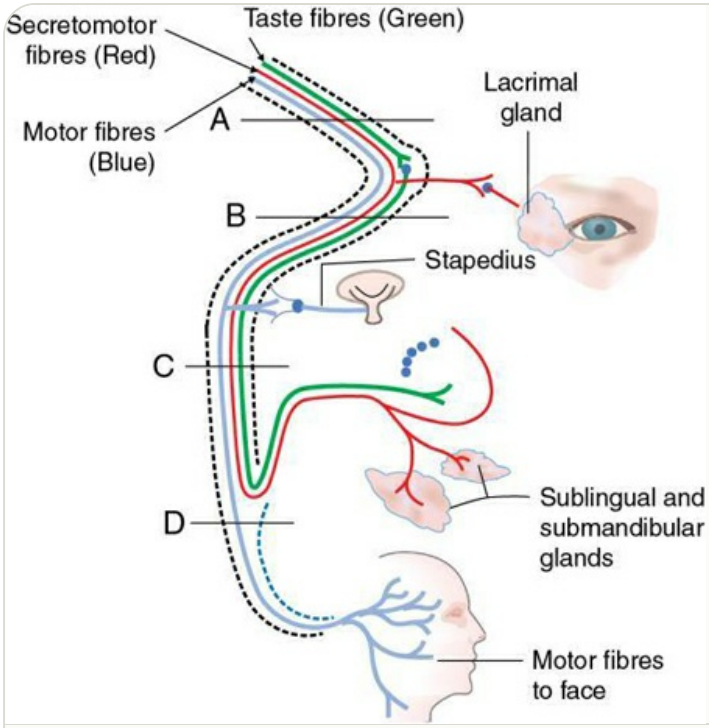


FIGURE 3 Topographical localisation of facial nerve lesions.
Dhingra, fig. 14.12

4 CHAPTER FOUR Causes

By site and by speed.

4.1 The common causes

Bell's palsy accounts for about 70% of acute lower motor neuron palsies. It affects about 30 people per 100,000 per year, peaks around 40 years and recurs in about 7%. **Ramsay Hunt syndrome**, caused by reactivation of varicella zoster in the geniculate ganglion, is the second commonest and carries a worse prognosis, with only about 20% recovering completely. The vesicles may appear late or not at all (*zoster sine herpete*), so a "Bell's palsy" with severe pain, hearing loss or vertigo deserves antiviral treatment.

Otological causes include acute otitis media in children, where the canal is often dehiscent and the prognosis is excellent with antibiotics and a myringotomy; cholesteatoma, which causes a palsy in under 1% of cases but in up to 50% of petrous apex disease and needs surgery; and necrotising otitis externa, with pain,

discharge and progressive cranial neuropathies in an elderly diabetic. **Trauma** includes temporal bone fractures, especially those crossing the otic capsule, and surgical injury in the middle ear, the mastoid, the cerebellopontine angle and the parotid.

Tumours may arise in the nerve (schwannoma and haemangioma), press on it (paraganglioma, meningioma, vestibular schwannoma) or invade it (parotid cancer, cutaneous squamous cell carcinoma with perineural spread). **Neurological and systemic causes** include stroke, Guillain–Barré syndrome, Lyme disease, sarcoidosis and, in children, congenital palsies such as Möbius syndrome and CHARGE.

WHERE THE LESION LIES

SITE	CLUES	EXAMPLES
Supranuclear	Forehead spared; limb or speech signs	Stroke, tumour
Pons	Sixth nerve palsy; long tract signs	Infarct, demyelination
Cerebellopontine angle and meatus	Hearing loss, tinnitus, imbalance; other cranial nerves	Vestibular schwannoma, meningioma
Temporal bone	Ear disease; hearing loss; vertigo; vesicles	Bell's palsy, Ramsay Hunt syndrome, cholesteatoma, fracture
Parotid and face	Mass; selective branch weakness; skin cancer	Parotid malignancy, surgery, trauma



FIGURE 4 Course of the facial nerve: meatal, labyrinthine, tympanic and mastoid segments, and its branches.

Dhingra, fig. 14.2

5 CHAPTER FIVE Investigation

What and when.

5.1 Tests

A typical Bell's palsy in an adult with a normal ear and no other neurological findings needs no tests at presentation, but a pure tone audiogram is useful in every patient and essential if there is hearing loss or vertigo. Blood glucose is checked when diabetes is possible, Lyme serology in endemic areas or with a tick bite, and a PCR swab of vesicles confirms zoster.

Imaging is needed for any atypical feature: slow onset, twitching, recurrence on the same side, no recovery by three to four months, other cranial neuropathies, hearing loss, ear disease or a parotid mass. Gadolinium-enhanced MRI covers the nerve from the brainstem to the parotid, and high-resolution CT shows the bony canal, fractures and cholesteatoma.

Electroneuronography compares the compound muscle action potential on the two sides in a complete palsy between day 3 and day 21. More than 90% degeneration predicts a poor recovery and is used in some centres to select patients after trauma for exploration. **Electromyography** after three weeks shows fibrillation potentials (degeneration with viable muscle) and polyphasic potentials (early reinnervation); electrical silence after a long palsy means the muscle is no longer viable.

CHAPTER SIX

6 Pathway

From weak face to plan.

THE WEAK FACE

1. Forehead moving?
   - → **spared**
     stroke pathway, neurology
   - → **paralysed**
     lower motor neuron, step 2
2. Eye care now
   - → **incomplete closure or poor Bell's phenomenon**
     lubricating drops by day, ointment and taping at night, ophthalmology if exposed

3 **Ear, parotid and nerves examined**

- → **vesicles, pain, hearing loss or vertigo**
  Ramsay Hunt — steroid and antiviral
- → **cholesteatoma or otitis media**
  CT and surgery or drainage
- → **parotid mass or skin cancer**
  imaging and fine needle aspiration
- → **other neurology or bilateral**
  neurology, Lyme serology

4 **Typical sudden palsy within 72 hours**

- → Bell's palsy — prednisolone within 72 hours, eye care, review

5 **Not recovering by three to four months, slow, twitching or recurrent**

- → MRI with gadolinium and CT of the temporal bone
- → refer to a facial nerve clinic

7 CHAPTER SEVEN Management

Principles for each group.

7.1 Bell's palsy

Prednisolone started within 72 hours improves the chance of full recovery; a typical adult course is 50 to 60 mg daily for 5 to 10 days, with or without a taper. Antivirals alone do not help, and adding them to steroids offers at most a small benefit, so they are reserved for severe palsy or suspected zoster. Every patient with incomplete closure needs eye protection. About 85% begin to recover within three weeks and most recover fully; complete palsy at onset, age over 60, severe pain, diabetes, hypertension and no recovery by three weeks predict a worse outcome.

Surgical decompression of the labyrinthine segment, the narrowest part of the canal at about 0.6 mm, remains controversial.

7.2 Ramsay Hunt syndrome

Treat with steroids and an antiviral such as aciclovir or valaciclovir, protect the eye, manage pain and vertigo, and refer for facial therapy early.

7.3 Trauma

An immediate complete palsy after a temporal bone fracture, with CT evidence of disruption of the canal or more than 90% degeneration on electroneuronography, is considered for exploration and repair. A delayed or incomplete palsy almost always recovers with steroids and eye care. A nerve known to have been cut at surgery is repaired at once.

7.4 Long-term weakness

Patients left with weakness need facial therapy, protection of the eye with gold or platinum weights in the upper lid and lower lid tightening, and botulinum toxin for synkinesis. Reanimation depends on whether the muscles are still viable, which is unlikely after 18 to 24 months: nerve repair, grafting or nerve transfer (masseteric or hypoglossal) while they are, and muscle transposition or free gracilis transfer once they are not.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - **Upper versus lower motor neuron:** the forehead and why.

- **Bell's palsy as a diagnosis of exclusion:** the features that make you look for another cause.
- **House–Brackmann grading** and the importance of eye closure.
- **Ramsay Hunt syndrome:** zoster sine herpete, antivirals, prognosis.
- **Electroneuronography and electromyography:** timing and meaning.
- **Steroids within 72 hours;** the limited role of antivirals.

8.2 Common pitfalls

COMMON PITFALLS

- Labelling a slowly progressive or twitching palsy as Bell's palsy.
- Forgetting the eye: an unprotected cornea can be lost while the face recovers.
- Never looking in the ear or feeling the parotid.
- Not reviewing the patient: a palsy that has not begun to recover by three to four months needs imaging.

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CLINICAL · APPROACH TO

Approach to nasal obstruction

Nasal obstruction is the commonest nasal complaint, affecting around a third of adults, and its causes fall into two families. **Inflammatory** obstruction from rhinitis and chronic rhinosinusitis fluctuates, alternates and is usually bilateral; **structural** obstruction from the septum and the nasal valves is fixed and often one-sided. Most patients have one or both, and the approach is to tell them apart, treat the mucosa before operating on the skeleton, and never miss the unilateral blockage with blood or a foul discharge that is a tumour or, in a child, a foreign body.

AT A GLANCE

- Fixed and one-sided: structural or a mass.
- Fluctuating and two-sided: mucosal.
- Children: adenoids, foreign body, choanal atresia.
- Unilateral blockage with bleeding or pain: a tumour until proved otherwise.

1 CHAPTER ONE How the nose blocks

Resistance, valves and mucosa.

1.1 The physiology that matters

The nose provides about two-thirds of total airway resistance, and most of it lies at the **internal nasal valve**, the slit between the septum, the caudal edge of the upper lateral cartilage and the head of the inferior turbinate. It is the narrowest part of the airway, and because resistance rises steeply as the radius falls (Poiseuille's law), a small narrowing here causes a large symptom. The valve is also richly supplied with receptors, so obstruction at this point feels worse than obstruction further back.

The mucosa of the turbinates is erectile tissue under sympathetic control. In about 80% of people it congests and decongests alternately on each side in the **nasal cycle**, every two to six hours, keeping total resistance constant. A patient who notices the cycle, or whose fixed deviation is unmasked by the congested phase, describes alternating blockage.

The sensation of airflow is carried by trigeminal cold receptors, not by the volume of air. This is why the subjective complaint and objective measures often disagree, why menthol makes the nose feel clearer, and why a nose that has lost too much turbinate can feel blocked although it is wide open (*empty nose syndrome*).

DO NOT MISS

- **Unilateral obstruction with blood-stained discharge, pain, facial swelling, numbness, loose teeth, diplopia or a neck node:** sinonasal or nasopharyngeal malignancy.
- **An adolescent boy with unilateral obstruction and heavy epistaxis:** juvenile angiofibroma — do not biopsy in clinic.
- **A child with unilateral foul discharge:** foreign body.
- **A newborn with cyclical cyanosis relieved by crying:** bilateral choanal atresia.
- **A midline nasal mass in a child:** dermoid, glioma or encephalocele — image before any biopsy.

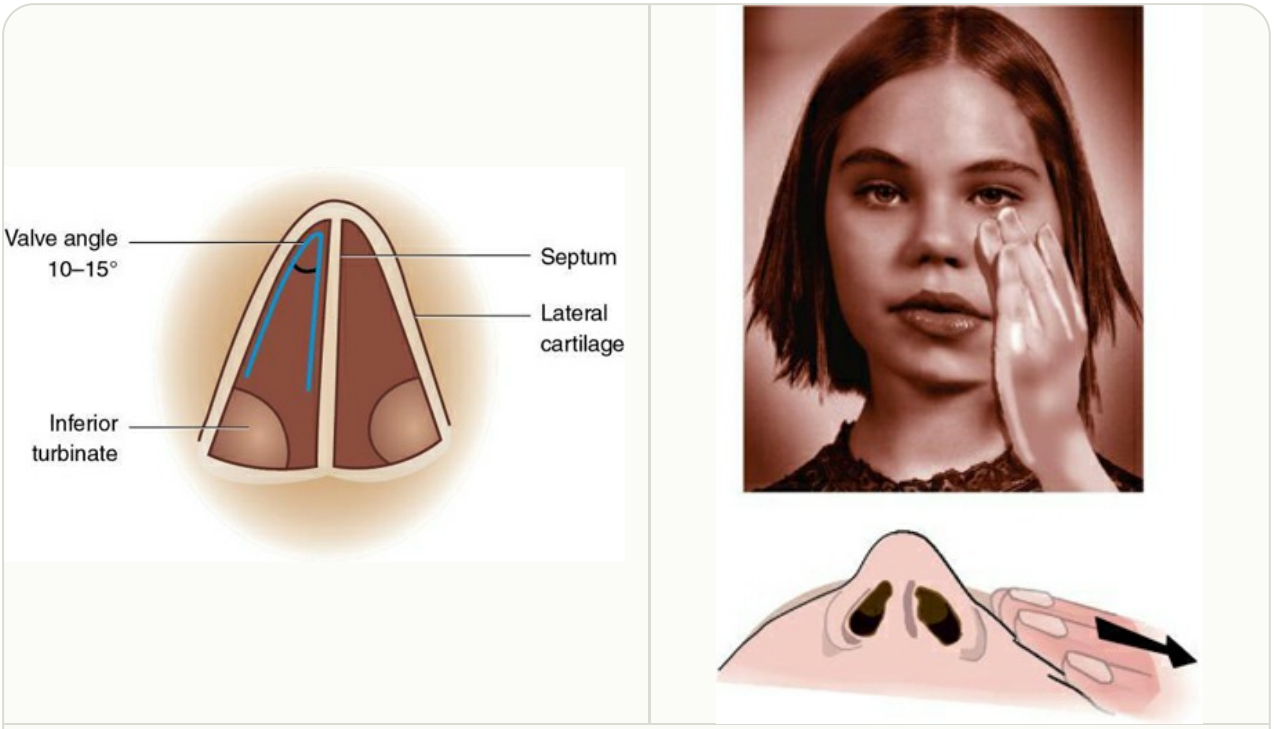


FIGURE 1 First, the internal nasal valve area. Second, Cottle's test: drawing the cheek laterally opens the valve.
Dhingra, figs 23.3 and 26.7

2 CHAPTER TWO History

Fixed or fluctuating?

2.1 The questions

Ask whether the blockage is one-sided or both, constant or variable, and whether it alternates. **Fixed, unilateral** obstruction points to the septum, a valve problem, a polyp, a tumour or a foreign body; **fluctuating, bilateral** obstruction points to rhinitis or chronic rhinosinusitis. Ask about onset (after trauma or surgery, or gradual), and whether the patient sniffs inward and the nostril collapses, which suggests valve collapse.

Ask about the company the symptom keeps: sneezing, itch and watery discharge in rhinitis; loss of smell, facial pressure and discoloured discharge in rhinosinusitis; bleeding, pain and crusting in tumours, granulomatous disease and septal perforation; and snoring and mouth breathing. Take a full drug history, including decongestant sprays (rhinitis medicamentosa), antihypertensives, the oral contraceptive pill and recreational cocaine, and ask about asthma, aspirin sensitivity, smoking and occupation. The **NOSE** score and **SNOT-22** quantify the symptom and its effect on life.

3 CHAPTER THREE Examination

Outside, then inside, then after decongestion.

3.1 The nose

Look at the external nose from the front, side and below for deviation, a twisted pyramid, a narrow or collapsing ala and a caudal septum dislocated into the nostril. Watch the alae on deep nasal inspiration. **Cottle's test**, in which the cheek is drawn laterally to open the valve, suggests a valve problem when it improves breathing; a modified version supports the valve from inside with a cotton bud or a curette.

Examine the vestibule, the septum and the turbinates with a headlight and speculum, then with an endoscope before and after topical decongestant. Mucosal swelling that shrinks well is reversible (rhinitis); a deviation, a spur or a polyp does not change. The endoscope also reveals middle meatal pus and polyps, the posterior septum, the choanae and the nasopharynx.

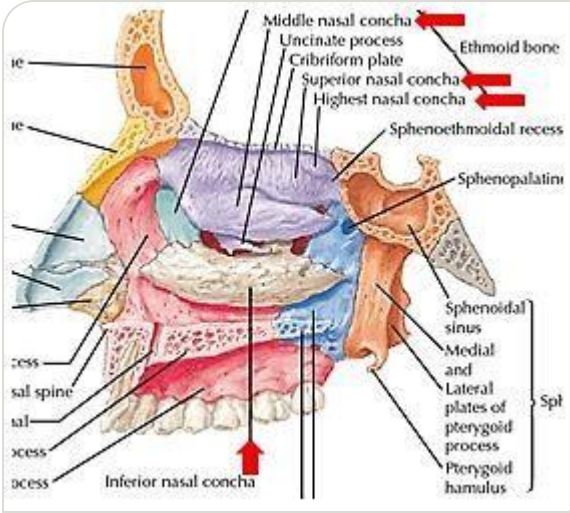


FIGURE 2 The lateral nasal wall: turbinates and meatuses.
Anatomy atlas · ENT Steps p. 99

3.2 Measuring the airway

Peak nasal inspiratory flow is the quickest objective test, normally around 120 to 140 L/min in adults. **Anterior rhinomanometry** measures resistance from flow and pressure and is the reference standard. **Acoustic rhinometry** maps cross-sectional area along the cavity; a large improvement after decongestion points to mucosal disease. CT is reserved for suspected sinus disease or before surgery, and imaging with MRI for any suspected tumour.

4 CHAPTER FOUR Causes

Inflammatory, structural, and the rest.

CAUSES OF NASAL OBSTRUCTION

GROUP	CAUSES	TYPICAL CLUE
Inflammatory	Allergic and non-allergic rhinitis; rhinitis medicamentosa; chronic rhinosinusitis with or without polyps	Bilateral, fluctuating, shrinks with decongestant
Septum	Deviation, spur, dislocated caudal septum, haematoma	Fixed, unilateral or S-shaped
Valves	Internal valve narrowing (high septal deviation, weak upper laterals, post-rhinoplasty); external valve collapse	Worse on inspiration; Cottle positive
Turbinates	Hypertrophy, concha bullosa	Seen on endoscopy, partly reversible
Masses	Polyps, antrochoanal polyp, inverted papilloma, malignancy, angiofibroma	Unilateral, progressive; blood
Granulomatous	Granulomatosis with polyangiitis, sarcoidosis, cocaine	Crusting, bleeding, perforation
Other	Adenoids, choanal atresia, foreign body, empty nose syndrome, pregnancy and hypothyroidism	Age and context

4.1 Rhinitis and rhinosinusitis

Allergic rhinitis affects about a quarter of people and chronic rhinosinusitis about one in ten. Allergic rhinitis mainly swells the inferior turbinates and responds well to decongestant; chronic rhinosinusitis affects the middle meatus and responds less. Both are common partners of a deviated septum, and treating the mucosa often settles symptoms without surgery.

4.2 The septum and the valves

Cottle's classification divides the septum into five areas from the nostril backwards. Deviations in areas 1 to 3, at or in front of the internal valve, cause the most obstruction; deviations further back matter less. A deviation at the valve or the caudal septum is not adequately corrected by a standard septoplasty and needs a **functional septorhinoplasty**, with spreader grafts for the internal valve and struts or alar grafts for the external valve.

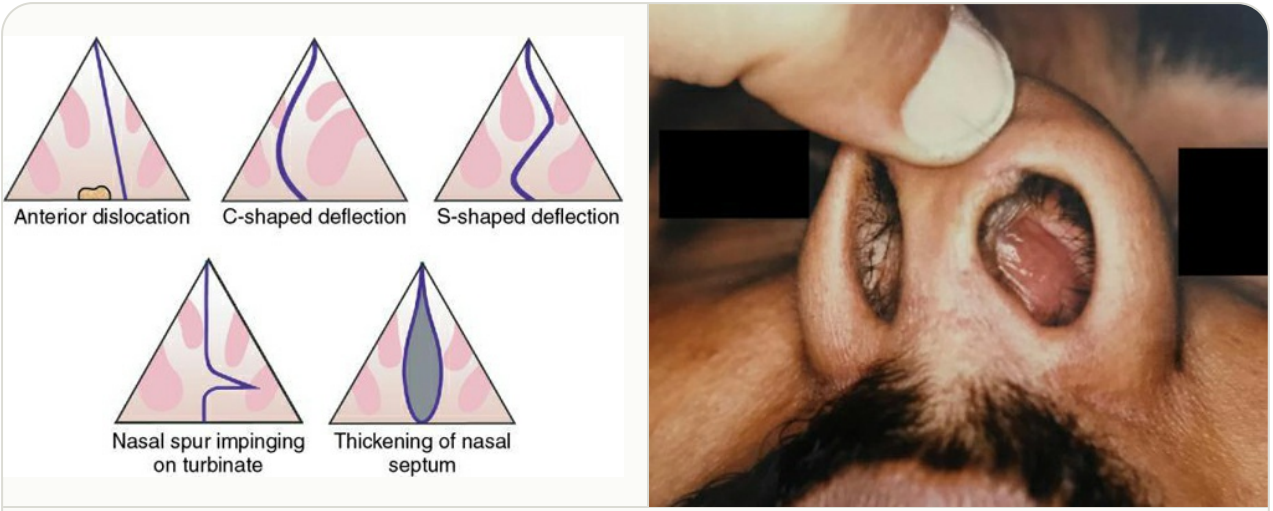


FIGURE 3 First, the types of septal deviation. Second, a C-shaped deviation nearly blocking one side.

Dhingra, figs 26.4 and 26.6

5 CHAPTER FIVE Pathway

Mucosa or skeleton?

THE BLOCKED NOSE

1 Red flags?

→ **unilateral with blood, pain or cranial nerve signs**
endoscopy, MRI and CT, biopsy unless angiofibroma suspected

→ **adolescent boy with bleeding**
contrast imaging first, no biopsy

→ **child with unilateral foul discharge**
remove the foreign body

2 Fixed or fluctuating?

→ **bilateral, fluctuating, shrinks with decongestant**
rhinitis — treat the mucosa

→ **smell loss, discharge, polyps**
chronic rhinosinusitis pathway

→ **fixed or unilateral**
step 3

3 Where is the narrowing?

→ **mid or posterior septum**
septoplasty

→ **high or caudal septum, collapsing valve**
functional septorhinoplasty

→ **turbinates**
medical first, then reduction

4 Still blocked after surgery or with a wide nose?

→ **recurrent deviation or valve collapse**
revision

→ **wide cavity with paradoxical obstruction**
empty nose syndrome — avoid further resection

CHAPTER SIX

6 Management

Treat the mucosa first.

6.1 Medical treatment

Every patient with an inflammatory component should have a proper trial of saline irrigation and a topical steroid spray, used correctly with the head bent forward and

the nozzle aimed away from the septum, for at least six to eight weeks. Antihistamines, allergen avoidance, a leukotriene antagonist in selected patients, and immunotherapy for confirmed allergy follow. Decongestant sprays are limited to a few days, because rebound congestion (*rhinitis medicamentosa*) follows longer use. Chronic rhinosinusitis is treated on its own pathway.

6.2 Surgery

A randomised trial has shown that septoplasty, with or without turbinate reduction, improves nasal obstruction more than continued medical treatment in patients with a deviated septum. **Septoplasty** corrects mid and posterior deviations through mucoperichondrial flaps, preserving an L-shaped strut of at least 1 cm dorsally and caudally. **Turbinate reduction** by submucosal diathermy, radiofrequency, outfracture or submucosal resection reduces the turbinates while preserving mucosa; its effect fades with time and it does not replace medical treatment. **Functional septorhinoplasty** treats the valve and the twisted nose, often through an external approach. Over-resection of the turbinates is avoided because of the risk of empty nose syndrome.

6.3 Children

In a child, adenoidal hypertrophy is the commonest cause of bilateral obstruction, with mouth breathing, snoring and hyponasal speech. Rhinitis is common, and a unilateral foul discharge is a foreign body until proven otherwise. A neonate who is an obligate nasal breather and blocked may have **choanal atresia**: bilateral atresia causes cyclical cyanosis relieved by crying and needs an oral airway, CT and endoscopic repair, while unilateral atresia presents later with a unilateral discharge. **Piriform aperture stenosis**, nasolacrimal duct cysts and midline masses (dermoid, glioma, encephalocele) are rarer neonatal causes that need imaging before intervention.

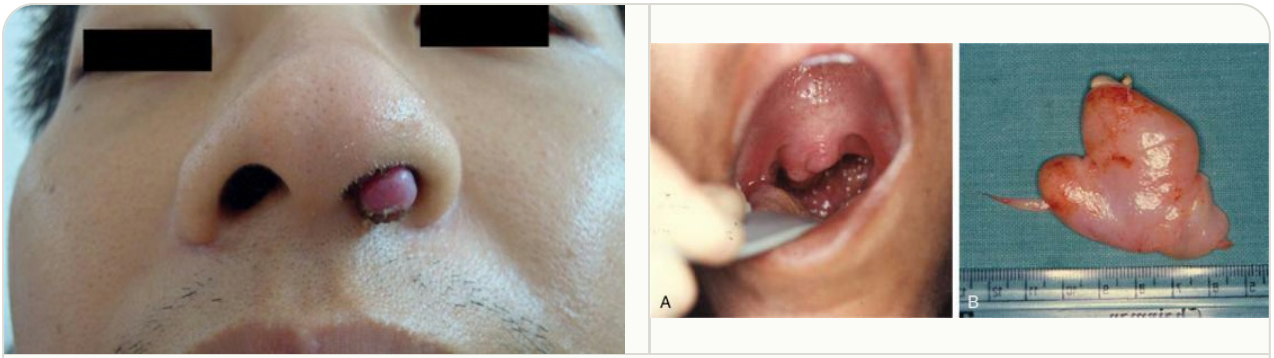


FIGURE 4 An antrochoanal polyp, a cause of unilateral obstruction in older children and young adults.

Dhingra, figs 32.1 and 32.2

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The internal nasal valve:** boundaries and why it matters.
- **Inflammatory versus structural obstruction:** the decongestant test.
- **Cottle's test and Cottle's areas.**
- **Objective tests:** peak nasal inspiratory flow, rhinomanometry, acoustic rhinometry.
- **Septoplasty versus functional septorhinoplasty;** the L-strut.
- **Red flags** for unilateral obstruction; choanal atresia in the newborn.

7.2 Common pitfalls

COMMON PITFALLS

- Operating on a deviated septum when the obstruction is rhinitis that was never treated.
- Performing a standard septoplasty for a high or caudal deviation that needs valve surgery.

- Biopsying a vascular mass in an adolescent boy in clinic.
- Over-reducing the turbinates.

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CLINICAL · APPROACH TO

Approach to a runny nose

A runny nose is usually rhinitis, and rhinitis is usually treatable. The rhinologist's task is to make the first and most important distinction: **bilateral** rhinorrhoea is almost always mucosal disease, allergic, non-allergic or infective, while **unilateral** clear rhinorrhoea is a cerebrospinal fluid leak until proven otherwise, and unilateral discoloured or bloody discharge is a foreign body or a tumour until the nose has been examined with an endoscope and, if needed, scanned.

AT A GLANCE

- Clear and watery from one side: test for CSF.
- Unilateral purulent in a child: foreign body.
- Gustatory rhinorrhoea is cholinergic: ipratropium.
- Rhinitis medicamentosa: stop the decongestant.

1 CHAPTER ONE Which side, and what colour?

The first two questions.

1.1 Sorting the discharge

RHINORRHOEA BY SIDE AND CHARACTER

PATTERN	THINK OF	KEY FEATURES
Bilateral, watery, with sneezing and itch	Allergic rhinitis	Seasonal or perennial triggers, itchy eyes, atopy
Bilateral, watery, no itch	Non-allergic rhinitis	Triggered by cold air, smells, food, alcohol; older patients
Bilateral, thick, discoloured	Acute or chronic rhinosinusitis	Blockage, pain, smell loss, pus in the middle meatus
Unilateral, clear, watery, salty, positional	CSF leak	After trauma or surgery, or spontaneous; worse bending forward

PATTERN	THINK OF	KEY FEATURES
Unilateral, foul, purulent	Foreign body (child), dental sinusitis, fungal ball	Smell noticed by others; dental history
Unilateral, blood-stained	Tumour	Obstruction, pain, numbness, proptosis

DO NOT MISS

- **Clear, watery, unilateral discharge**, especially after head injury or sinus surgery, worse on bending forward or straining, or with a salty taste: collect it for beta-2 transferrin or beta-trace protein.
- **Any history of meningitis** in a patient with a runny nose: a CSF leak until proven otherwise.
- **Blood-stained unilateral discharge in an adult:** endoscopy and imaging for a tumour.
- **Unilateral foul discharge in a child:** foreign body.

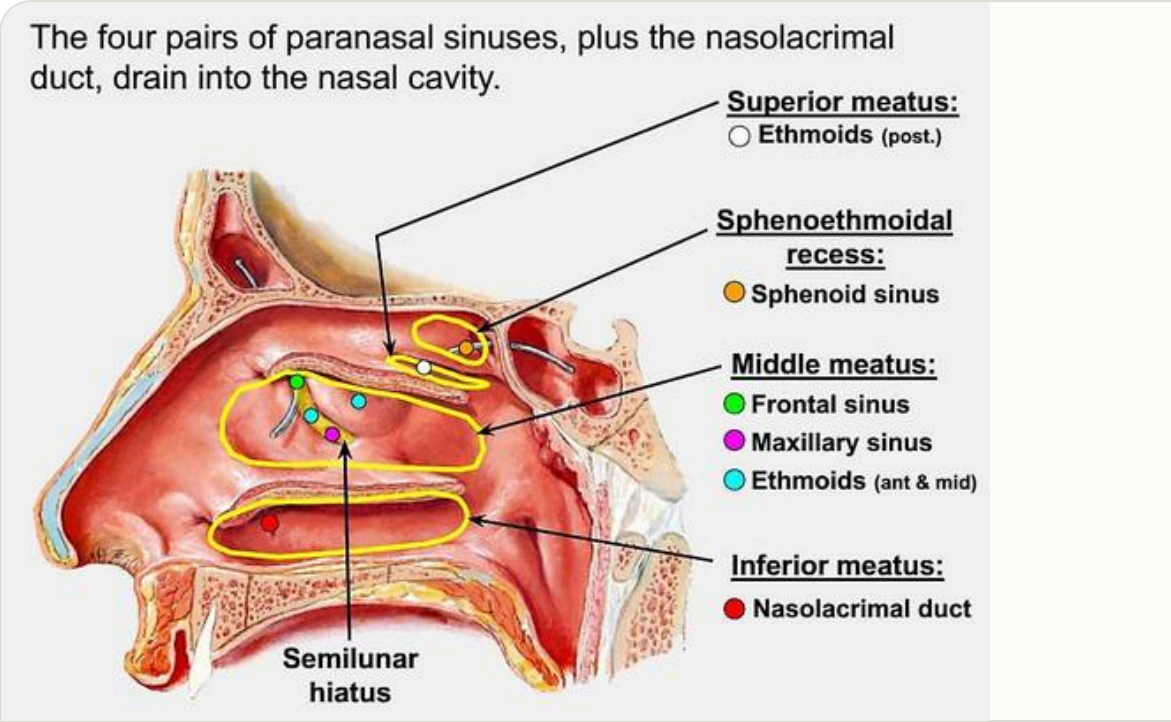


FIGURE 1 Where each sinus and the nasolacrimal duct drains into the meatuses: the site of discharge on endoscopy points to its source.

Anatomy atlas · ENT Steps p. 101

2 CHAPTER TWO Bilateral rhinorrhoea

Mostly mucosal.

2.1 Allergic rhinitis

Allergic rhinitis is an IgE-mediated inflammation producing sneezing, watery discharge, itch of the nose and eyes, and blockage. The **ARIA** classification divides it by duration into *intermittent* (fewer than four days a week or for less than four consecutive weeks) and *persistent*, and by severity into *mild* and *moderate to severe*, depending on effects on sleep, daily activities, work and troublesome symptoms. It is diagnosed from the history and confirmed with skin-prick tests or specific IgE. A subgroup has *local allergic rhinitis*, with typical symptoms and negative skin tests but IgE in the nasal mucosa.

Treatment follows the ARIA steps: allergen avoidance and saline; a non-sedating oral or topical antihistamine for mild intermittent disease; an intranasal steroid, or combined steroid and antihistamine spray, for persistent or moderate to severe disease; and **allergen immunotherapy**, sublingual or subcutaneous, for selected patients whose allergy is confirmed and who do not respond. Leukotriene antagonists help some patients with asthma. Depot and intranasal steroid injections are not recommended.

2.2 Non-allergic rhinitis

Non-allergic rhinitis produces similar symptoms without atopy. It includes **vasomotor** (idiopathic) rhinitis, triggered by temperature change, strong smells and alcohol; **gustatory** rhinitis with eating, especially hot or spicy food; **rhinitis of the elderly**, a profuse clear drip; **hormonal** rhinitis in pregnancy and hypothyroidism; **drug-induced** rhinitis from decongestant overuse, antihypertensives, the oral contraceptive pill, aspirin and NSAIDs; **occupational** rhinitis; and **NARES** (non-allergic rhinitis with eosinophilia syndrome), in which nasal eosinophilia without atopy may precede polyps, asthma and aspirin sensitivity.

When watery discharge dominates, **ipratropium bromide** spray, an anticholinergic, used before meals or triggers, is the drug of choice. Intranasal steroids help NARES and congestion. For intractable cases, the parasympathetic supply can be interrupted surgically: **vidian neurectomy** is effective but risks a dry eye, so the more selective **posterior nasal nerve** procedures (neurectomy, cryoablation or radiofrequency) are now preferred.

2.3 Infective rhinitis and rhinosinusitis

The common cold is the commonest cause of all and settles in about ten days. Discoloured discharge with blockage, facial pain or smell loss for more than twelve weeks is chronic rhinosinusitis and is managed on its own pathway.

3 CHAPTER THREE Unilateral rhinorrhoea

Leak or lesion?

3.1 CSF rhinorrhoea

CSF rhinorrhoea is a communication between the subarachnoid space and the nose. About 80 to 90% are **traumatic**, from head injury or surgery; the rest are **non-traumatic**, and many of these are *spontaneous* leaks in women with obesity and raised intracranial pressure (idiopathic intracranial hypertension), often through the cribriform plate or a lateral sphenoid recess. Spontaneous leaks have the highest recurrence rates.

The discharge is clear, watery and unilateral, worse on bending forward (the *teapot sign*) or straining, and may run backwards when lying down. It does not cause sneezing or itchy eyes. Diagnosis is confirmed by **beta-2 transferrin** or **beta-trace protein** in the fluid; the "halo" and glucose tests are unreliable. **High-resolution CT** shows the bony defect, MRI shows any encephalocele or herniated meninges, and CT cisternography or intrathecal fluorescein at surgery helps find small or intermittent leaks.

Traumatic leaks are first treated conservatively, with bed rest, head elevation, stool softeners and avoidance of nose blowing, and a lumbar drain if needed; many close within a week. Persistent and spontaneous leaks are repaired endoscopically with a graft or a nasoseptal flap, and spontaneous leaks need their raised pressure treated to prevent recurrence. Pneumococcal vaccination is recommended.

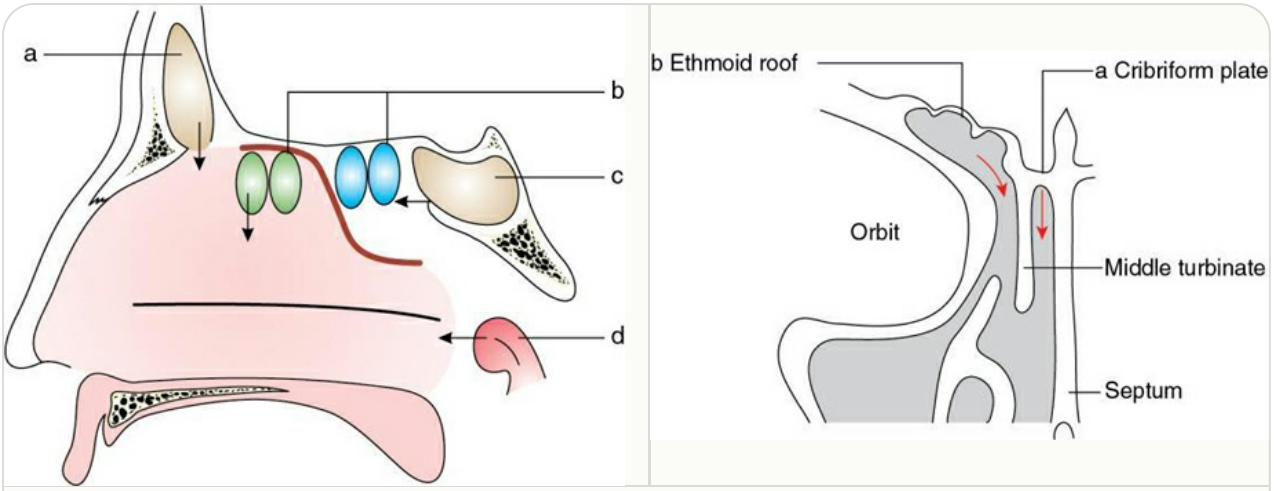


FIGURE 2 Sites of CSF leakage into the nose and ear.
Dhingra, figs 29.4 and 29.5

3.2 Other unilateral causes

A **foreign body** in a child, or after dental or sinus surgery in an adult, gives a foul unilateral discharge. **Dental (odontogenic) sinusitis** and a **fungal ball** affect one maxillary sinus, often with a foul smell. **Choanal atresia** occasionally presents with unilateral discharge. **Tumours**, inverted papilloma and sinonasal malignancy, produce discharge with blockage, bleeding and pain. Unilateral watery discharge with tearing during attacks of severe orbital pain is **cluster headache**.

4 CHAPTER FOUR Pathway

Discharge to diagnosis.

THE RUNNY NOSE

1 Which side?

→ **bilateral**
step 2

→ **unilateral**
step 3

2 Bilateral — itch, sneezing, triggers?

→ **atopic, itchy**
allergic rhinitis — avoidance, antihistamine, steroid spray, immunotherapy

→ **watery, triggered by cold, food or age**
non-allergic — ipratropium

→ **discoloured with smell loss over 12 weeks**
chronic rhinosinusitis

→ **decongestant overuse**
stop it

3 Unilateral — clear and watery?

→ **yes**
collect for beta-2 transferrin, CT for defect, MRI

→ **foul**
endoscopy, remove foreign body, dental review, CT

→ **bloody**
endoscopy and imaging, biopsy

4 CSF leak confirmed

→ **traumatic**
conservative first, repair if persisting

→ **spontaneous**
endoscopic repair and treat raised pressure

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Unilateral clear rhinorrhoea is CSF until proven otherwise;** beta-2 transferrin.
- **The ARIA classification** and stepwise treatment of allergic rhinitis.
- **Non-allergic rhinitis subtypes,** NARES and ipratropium.
- **Vidian and posterior nasal nerve surgery:** what they do and the dry eye risk.
- **Spontaneous CSF leak** and idiopathic intracranial hypertension.

5.2 Common pitfalls

COMMON PITFALLS

- Treating a unilateral watery discharge as rhinitis for months.
- Relying on glucose testing to diagnose CSF.
- Giving steroid sprays for rhinorrhoea in the elderly, where ipratropium works better.
- Repairing a spontaneous leak without addressing raised intracranial pressure.

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CLINICAL · NOSE & SINUSES

Epistaxis

Bleeding from the nose is a sign, not a diagnosis. Most bleeds are small, anterior and self-limiting; a few are posterior, brisk and dangerous. The skill lies in telling the two apart, and in climbing the ladder of treatment one rung at a time.

AT A GLANCE

- Resuscitate while you stop the bleeding.
- Pressure, vasoconstrictor, cautery, then packing.
- Then ligation or embolisation.
- Check anticoagulants and blood pressure.
- Unilateral bleeding with blockage needs a scope.
- SPA dominant in 85%; landmark the crista ethmoidalis; electrocautery beats silver nitrate.

1 CHAPTER ONE Overview

A common bleed with an uncommon tail.

1.1 Definition

Epistaxis is bleeding from inside the nose. The blood may run forward out of the nostril, backward into the nasopharynx and throat, or both. It is always a sign of something else, never a final diagnosis, and Dhingra's rule holds on every shift: an attempt should always be made to find its local or systemic cause.

1.2 Epidemiology

Most people have at least one nosebleed in their lives, and Lee estimates that around one in six of the population bleeds in any given year. Only a small fraction ever seek medical help. The distribution is bimodal: children and young adults, who bleed from the front of the septum, and adults past middle age, who make up most hospital admissions and most posterior bleeds.

1.3 Why it matters

The patients who reach the emergency department are older, often hypertensive, frequently on an anticoagulant or antiplatelet drug, and bleeding from further back in the nose, where blood runs silently down the pharynx and is easy to underestimate. In them, epistaxis is as much a resuscitation problem as a nasal one.

CHAPTER TWO

2 Applied anatomy

Two carotid systems meeting in the nose.

2.1 Arterial supply

The nasal cavity draws blood from both the internal and the external carotid arteries, and the two systems anastomose freely on the septum and the lateral wall. That double supply is why nosebleeds can be persistent, and why ligating one vessel sometimes isn't enough.

2.1.1 Internal carotid system

- **Anterior ethmoidal artery,** from the ophthalmic artery, entering the nose through the roof.
- **Posterior ethmoidal artery,** also from the ophthalmic, smaller and further back.

These supply the upper septum and lateral wall above the middle turbinate.

2.1.2 External carotid system

- **Sphenopalatine artery,** terminal branch of the maxillary artery: the main supply of the nose.
- **Greater palatine artery,** from the maxillary, reaching the septum through the incisive canal.
- **Superior labial artery,** from the facial, supplying the front of the septum and vestibule.

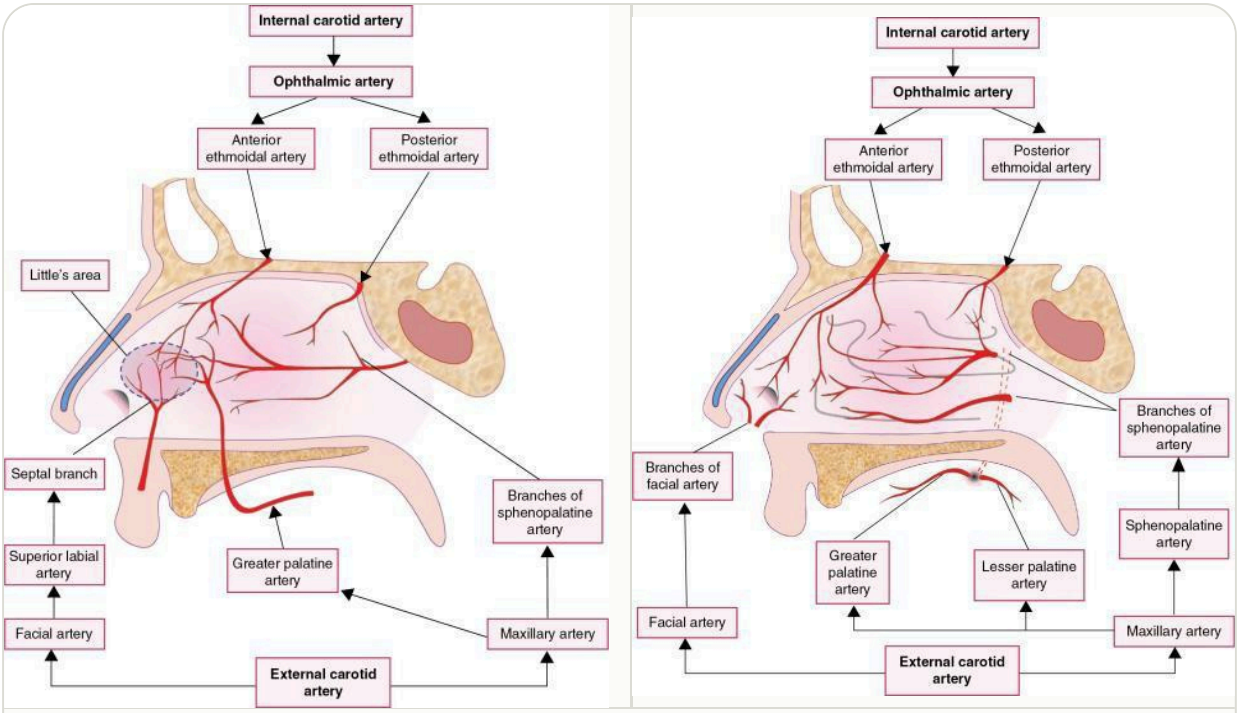


FIGURE 1 Arterial supply of the nasal septum (first) and the lateral nasal wall (second). The anterior ethmoidal, septal branch of the sphenopalatine, greater palatine and superior labial arteries converge on Little's area.

Dhingra, figs 33.1 and 33.2

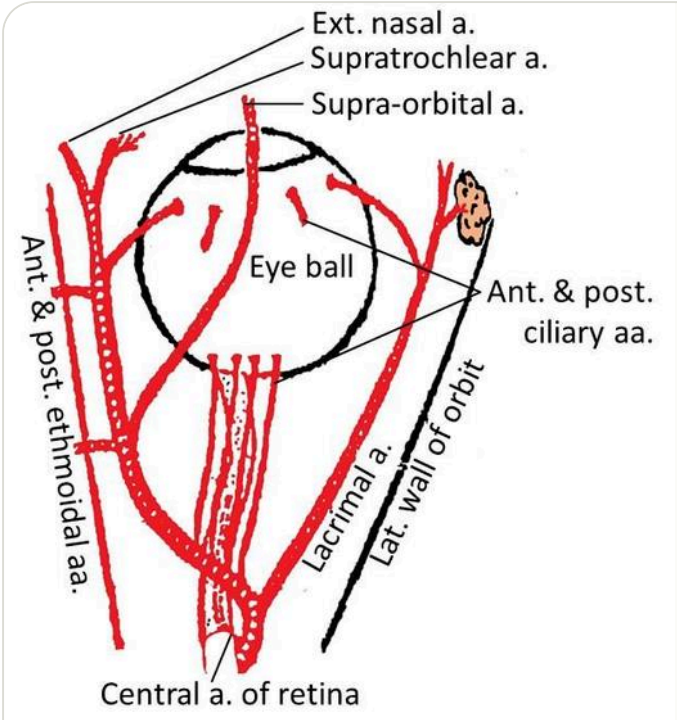


FIGURE 2 The ophthalmic artery giving off the anterior and posterior ethmoidal arteries in the orbit, on their way to the roof of the nose.

Anatomy atlas · Doss p. 100

2.2 Little's area and Kiesselbach's plexus

On the anteroinferior septum, just above the vestibule, four arteries meet to form *Kiesselbach's plexus*; the region it occupies is *Little's area*.

2.2.1 The four contributors

- Anterior ethmoidal (internal carotid)
- Septal branch of the sphenopalatine (external carotid)
- Greater palatine (external carotid)
- Septal branch of the superior labial (external carotid)

2.2.2 Why it bleeds

It sits exactly where inspired air dries and cracks the mucosa, where the mucosa is thin and closely bound to cartilage, and where a fingernail reaches. About nine in ten bleeds start here.

2.3 The posterior territory

2.3.1 Sphenopalatine artery

It enters the nose through the sphenopalatine foramen, just behind the *crista ethmoidalis* of the palatine bone, and divides into the posterior lateral nasal and posterior septal arteries. Bleeding from these branches is the classic posterior epistaxis.

2.3.2 Woodruff's plexus

A plexus on the lateral wall below the posterior end of the inferior turbinate, and the other classic posterior bleeding site in adults.

2.3.3 Ethmoidal territory

Bleeding above the level of the middle turbinate points to the ethmoidal arteries and therefore to the internal carotid system, which matters because embolisation cannot safely reach it.

WHERE THE BOOKS DIFFER

Dhingra	Lee
Woodruff's plexus is a plexus of veins below the posterior end of the inferior turbinate.	Woodruff's is the naso-nasopharyngeal plexus over the posterior centimetre of the

nasal floor, inferior meatus and middle meatus.

For the exam, give Dhingra's definition, and add that posterior arterial bleeds come from sphenopalatine branches.

2.4 Venous drainage

Veins follow the arteries. Dhingra highlights the *retrocolumellar vein*, running vertically just behind the columella across the floor of the nose, as a common source of venous bleeding in young people.

3 CHAPTER THREE Aetiology

Local, systemic, or neither.

CAUSES OF EPISTAXIS

- LOCAL **Trauma** inflammation and drying, structural, neoplastic, iatrogenic
- SYSTEMIC **Drugs** coagulation and platelet disorders, vascular disorders, cardiovascular, organ failure
- IDIOPATHIC **No cause found** despite assessment

3.1 Local causes

3.1.1 Trauma

The commonest cause overall. Digital trauma (nose picking) dominates in children; in adults add forceful blowing, blows to the face, fractures of the nasal bones, midface and skull base, and foreign bodies.

3.1.2 Inflammation and drying

Dry, heated air in winter, acute viral rhinitis and sinusitis, rhinitis sicca, atrophic rhinitis, and the crust-forming granulomatous infections all leave a fragile, cracked mucosa.

3.1.3 Structural

A septal spur or deviation creates turbulent, drying airflow over its edge; a septal perforation bleeds from its crusted rim.

3.1.4 Neoplastic

- **Benign:** juvenile nasopharyngeal angiofibroma, haemangioma, inverted papilloma.
- **Malignant:** carcinoma of the nose, sinuses and nasopharynx.

Any unilateral, recurrent bleed, especially with obstruction, must be examined endoscopically.

3.1.5 Iatrogenic and topical

Nasal and sinus surgery, nasogastric and nasotracheal tubes, nasal oxygen cannulae, topical steroid and decongestant sprays, and intranasal cocaine.

3.2 Systemic causes

The logic is simple: anything that impairs clotting, platelets or vessel walls will show up in the nose.

3.2.1 Drugs and supplements

Warfarin and the direct oral anticoagulants, aspirin, clopidogrel and other antiplatelets, NSAIDs, and, as Lee reminds us, over-the-counter vitamin E, fish oil, garlic, ginkgo and ginseng.

3.2.2 Coagulation and platelet disorders

Haemophilia, von Willebrand disease (the commonest inherited bleeding disorder), thrombocytopenia, leukaemia and aplastic anaemia.

3.2.3 Vascular disorders

Hereditary haemorrhagic telangiectasia (see
8.3
), scurvy, and the vessel fragility of old age.

3.2.4 Cardiovascular

Hypertension and atherosclerosis are associated with posterior bleeds in older adults. Whether raised pressure causes the bleed or simply keeps it going is still debated; either way, it should be measured and treated.

3.2.5 Organ failure

> Liver failure (reduced clotting factors) and renal failure (uraemic platelet dysfunction).

3.3 Idiopathic

In a large share of patients no specific cause is found despite a careful history, examination and blood tests. The label is earned only after the search, not instead of it.

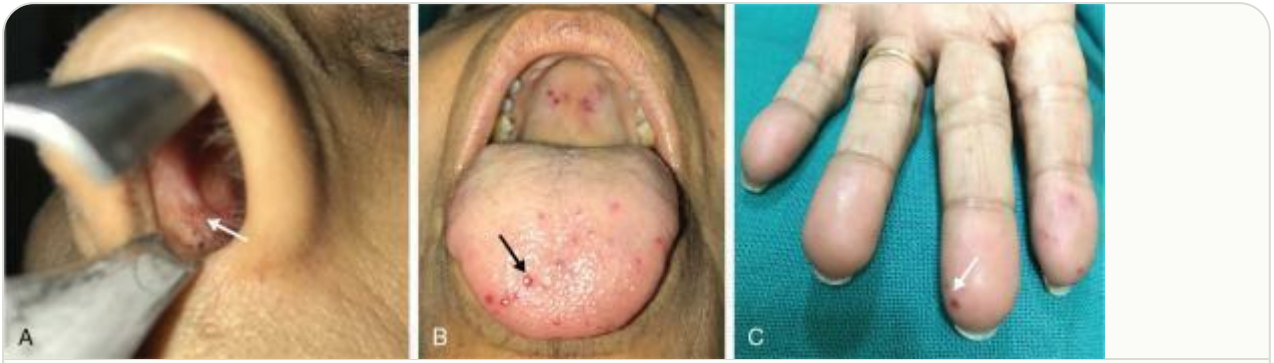


FIGURE 3 Hereditary haemorrhagic telangiectasia: lesions on the nasal septum, tongue, palate and fingertips (arrows). Always look at the lips, tongue and fingers of a recurrent bleeder.
Dhingra, fig. 33.3

RED FLAGS

- **Adolescent boy, unilateral bleeding and obstruction:** juvenile angiofibroma until proven otherwise. No biopsy in clinic.
- **Adult with unilateral bleeding, obstruction, crusting or facial pain:** endoscopy for a sinonasal tumour.
- **A child under two:** rare; think of non-accidental injury or a bleeding disorder.
- **Massive or recurrent bleeding after trauma or skull base surgery:** carotid pseudoaneurysm; needs angiography.

4 CHAPTER FOUR Classification

The distinction that drives everything that follows.

4.1 By site: anterior or posterior

With the patient sitting up, an anterior bleed runs mostly out of the nostril. A posterior bleed runs back into the throat, often from both sides, and no source is seen on anterior rhinoscopy.

TABLE

	ANTERIOR	POSTERIOR
Frequency	Common, about 90%	Less common
Site	Little's area or the front of the lateral wall	Posterosuperior cavity; often hard to localise
Age	Children and young adults	Usually over 40
Cause	Mostly trauma, dryness	Often spontaneous; hypertension, atherosclerosis, anticoagulation
Severity	Usually mild; pressure, cautery or an anterior pack	Often severe; admission, posterior pack or surgery

4.2 By severity

There is no universal scale; in practice the useful split is whether first aid is enough.

4.2.1 Minor

Stops with pressure or a single cautery, haemodynamically stable, can go home.

4.2.2 Major

Continues despite first aid, needs packing, shows haemodynamic change or a falling haemoglobin, or occurs in a frail or anticoagulated patient. These patients are admitted.

4.3 By pattern: single or recurrent

Recurrence changes the question from how to stop it to why it keeps happening. Recurrent unilateral bleeding needs endoscopy; recurrent bilateral bleeding, especially with a family history, needs HHT to be considered.

5 CHAPTER FIVE Clinical evaluation

Resuscitate first, then find the bleeding point, then ask why.

5.1 Immediate assessment

The first question is not where the blood is coming from, but how much has been lost and whether the airway is safe. A posterior bleed can fill the stomach before it fills a basin, and swallowed blood that returns as dark vomit is easily mistaken for haematemesis.

ON THE WARD: THE FIRST FIVE MINUTES

1. Sit the patient up and leaning forward. Check pulse, blood pressure and saturation; secure IV access and send a full blood count, group and save, and clotting if anticoagulated or bleeding heavily.
2. Firm pressure on the soft lower third of the nose for at least five minutes without peeking; many units ask for 10 to 15. Ice in the mouth adds a little reflex vasoconstriction.
3. Suction the clots, spray topical vasoconstrictor and anaesthetic, and look with a headlight and speculum.
4. Ask about anticoagulants, antiplatelets, previous bleeds, family history and alcohol, and which side bled first.
5. Don't stop or reverse anticoagulants for a bleed that local measures can control. Discuss with the prescribing team if it is life-threatening.

5.2 History

5.2.1 The bleed

Side of onset, duration, estimated volume, whether it runs forward or into the throat, and previous episodes and their treatment.

5.2.2 Local symptoms

Obstruction, discharge, crusting, facial pain or numbness, and trauma.

5.2.3 Systemic background

Medications including supplements, hypertension, liver or renal disease, easy bruising, and a family history of bleeding or nosebleeds.

5.3 Examination

5.3.1 General

Pallor, pulse and blood pressure, signs of liver disease, bruising, and telangiectasias on the lips, tongue and fingertips.

5.3.2 Anterior rhinoscopy

After clot removal and vasoconstriction: inspect Little's area first, then the front of the lateral wall. The oropharynx is checked for blood running down from behind.

5.3.3 Nasal endoscopy

For recurrent bleeds, for bleeding not localised on rhinoscopy, and for any unilateral bleed with suspicious symptoms. It finds points hidden lateral to the turbinates or high in the nose and allows directed cautery instead of a blind pack.

5.4 Investigations

5.4.1 Blood tests

Not needed for a minor anterior bleed in a healthy patient. For significant bleeds: full blood count, group and save or crossmatch, clotting with INR in warfarinised patients, and liver and renal function where relevant.

5.4.2 Imaging

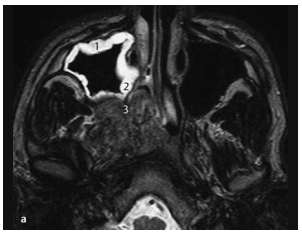
CT or MRI when a tumour is suspected; for juvenile angiofibroma, contrast imaging replaces biopsy.

5.4.3 Angiography

Diagnostic and therapeutic: for suspected pseudoaneurysm, vascular tumours, and as the first step of embolisation.

IMAGING IN THE APP 1 case

Juvenile angiofibroma, the tumour to think of in a teenage boy with one-sided bleeding.



Angiofibroma, Juvenile

Tumours · 12 images

CHAPTER SIX

6 Local management

The ladder of control.

6.1 The treatment ladder

1. **First aid and topical vasoconstriction**
   Pressure on the soft nose, leaning forward; pledgets of vasoconstrictor and local anaesthetic.
2. **Cautery of a visible point**
   Silver nitrate or bipolar, to the bleeding point only.
3. **Anterior packing**
   Ribbon gauze, expanding sponge, or resorbable material.
4. **Posterior packing**
   Postnasal pack, Foley or dual balloon; admit and monitor.
5. **Endoscopic cautery**
   Directed at the source under local or general anaesthesia.

6 **Ligation or embolisation**
For bleeding that persists despite packing.

6.2 First aid

Trotter's method: the patient sits up, leaning slightly forward over a basin, breathing through the mouth, while the soft part of the nose is pinched firmly against the septum. Leaning forward stops blood being swallowed and lets you see how much is lost. Cold compresses add reflex vasoconstriction.

6.3 Cautery

Cautery works when you can see what you are treating. Anaesthetise first, and treat only the bleeding point.

6.3.1 Chemical

Roll a silver nitrate stick over a small ring around the vessel, edge first and then inwards, before touching the vessel itself. Best for small, visible anterior points in a field that is not actively pouring.

6.3.2 Electrocautery

Bipolar diathermy, usually under endoscopic view, is more effective for brisker or deeper points and for posterior sites.

PEARL

Never cauterise both sides of the septum at the same sitting: the mucoperichondrium on both sides may necrose and leave a perforation.

6.4 Anterior packing

6.4.1 Ribbon gauze

About a metre of ribbon gauze, 2.5 cm wide in adults and 12 mm in children, soaked in liquid paraffin or bismuth iodoform paraffin paste and laid in layers so that it presses evenly on the whole cavity. Usually removed after 24 hours; antibiotics are given if it is left longer.

6.4.2 Expanding sponges and balloons

Compressed sponges that swell when wetted, and inflatable balloon packs: quicker to insert, more comfortable, and now the standard in most emergency departments.

6.4.3 Resorbable packs

Preferred when the patient is anticoagulated or has a bleeding disorder, because removing a non-resorbable pack can restart the bleed.

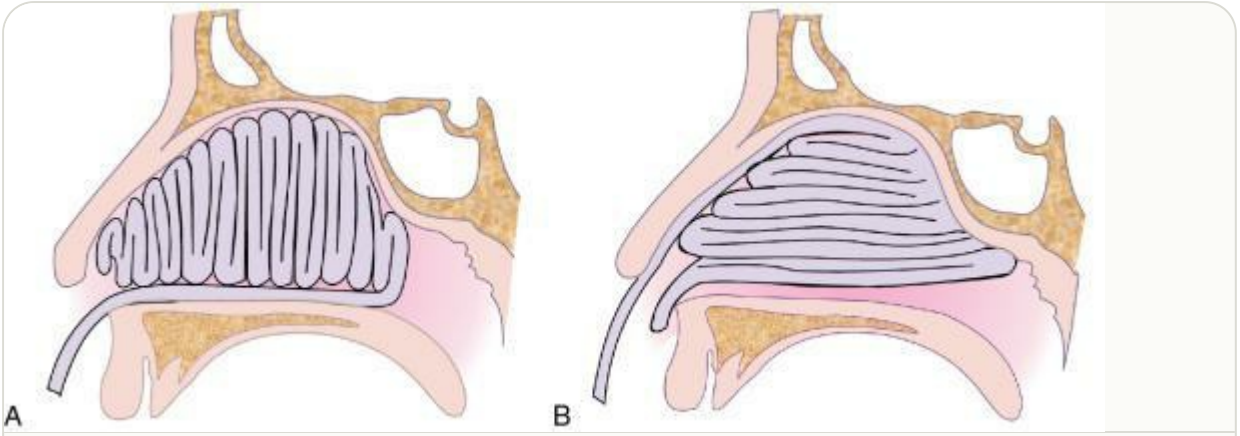


FIGURE 4 Anterior nasal packing. (A) Vertical layers from the back forwards. (B) Horizontal layers from the floor upwards.
Dhingra, fig. 33.4

6.5 Posterior packing

When blood runs back into the pharynx and an anterior pack fails, the choana must be blocked.

6.5.1 Postnasal gauze pack

A rolled gauze cone with three silk ties is drawn into the nasopharynx on a catheter passed through the nose and out of the mouth, then pushed into place with a finger. An anterior pack goes in front, and two ties are secured over a dental roll at the columella. The third tie is left in the oropharynx, taped to the cheek, to pull the pack out later.

6.5.2 Foley catheter

A 12 to 14 F Foley is passed until the balloon lies in the nasopharynx, inflated with saline (not air), drawn forward against the choana and backed by an anterior pack. Protect the ala and columella from pressure.

6.5.3 Dual-balloon devices

Commercial catheters with a posterior and an anterior balloon, and a central tube that preserves a nasal airway.

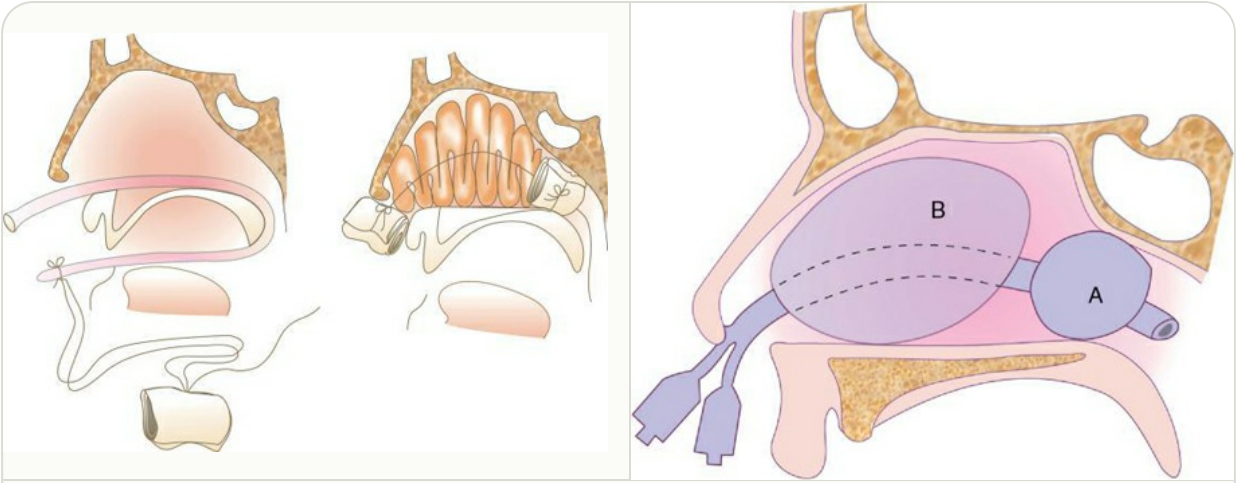


FIGURE 5 First, inserting a postnasal pack on a catheter. Second, a dual-balloon catheter with the posterior balloon at about 10 mL and the anterior at about 30 mL.
Dhingra, figs 33.5 and 33.6

6.5.4 Care of the packed patient

- **Admit and monitor:** posterior packs fail in about a third.
- **Oxygen:** bilateral packs can cause hypoxia through the nasopulmonary reflex, especially in the elderly.
- **Watch for:** alar and columellar necrosis, sinusitis, otitis media, aspiration and toxic shock syndrome.
- **Antibiotics** are traditionally given while a non-resorbable pack is in, though the evidence for routine use is weak.
- **Explain the pack:** what type it is, when it comes out, and what should bring them back.

6.6 Medical adjuncts

6.6.1 Blood pressure

Treat pain and anxiety first; control persistent hypertension, but avoid dropping the pressure sharply.

6.6.2 Tranexamic acid

Topical tranexamic acid did not reduce the need for packing in the NoPAC trial; it should not delay standard treatment.

GUIDELINE UPDATE

The **AAO-HNS Nosebleed guideline (2020)**: compression for at least five minutes; packing if the site cannot be seen; resorbable packs on anticoagulants or antiplatelets; endoscopy for recurrent or difficult bleeds; cautery confined to the bleeding site; ligation or embolisation when these fail; and first-line local treatment before transfusion, reversal or stopping anticoagulants unless the bleed is life-threatening.

CHAPTER SEVEN

7 Surgery and intervention

Taking out the supply: the vessel follows the site.

7.1 Choosing the vessel

PERSISTENT BLEEDING

- SOURCE SEEN **Endoscopic cautery** under local or general anaesthesia
- POSTERIOR, BELOW THE MIDDLE TURBINATE **Sphenopalatine ligation** or embolisation
- ABOVE THE MIDDLE TURBINATE **Ethmoidal ligation** not embolisation
- SEPTAL DEVIATION **Septal surgery** to reach the bleed

7.2 Endoscopic sphenopalatine artery ligation

7.2.1 Indications

Posterior bleeding that persists or recurs despite adequate packing. TESPAL is now the operation of choice and has replaced the older open procedures.

7.2.2 Technique

Under general anaesthesia, a mucosal incision on the lateral wall in front of the crista ethmoidalis; a mucoperiosteal flap is raised back to the crista, behind which the artery and its branches emerge from the foramen. They are clipped or coagulated, and the flap is replaced.

7.2.3 Outcomes and complications

Lee quotes control in 95 to 98%, usually with discharge the next day. Complications are minor: crusting, adhesions and numbness of the palate or nose.

7.3 Ethmoidal artery ligation

7.3.1 Approach

An external (Lynch) incision at the medial canthus, raising the periosteum of the medial orbital wall, usually for high bleeds after trauma.

7.3.2 The 24-12-6 rule

The anterior ethmoidal foramen lies about 24 mm behind the anterior lacrimal crest, the posterior foramen about 12 mm further back, and the optic canal about 6 mm beyond that. The posterior artery is approached with great care, or left alone.

7.4 Embolisation

7.4.1 Method

The internal maxillary artery is catheterised by the femoral route and blocked with particles, gelfoam or coils, often bilaterally because of cross-flow.

7.4.2 Who it suits

Patients unfit for anaesthesia, vascular tumours and traumatic pseudoaneurysms. It cannot treat the ethmoidal arteries, which arise from the internal carotid system.

7.4.3 Outcomes and risks

Success around 88%. Complications include stroke, blindness, facial pain and skin necrosis.

7.5 Septal surgery

Raising a mucoperichondrial flap divides the vessels feeding a recurrent septal bleed; septoplasty also straightens a deviation that hides a bleeding point or

prevents adequate packing.

7.6 Historical operations

7.6.1 Maxillary artery ligation

Through a Caldwell-Luc approach and the posterior antral wall. Failure in 10 to 15%, with infraorbital numbness, oroantral fistula and dental injury.

7.6.2 External carotid ligation

Above the origin of the superior thyroid artery. Too proximal to be reliable, given the collateral flow; of examination interest now.

PEARLS

- Bleeding above the middle turbinate is ethmoidal: embolisation won't help, think ligation.
- The crista ethmoidalis is the landmark for the sphenopalatine foramen: the artery lies just behind it.
- If endoscopic ligation fails, look for an ethmoidal source or a missed accessory branch before repacking.

OPERATIVE GUIDES IN THEATRE 4 operations

Anterior and Posterior
Anterior and posterior ethmoid artery ligation
Sphenopalatine Artery Ligation for Epistaxis
Sphenopalatine artery ligation
Surgical Management of Hereditary Hemorrhagic Telangiectasia
HHT
Juvenile Angiofibroma
Angiofibroma

8 CHAPTER EIGHT Special situations

Children, the anticoagulated, HHT, and the bleed after trauma.

8.1 Children

8.1.1 Typical picture

Bleeding from Little's area, from crusting and picking, that stops on its own.

8.1.2 Treatment

Antiseptic cream such as chlorhexidine with neomycin to the vestibule twice daily for four weeks settles most cases; silver nitrate for a visible vessel that keeps bleeding. Avoid creams containing arachis oil in peanut allergy.

8.1.3 When to investigate

Recurrent bleeding with easy bruising or a family history needs a clotting screen including von Willebrand studies. Under two years of age, think of a bleeding disorder or non-accidental injury.

8.2 The anticoagulated patient

Patients on warfarin, a direct oral anticoagulant or dual antiplatelets bleed longer and re-bleed more often, but stopping their drugs risks stroke or stent thrombosis. Control the bleed locally first with resorbable packing and check the INR in warfarinised patients. Reversal and withholding are for life-threatening bleeds, decided with the prescribing team.

8.3 Hereditary haemorrhagic telangiectasia

Recurrent bleeding from multiple telangiectasias, usually from childhood, with a family history. Repeated packing and aggressive cautery make things worse; laser, bevacizumab, septodermoplasty and Young's procedure are the tools. See the HHT topic for the full account.

8.4 Trauma and pseudoaneurysm

Delayed, massive or recurrent bleeding days to weeks after head injury or skull base surgery suggests a pseudoaneurysm of the internal carotid artery. It is a neurovascular emergency: pack, resuscitate and arrange urgent angiography.

9 CHAPTER NINE Complications and outcome

Of the bleed and of its treatment.

9.1 Of the bleed

Hypovolaemic shock, anaemia, aspiration, and myocardial ischaemia in patients with coronary disease.

9.2 Of treatment

- **Cautery:** septal perforation.
- **Packing:** pressure necrosis, sinusitis, otitis media, hypoxia, toxic shock syndrome.
- **Surgery and embolisation:** see 7.2 to 7.4 .

9.3 Prognosis

Most bleeds settle with simple measures. Recurrence is common in the elderly, the anticoagulated and those with HHT, which is why the cause matters as much as the cure.

PRACTICAL POINTS (OXFORD TEXTBOOK)

About 95% of bleeds arise from Little's area, but the **sphenopalatine artery** is the dominant supply to the nose in about 85% of people (the rest are anterior ethmoid-dominant), which is why endoscopic sphenopalatine ligation works so well for posterior bleeds. At surgery the foramen lies just behind and below the **crista ethmoidalis**, a bony ridge on the lateral wall near the posterior end of the middle turbinate, and the artery often divides into several branches before emerging, all of which must be controlled. Risks to explain include rebleeding, crusting, palatal numbness, adhesions, epiphora and, very rarely, visual loss.

Where available, **electrocautery** is preferred to silver nitrate because it fails less often and reduces packing, admission and pain; after silver nitrate, irrigate with saline to

prevent skin burns. Ice on the nasal bridge has no proven benefit. A patient whose anterior bleed is stopped by cautery can go home with antiseptic or antibiotic cream for two weeks; after a significant posterior bleed, observe for 6 to 12 hours; anyone with a pack is admitted. Many specialists give antibiotics when a pack stays more than 48 hours. For warfarin with major bleeding, give intravenous vitamin K and prothrombin complex concentrate (fresh frozen plasma only if that is unavailable); for minor bleeding with a high INR, withhold warfarin and give oral vitamin K. Ask about "natural blood thinners" such as garlic, ginger, ginkgo, turmeric and vitamin E.

10 CHAPTER TEN For the examination

What the examiner wants to hear.

10.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Little's area:** the four arteries, their parent vessels, and why it bleeds.
- **Trotter's method:** position, mouth breathing, and pinching the soft nose.
- **Anterior versus posterior:** the five-row comparison.
- **Postnasal pack:** technique, the third tie, and why the patient is admitted.
- **Which artery for which site,** and why ethmoidal bleeds can't be embolised.
- **Teenage boy, unilateral recurrent bleed:** angiofibroma, and no biopsy in clinic.

10.2 Common pitfalls

COMMON PITFALLS

- Underestimating blood loss in an elderly patient who is swallowing blood.
- Packing blindly before suctioning and looking, when a cauterisable point was visible.
- Cauterising both sides of the septum at once.
- Stopping anticoagulation reflexively for a bleed a pack would have controlled.

- Repeated packing in recurrent bilateral bleeds that are really HHT.

Written for this app from Oxford Textbook of Otolaryngology ch. 27, 77, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, 25, Dhingra ch. 33 and K. J. Lee ch. 31. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · APPROACH TO

Approach to postnasal drip and catarrh

Postnasal drip, or catarrh, is the feeling of mucus running down the back of the nose and throat, with the urge to clear the throat and sometimes a cough. It is one of the most common and most frustrating complaints in rhinology, because the sensation often outlasts any visible mucus. The approach is to find and treat what can be treated, the nose, the sinuses and reflux, to recognise the patients whose problem is a hypersensitive throat rather than excess mucus, and to make sure that nothing serious in the nasopharynx, larynx or chest has been missed.

AT A GLANCE

- Endoscope the nose and nasopharynx.
- Treat rhinosinusitis first: saline and steroid spray for 2 to 3 months.
- Catarrh without findings: explain, avoid antibiotics.
- Persistent unilateral signs: biopsy.

1 CHAPTER ONE What the patient means

Mucus, sensation or cough.

1.1 Definition and mechanism

Normally, a litre or so of mucus a day is carried backwards by the cilia and swallowed without being noticed. A sensation of postnasal drip arises when the volume rises, when the mucus becomes thicker or more irritant, when mucociliary clearance slows, or when the sensory nerves of the pharynx become hypersensitive. Many patients have more than one of these. In chest medicine, a cough attributed to the nose is called *upper airway cough syndrome*, but many patients with postnasal drip have no cough, and many with chronic cough have a sensitised cough reflex rather than a nasal cause.

DO NOT MISS

- **Unilateral nasal symptoms, blood-stained mucus or a neck lump:** nasopharyngeal and sinonasal tumour.

- **Hoarseness, dysphagia, weight loss or pain on swallowing:** laryngeal or pharyngeal malignancy.
- **A smoker with a cough that has changed, haemoptysis or breathlessness:** chest X-ray and respiratory referral.
- **Persistent cough in a patient taking an ACE inhibitor:** the drug.

2 CHAPTER TWO History

Four sources to explore.

2.1 The questions

Ask what exactly is felt: mucus that can be spat out, or a sensation that nothing relieves; and whether the mucus is clear, thick or discoloured. Ask about the nose, blockage, discharge, sneezing, smell loss and facial pressure, which point to rhinitis or rhinosinusitis. Ask about heartburn, a sour taste, hoarseness worse in the morning, globus and throat clearing, which point to reflux. Ask about cough, its duration (chronic cough lasts eight weeks or more), smoking, asthma and medication, including ACE inhibitors and drugs that dry the mucosa (antidepressants, anxiolytics, anticholinergics). The *Reflux Symptom Index* and *SNOT-22* quantify the symptoms.

3 CHAPTER THREE Examination

Look at both ends.

3.1 Nose, nasopharynx and larynx

Examine the nose with an endoscope, looking for mucopurulent discharge in the middle meatus or sphenoethmoidal recess, polyps, adenoidal tissue, a Thornwaldt cyst and any nasopharyngeal mass. Pass the scope down to the larynx and look for signs associated with reflux (posterior commissure thickening, oedema, erythema) and for any lesion that explains the throat symptoms. Examine the neck. A CT of the

sinuses is arranged only if chronic rhinosinusitis is suspected and not settling on medical treatment.

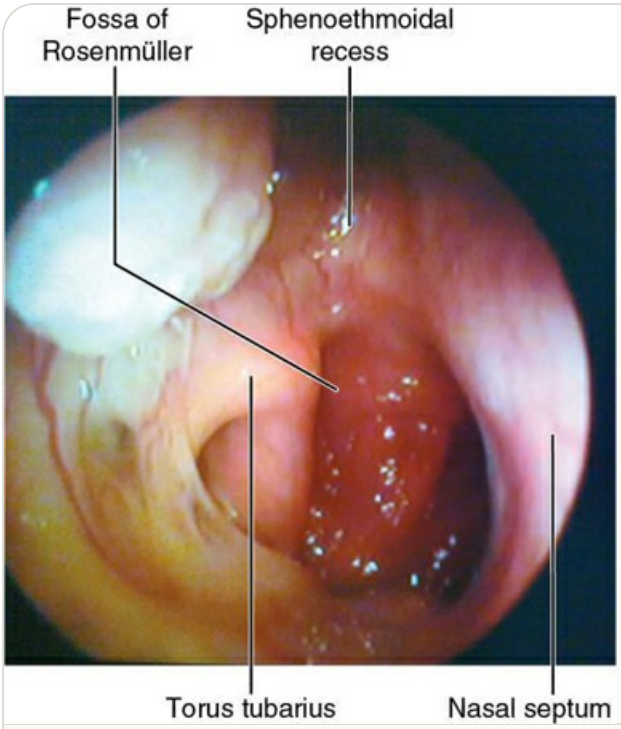


FIGURE 1 The region of the Eustachian tube, sphenoethmoidal recess, choana and fossa of Rosenmüller, examined at endoscopy.

Dhingra, fig. 89.1

4 CHAPTER FOUR Causes

Where the drip comes from.

CAUSES OF POSTNASAL DRIP

SOURCE	EXAMPLES	CLUE
Nose	Allergic and non-allergic rhinitis, rhinitis of the elderly	Sneezing, watery discharge, triggers
Sinuses	Chronic rhinosinusitis, dental sinusitis, fungal ball	Pus in the meatus, smell loss, unilateral
Nasopharynx	Adenoiditis, Thornwaldt cyst, tumour	Endoscopic findings
Clearance	Primary ciliary dyskinesia, cystic fibrosis, dry atrophic nose, after sinus surgery	Thick mucus, crusting, lifelong symptoms

SOURCE	EXAMPLES	CLUE
Throat and reflux	Laryngopharyngeal reflux, globus	Throat clearing, hoarseness, normal nose
Sensory	Cough hypersensitivity, laryngeal hypersensitivity	Cough triggered by talking, cold air, scents
Drugs	ACE inhibitors, drying drugs	Medication history

5 CHAPTER FIVE Management

Treat what is found.

5.1 When a cause is found

Rhinitis is treated with saline irrigation and a steroid spray, with an antihistamine for allergy and ipratropium for watery discharge. **Chronic rhinosinusitis** follows its own pathway: saline, steroids, a long course of a macrolide in selected patients, and surgery if medical treatment fails. **Reflux** symptoms are managed with diet and lifestyle changes, alginates and, for typical acid symptoms, a proton pump inhibitor trial; evidence for proton pump inhibitors in throat symptoms alone is weak. Any drying or causative medication is reviewed with the prescriber.

5.2 When the sensation persists

Many patients have a normal nose and larynx and still feel the drip. Explanation, reassurance that nothing serious has been found, adequate hydration and steam inhalation help. Throat clearing perpetuates the irritation, so **speech and language therapy** for laryngeal hygiene and cough suppression techniques is effective. In refractory chronic cough, a neuromodulator such as low-dose amitriptyline or **gabapentin** may be tried under specialist care.

POSTNASAL DRIP

1 Red flags?

→ unilateral symptoms, blood, neck lump, hoarseness or dysphagia: full endoscopic examination and imaging

→ **smoker with changed cough**
chest X-ray

2 Nasal endoscopy

→ **rhinitis**
saline and steroid spray, ipratropium if watery

→ **pus or polyps**
chronic rhinosinusitis pathway

→ **nasopharyngeal lesion**
biopsy

3 Throat and larynx

→ **reflux features**
lifestyle, alginate, trial of proton pump inhibitor

→ **normal**
step 4

4 Normal nose and larynx

→ review drugs

→ explanation and hydration

→ speech therapy for throat clearing

→ neuromodulator for refractory cough

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The mechanisms:** volume, viscosity, clearance, hypersensitivity.
- **Upper airway cough syndrome** and cough hypersensitivity.
- **The red flags** that need endoscopy and imaging.
- **Treating the nose, the sinuses and reflux;** speech therapy when nothing is found.

6.2 Common pitfalls

COMMON PITFALLS

- Repeating antibiotics for "catarrh" with no evidence of infection.
- Missing the ACE inhibitor in a patient with chronic cough.
- Operating on the sinuses for postnasal drip alone when the CT and endoscopy are normal.

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CLINICAL · APPROACH TO

Approach to facial pain and headache

Patients and doctors alike blame the sinuses for facial pain, but most people referred with "sinus headache" do not have sinus disease. Rhinosinusitis rarely causes pain without nasal blockage, discharge or smell loss, and in the absence of pus or inflammation on endoscopy the diagnosis is very unlikely. The approach is to take a structured pain history, examine the nose with an endoscope, avoid CT as a screening test, and recognise the primary headaches and neuralgias that are the real causes for most patients: migraine, tension-type headache, midfacial segment pain, the trigeminal autonomic cephalalgias and trigeminal neuralgia.

AT A GLANCE

- No nasal symptoms and normal endoscopy: it is not sinusitis.
- Midfacial segment pain responds to low-dose amitriptyline.
- Trigeminal neuralgia: carbamazepine and MRI.
- Numbness with pain = perineural tumour until imaged.

1 CHAPTER ONE The pain history

Most of the diagnosis.

1.1 A structured history

Use the same headings for every patient. **Site**: unilateral or bilateral, and the dermatome. **Character**: throbbing, pressing, stabbing or burning. **Severity**, out of ten: migraine and cluster headache often reach ten. **Timing**: how long each attack lasts, how often attacks come, and whether there are clusters and remissions. **Associated features**: nausea, photophobia and aura (migraine); a red, watering eye, rhinorrhoea, blockage and a drooping lid on the side of the pain (trigeminal autonomic cephalalgias); nasal symptoms (rhinosinusitis). **Triggers**: alcohol in cluster headache, touch or chewing in trigeminal neuralgia, the menstrual cycle in migraine. **Relieving factors** and what has been tried, including analgesic overuse, which itself causes daily headache.

Rhinosinusitis pain accompanies an attack of nasal symptoms and settles when they do. Pain that has not changed with years of antibiotics and nasal sprays is unlikely to be sinogenic.

RED FLAGS IN HEADACHE AND FACIAL PAIN

- **Thunderclap headache**, new headache with fever, neck stiffness, confusion or focal neurological signs: emergency imaging and neurology.
- **New headache over 50 with jaw claudication, scalp tenderness or visual symptoms:** giant cell arteritis — ESR, CRP and steroids the same day.
- **Headache worse lying down, on waking or with straining, with visual obscurations or pulsatile tinnitus:** raised intracranial pressure.
- **Facial numbness, diplopia, proptosis, a loose tooth or a nasal mass:** sinonasal malignancy.
- **Acute sinusitis with periorbital swelling, reduced vision or severe headache:** orbital or intracranial complication.

2 CHAPTER TWO Examination and tests

Endoscopy first.

2.1 The examination

Examine the nose with an endoscope: mucopus or oedema in the middle meatus supports rhinosinusitis, and its absence makes it very unlikely. Examine the cranial nerves, especially the trigeminal nerve and the eyes, the temporomandibular joints, the teeth, the temporal arteries and the neck muscles.

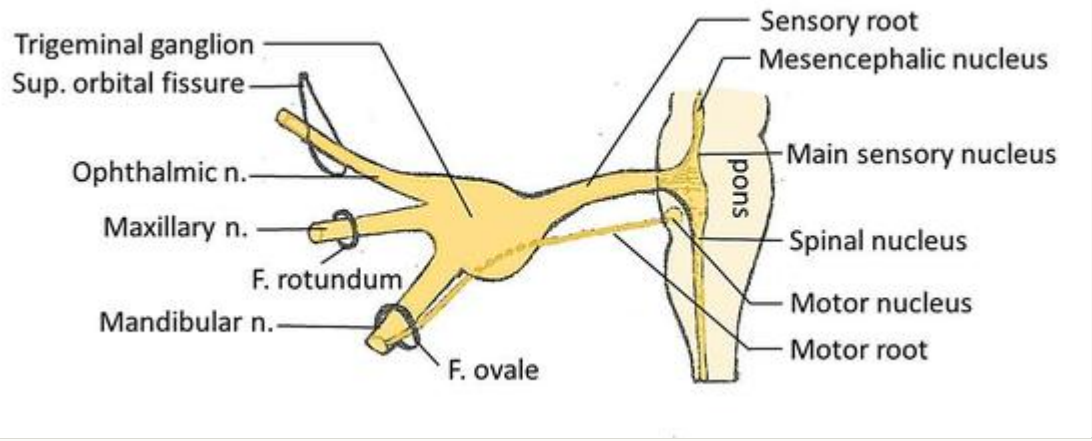


FIGURE 1 The trigeminal ganglion and its divisions, the pathway for most facial pain.
Anatomy atlas · Doss p. 148

2.2 Investigation

CT of the sinuses is not a screening test for facial pain, because incidental mucosal thickening is found in about a third of people without symptoms. It is used when malignancy, fungal disease or a complication is suspected, and in chronic rhinosinusitis that has failed medical treatment when surgery is planned. Dental radiographs and CT show odontogenic disease. MRI is used for trigeminal neuralgia, to look for a compressing vessel or demyelination, and for atypical or neurological features.

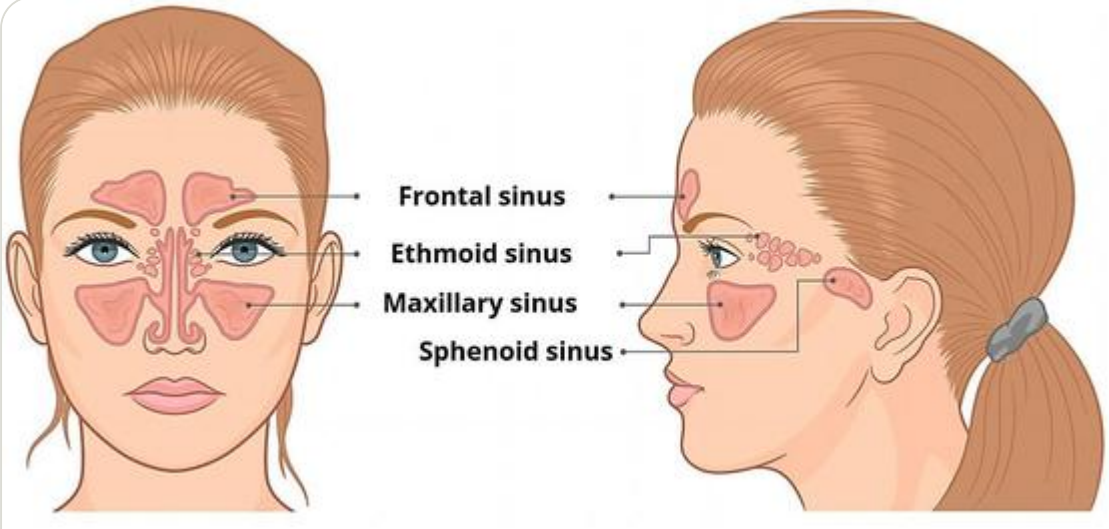


FIGURE 2 The paranasal sinuses.
Anatomy atlas · ENT Steps p. 126

3 CHAPTER THREE The causes

Recognising each.

FACIAL PAIN AND HEADACHE — FEATURES

CONDITION	PAIN AND TIMING	ASSOCIATED FEATURES	FIRST-LINE TREATMENT
Rhinosinusitis	Pressure with nasal symptoms; settles when they do	Blockage, discharge, smell loss; pus on endoscopy	Treat the sinuses
Migraine	Unilateral, throbbing, 4 to 72 hours	Nausea, photophobia, aura in about a quarter; may cause nasal congestion	Triptan acutely; propranolol, topiramate or amitriptyline to prevent
Tension-type headache	Bilateral, band-like pressing, mild to moderate	Few; stress, poor sleep	Simple analgesia, amitriptyline
Midfacial segment pain	Symmetrical pressure over the nasal bridge, cheeks or between the eyes	Normal endoscopy; hyperaesthesia of the skin	Low-dose amitriptyline
Cluster headache	Severe, orbital, 15 to 180 minutes, up to eight a day, in bouts	Ipsilateral tearing, rhinorrhoea, ptosis, restlessness; alcohol trigger	High-flow oxygen, subcutaneous sumatriptan; verapamil to prevent
Paroxysmal hemicrania	Like cluster but 2 to 30 minutes, many a day	Autonomic features	Indometacin (diagnostic response)
SUNCT and SUNA	Seconds to minutes, very frequent	Conjunctival injection and tearing	Lamotrigine
Trigeminal neuralgia	Electric shocks for seconds, usually V2 or V3	Triggered by touch, chewing, washing	Carbamazepine; MRI; microvascular decompression
Giant cell arteritis	Temporal, new, over 50	Jaw claudication, visual loss, raised ESR	Steroids at once

CONDITION	PAIN AND TIMING	ASSOCIATED FEATURES	FIRST-LINE TREATMENT
Temporomandibular disorder	Preauricular, with chewing	Clicking, tender joint and muscles	Soft diet, splint, physiotherapy

3.1 Midfacial segment pain

Midfacial segment pain is a form of tension-type headache affecting the middle third of the face. Patients describe a symmetrical pressure or tightness across the bridge of the nose, behind the eyes or over the cheeks, often feel their nose is blocked although it is not, and may have tender skin over the area. Endoscopy is normal, and the CT is normal or shows incidental changes. It responds to low-dose amitriptyline, starting at 10 mg at night and increased if needed, given for at least six weeks before its effect is judged and continued for about six months. Sinus surgery does not help.

3.2 Migraine and the nose

Migraine affects about one in seven people. Attacks can cause nasal congestion, discharge and tearing, and facial pain may be the only symptom, so migraine is commonly misdiagnosed as sinusitis. Acute attacks are treated with analgesics and triptans, used on no more than a few days a week to avoid medication-overuse headache. Prevention uses propranolol, topiramate (contraindicated in pregnancy) or amitriptyline, then candesartan, botulinum toxin for chronic migraine, or CGRP-targeted antibodies for refractory cases.

3.3 Contact point headache

The idea that a septal spur touching a turbinate causes pain is controversial. Most patients with contact points have no pain, and most with pain and contact points improve with medical treatment for midfacial segment pain. Surgery is considered only when a temporary local anaesthetic block abolishes the pain and other causes have been treated.

4 CHAPTER FOUR Pathway

Sinus or not?

FACIAL PAIN

1 Red flags?

- → **thunderclap, neurological signs, fever**
  emergency imaging
- → **over 50 with jaw claudication**
  ESR, CRP, steroids today
- → **numbness, diplopia, nasal mass**
  imaging and biopsy

2 Nasal symptoms with the pain?

- → **yes, pus or polyps on endoscopy**
  rhinosinusitis pathway, CT if surgery considered
- → **no nasal symptoms, normal endoscopy**
  step 3

3 Pattern of the pain

- → **unilateral throbbing with nausea**
  migraine
- → **symmetrical pressure over the midface**
  midfacial segment pain — amitriptyline
- → **short, severe, with ipsilateral autonomic features**
  cluster headache or another trigeminal autonomic cephalalgia
- → **electric shocks triggered by touch**
  trigeminal neuralgia — MRI, carbamazepine

4 Not improving

- → review analgesic overuse

→ refer to neurology or a pain clinic

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Why most "sinus headache" is not sinusitis:** pain needs nasal symptoms and endoscopic signs.
- **A structured pain history** with timing and associated features.
- **Midfacial segment pain:** features and amitriptyline.
- **The trigeminal autonomic cephalalgias:** cluster headache, paroxysmal hemicrania, SUNCT and SUNA.
- **Why CT is not a screening test** for facial pain.
- **Red flags,** especially giant cell arteritis and raised intracranial pressure.

5.2 Common pitfalls

COMMON PITFALLS

- Operating on the sinuses for facial pain with a normal endoscopy and incidental CT changes.
- Missing cluster headache because the patient also has a runny nose.
- Giving repeated antibiotics for recurrent "sinusitis" that is migraine.
- Missing medication-overuse headache.

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CLINICAL · APPROACH TO

Approach to loss of smell

Loss of smell was a neglected symptom until COVID-19 made it famous. It matters more than patients and doctors once thought: it removes a warning system for smoke, gas and spoiled food, flattens the taste of food (most "taste" is smell reaching the nose from the mouth), affects mood and relationships, and in older people predicts both neurodegenerative disease and death. The approach divides causes into **conductive** loss, where odours cannot reach the olfactory cleft, and **sensorineural** loss, where the receptors or their central pathways are damaged, and looks for the treatable causes of each.

AT A GLANCE

- Conductive: polyps, CRS — oral steroid trial helps.
- Sensorineural: post-viral, head injury, age, Parkinson's.
- Psychophysical testing (Sniffin' Sticks, UPSIT).
- Olfactory training twice daily for months.

1 CHAPTER ONE Terms and anatomy

What can go wrong, and where.

1.1 Definitions

Anosmia is total loss of smell; **hyposmia** is a reduced sense of smell. **Parosmia** is a distorted perception of a familiar odour, usually unpleasant (coffee smells of sewage); **phantosmia** is perceiving a smell that is not there. Parosmia is often a sign that the olfactory nerve is regenerating and carries a reasonable prognosis.

1.2 How smell works

The olfactory epithelium lies in the narrow olfactory clefts, on the upper septum, the superior turbinate and the cribriform plate. About 10 to 15% of inspired air reaches it with each breath, more with sniffing. The receptor neurons send axons through the cribriform plate to the olfactory bulb and regenerate throughout life

from basal stem cells, which is why recovery is possible. **Retronasal** smell, from food in the mouth reaching the nose, makes up most of what patients call flavour, so a patient with anosmia often complains of losing taste.

DO NOT MISS

- **Unilateral loss with nasal blockage or bleeding:** a sinonasal tumour, including olfactory neuroblastoma.
- **Loss with headache, personality change or visual symptoms:** a frontal or olfactory groove meningioma.
- **Phantosmia in brief episodes:** temporal lobe seizures.
- **Gradual loss in an older adult with tremor or memory problems:** Parkinson's disease or dementia, in which smell loss can come years first.
- **Absent smell from birth with delayed puberty:** Kallmann syndrome.

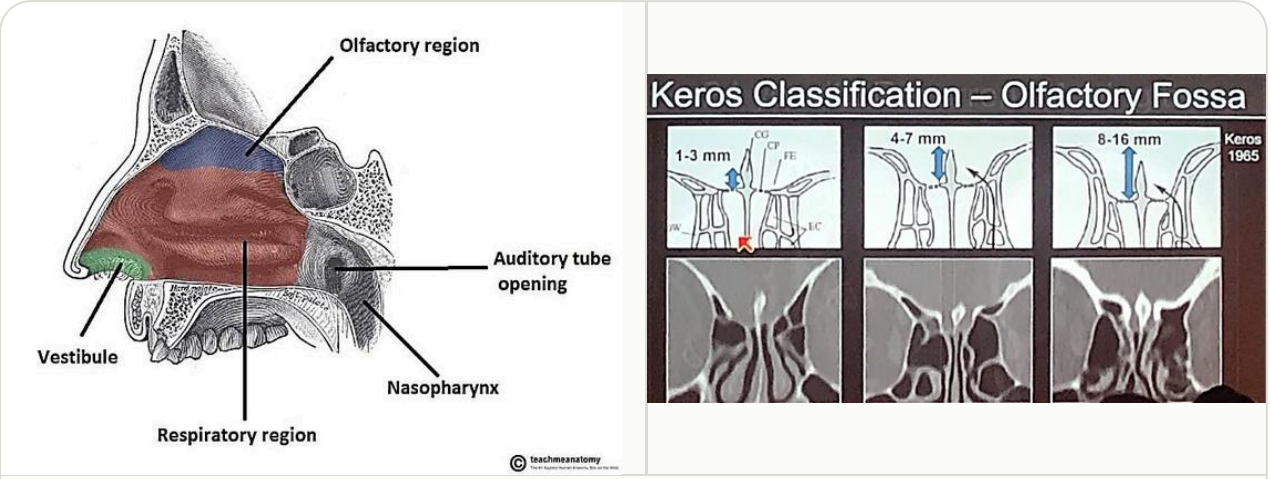


FIGURE 1 First, the olfactory region of the nasal cavity. Second, the Keros classification of the olfactory fossa.

Anatomy atlas · ENT Steps p. 97, ENT Steps p. 143

2 CHAPTER TWO History and examination

Pattern and cause.

2.1 History

Ask whether the loss came on suddenly or gradually, whether it fluctuates, and what preceded it: a cold or COVID-19, a head injury, nasal surgery. **Fluctuating** loss

that improves with steroids is conductive and inflammatory, typically chronic rhinosinusitis with polyps. **Sudden** loss after a viral infection or a blow to the head is sensorineural. Ask about distortions and phantom smells, nasal symptoms, smoking, chemical exposure, drugs, neurological symptoms and family history, and about safety at home (smoke alarms, gas detectors, food dates).

2.2 Examination and testing

Examine the nose with an endoscope, paying particular attention to the olfactory clefts, which may be blocked by polyps or oedema when the rest of the nose looks normal. Examine the cranial nerves.

Smell should be measured, not estimated. Validated tests include the 40-item **University of Pennsylvania Smell Identification Test (UPSIT)**, the **Sniffin' Sticks** (threshold, discrimination and identification, giving a TDI score) and short screening tests. Smell tests also detect malingering, because a patient who scores well below chance must be recognising the odours to avoid them.

2.3 Imaging

MRI of the anterior skull base and brain is indicated for unexplained loss, unilateral loss, or neurological symptoms; it may show reduced olfactory bulb volume. CT of the sinuses is indicated when chronic rhinosinusitis or a sinonasal mass is suspected.

3 CHAPTER THREE Causes

Conductive and sensorineural.

CAUSES OF SMELL LOSS

TYPE	CAUSE	FEATURES
Conductive	Chronic rhinosinusitis with polyps	Fluctuating, improves with steroids; commonest cause in clinic

TYPE	CAUSE	FEATURES
Conductive	Allergic rhinitis, septal deviation, tumours	Blockage; unilateral if a mass
Sensorineural	Post-viral, including COVID-19	Sudden after infection; parosmia common
Sensorineural	Head injury	Shearing of olfactory fibres at the cribriform plate; worse with occipital blows
Sensorineural	Neurodegeneration	Parkinson's (about 90% at diagnosis), Alzheimer's
Sensorineural	Ageing (presbyosmia)	Common over 65
Sensorineural	Toxins and drugs	Smoking, chemicals, radiotherapy, some drugs
Sensorineural	Congenital	Kallmann syndrome, isolated congenital anosmia
Other	Idiopathic	About a fifth of specialist clinic patients

3.1 Post-viral and COVID-19 smell loss

Viral damage to the supporting cells and receptor neurons is the commonest sensorineural cause. COVID-19 typically caused a sudden, complete anosmia that patients noticed sharply, and most recovered within weeks, but a substantial minority were left with lasting loss and parosmia.

3.2 Post-traumatic smell loss

Head injury shears the olfactory fibres at the cribriform plate or bruises the olfactory bulbs and frontal lobes. Recovery is less likely than after viral loss; only about a third to a half notice improvement over years.

4 CHAPTER FOUR Management

Treat the cause; train the nose.

4.1 Conductive loss

Chronic rhinosinusitis with polyps is treated with saline and topical steroids, preferably as drops in the head-down position, with a short course of oral steroids that also shows whether the loss is reversible. Endoscopic sinus surgery opens the olfactory clefts, and biologics (such as dupilumab) can restore smell in severe polyp disease.

4.2 Sensorineural loss

Olfactory training is the best-supported treatment: the patient sniffs four strong odours (traditionally rose, lemon, eucalyptus and clove) for about 20 seconds each, twice a day, for at least three to six months. Changing the odours every few months may improve results. Oral steroids are most likely to help early after a viral or traumatic loss, and topical steroids when the olfactory cleft is swollen. Other treatments, including omega-3 supplements, vitamin A drops and platelet-rich plasma, have limited evidence.

Every patient needs safety advice: smoke and gas detectors, checking food dates, and care with personal hygiene. Recovery after a viral loss can take years, and parosmia often signals regeneration.

SMELL LOSS

1 Red flags?

→ **unilateral with blockage or bleeding**
endoscopy and imaging, biopsy

→ **neurological symptoms**
MRI and neurology

2 Endoscopy of the olfactory clefts

→ **polyps or oedema**
conductive — steroids, chronic rhinosinusitis pathway, surgery or biologic

→ **normal**
sensorineural, step 3

3 Measure smell

→ **anosmia or hyposmia confirmed**
identify the cause — viral, trauma, ageing, neurological, congenital

→ **scores below chance**
consider non-organic loss

4 Sensorineural loss

→ olfactory training for at least three to six months

→ oral and topical steroids early

→ safety advice

→ MRI if no clear cause

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Conductive versus sensorineural loss;** the olfactory cleft.
- **Anosmia, hyposmia, parosmia and phantosmia.**
- **Validated smell tests:** UPSIT and Sniffin' Sticks.
- **Olfactory training:** the method and evidence.
- **Smell loss as an early sign** of Parkinson's disease.

- **Safety advice** for patients with anosmia.

5.2 Common pitfalls

COMMON PITFALLS

- Not examining the olfactory clefts.
- Telling a patient with post-viral loss that nothing can be done, when training helps.
- Missing a unilateral tumour behind "polyps".

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CLINICAL · APPROACH TO

Approach to nasal crusting

Some crusting is part of a healthy nose in a dry climate, but persistent crusting means that something has damaged the mucosa or the mucociliary clearance. The causes range from the simple, such as a dry environment, nose picking and recent surgery, through septal perforation and atrophic rhinitis, to diseases that must not be missed: granulomatosis with polyangiitis, sarcoidosis, tuberculosis, cocaine-induced destruction and malignancy. The approach separates local from systemic causes, takes biopsies and blood tests when the cause is not obvious, and relies on regular saline irrigation for almost everyone.

AT A GLANCE

- Post-surgical crusting is common and self-limiting.
- Perforation plus crusting: GPA, cocaine, TB, sarcoid, tumour.
- Bloods: ANCA, ACE, syphilis serology.
- Douching is the treatment for most.

1 CHAPTER ONE Why the nose crusts

Mucosa, airflow, clearance.

1.1 Mechanisms

Crusts form when mucus dries faster than it is cleared. This happens when the mucosa is damaged and replaced by squamous or scarred epithelium, when airflow is abnormally fast or turbulent across one area (the edge of a perforation, a wide postoperative cavity), when the cilia fail, or when the mucosa is inflamed by a granulomatous or vasculitic disease. Crusts then harbour bacteria, cause obstruction, bleeding and odour, and damage the mucosa further.

DO NOT MISS

- **Crusting with systemic symptoms** (cough, haemoptysis, arthralgia, rash, red eyes, deafness, kidney disease, weight loss): granulomatosis with polyangiitis, EGPA or sarcoidosis — ANCA, ACE, urinalysis, chest imaging, biopsy.

- **Crusting with a perforation or palatal defect in a cocaine user:** cocaine-induced midline destructive lesion; levamisole adds a vasculitis.
- **Unilateral crusting with ulceration, a mass or bleeding:** NK/T-cell lymphoma, squamous carcinoma.
- **Collapse of the nasal bridge (saddle nose):** granulomatosis with polyangiitis, relapsing polychondritis, cocaine, previous surgery.

CHAPTER TWO

2 Assessment

History, endoscopy, tests.

2.1 History

Ask how long the crusting has lasted, whether it followed surgery, and whether there is bleeding, a whistle, a bad smell (often noticed by others more than by the patient), blockage or smell loss. Ask about digital trauma, nasal sprays and drugs (including long-term decongestants and intranasal cocaine), occupational exposure to chrome compounds and industrial dusts, radiotherapy and nasogastric tubes. A full systems review looks for features of vasculitis, sarcoidosis and tuberculosis, and for chest disease, infertility and childhood infections suggesting ciliary dyskinesia or cystic fibrosis.

2.2 Examination and investigation

Examine the nose with an endoscope, before and after the crusts are softened and removed, looking for perforation, granulations, ulceration, a cobblestoned or yellow nodular mucosa (sarcoidosis), a wide cavity, and synechiae. Look at the palate, the eyes, the ears and the skin.

When the cause is not obvious, blood tests include a full blood count, renal function, inflammatory markers, **ANCA** (c-ANCA directed against proteinase-3 in granulomatosis with polyangiitis, p-ANCA against myeloperoxidase in other vasculitides), serum ACE and calcium, and, as appropriate, syphilis and tuberculosis tests. Urinalysis looks for blood and protein from renal involvement. A **biopsy** is taken from the edge of active disease, not from the crusts, when malignancy or

vasculitis is suspected; it may need repeating, because a single biopsy is often non-diagnostic. CT shows bony destruction and sinus disease.

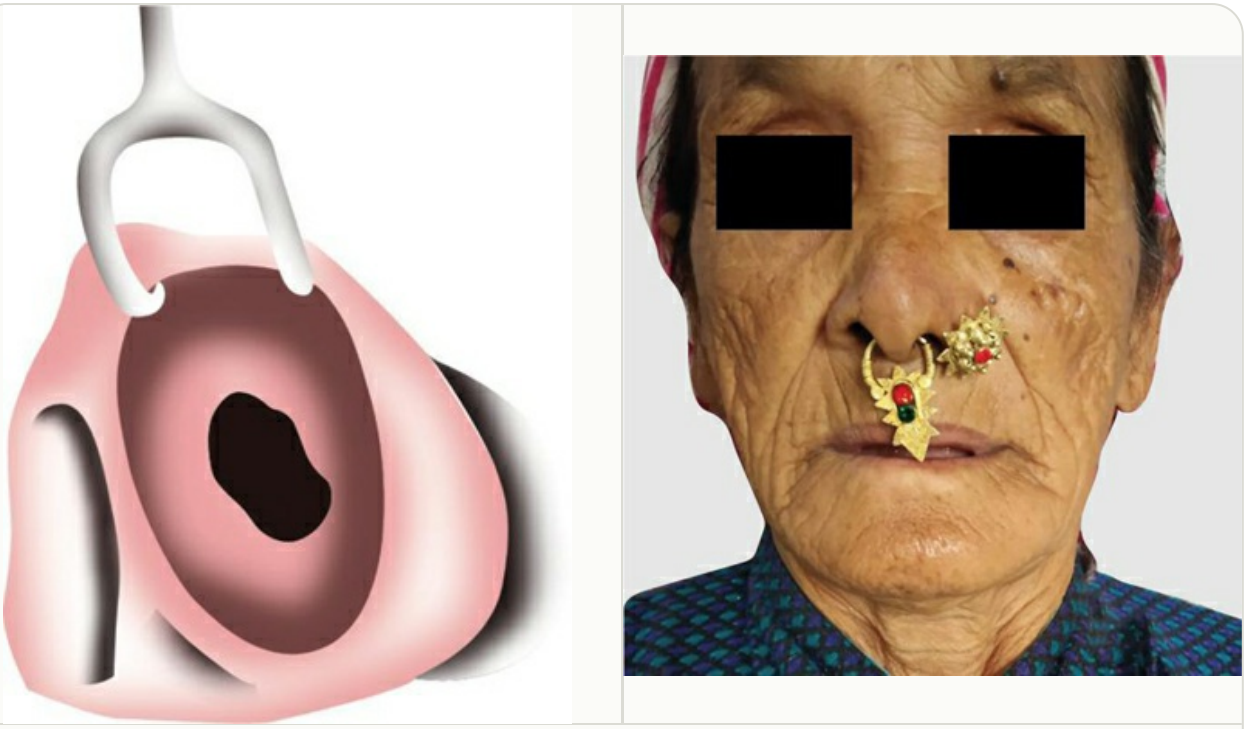


FIGURE 1 First, a septal perforation. Second, a septum deliberately perforated for an ornament.
Dhingra, figs 26.10 and 26.11

3 CHAPTER THREE Causes

Local and systemic.

CAUSES OF NASAL CRUSTING

GROUP	CAUSES	CLUES
Environmental and local	Dry air, digital trauma, decongestant overuse	Anterior septum, picking, Little's area
Surgical	After sinus surgery or turbinate reduction; empty nose syndrome	History of surgery; wide cavity
Septal perforation	Surgery, trauma, picking, cocaine, chrome workers, vasculitis	Whistling, bleeding, crusts around the edge
Atrophic rhinitis	Primary (ozaena); secondary after surgery, radiotherapy, granulomatous disease	Wide cavity, foul crusts, smell loss

GROUP	CAUSES	CLUES
Vasculitis	Granulomatosis with polyangiitis, EGPA	ANCA, systemic features, saddle nose
Granulomatous	Sarcoidosis, tuberculosis, leprosy, syphilis, rhinoscleroma	Yellow nodules, systemic features, culture
Drugs	Cocaine, levamisole	Destruction, palatal perforation
Malignancy	NK/T-cell lymphoma, squamous carcinoma	Unilateral, progressive, ulceration
Clearance	Primary ciliary dyskinesia, cystic fibrosis	Lifelong symptoms, chest disease

3.1 Atrophic rhinitis

Primary atrophic rhinitis (ozaena) occurs mainly in women in tropical and low-income settings. The mucosa and turbinates atrophy, the cavities widen, and greenish, foul-smelling crusts form, with anosmia, so the patient does not notice the odour that others do. *Klebsiella ozaenae* is often cultured. **Secondary atrophic rhinitis** follows excessive surgery, radiotherapy, trauma and granulomatous disease. **Empty nose syndrome** is a related condition after over-resection of the turbinates: the cavity is wide and clear, yet the patient feels blocked and short of breath.

3.2 Granulomatosis with polyangiitis

This necrotising vasculitis of small vessels affects the upper airway in most patients and may be limited to it. Nasal features are crusting, bleeding, obstruction, septal perforation and saddle collapse; there may also be otitis media with effusion, sensorineural hearing loss, subglottic stenosis, lung nodules and glomerulonephritis. c-ANCA supports the diagnosis, which is confirmed by biopsy and managed with rheumatology and renal teams.

4 CHAPTER FOUR Management

Irrigate, treat the cause.

4.1 General measures

Regular high-volume **saline irrigation**, several times a day, is the foundation for every cause. Lubricating ointments or oils and humidification help, and glucose-glycerine drops or warm alkaline douches soften crusts in atrophic rhinitis. Topical antibiotic ointment helps when the crusts are infected or *Staphylococcus aureus* carriage is a problem. Decongestant and steroid sprays are reviewed and stopped if they worsen crusting.

4.2 Specific treatment

Septal perforations that cause symptoms are treated with irrigation, a silicone button or surgical repair. **Atrophic rhinitis** is treated with irrigation and, for Klebsiella infection, appropriate antibiotics; surgery to narrow the cavity (Young's operation, closing the nostrils for months, or submucosal implants) is reserved for severe cases. **Vasculitis and sarcoidosis** are treated systemically with steroids and immunosuppression or biologics such as rituximab. **Cocaine-related disease** requires abstinence before any reconstruction. **Malignancy** is treated on oncological principles.

NASAL CRUSTING

1 Red flags?

→ **systemic symptoms, saddle nose or perforation**
ANCA, ACE, urinalysis, chest imaging, biopsy

→ **unilateral ulceration or mass**
biopsy and imaging

2 Clean and examine

→ **perforation**
perforation pathway

- **wide cavity after surgery**
  secondary atrophic rhinitis or empty nose
- **foul green crusts with wide cavities**
  primary atrophic rhinitis — culture
- **anterior septum only**
  dryness, picking, decongestants

3 Treat

- saline irrigation for all
- treat the cause
- stop drugs that worsen crusting
- repeat biopsy if disease progresses without a diagnosis

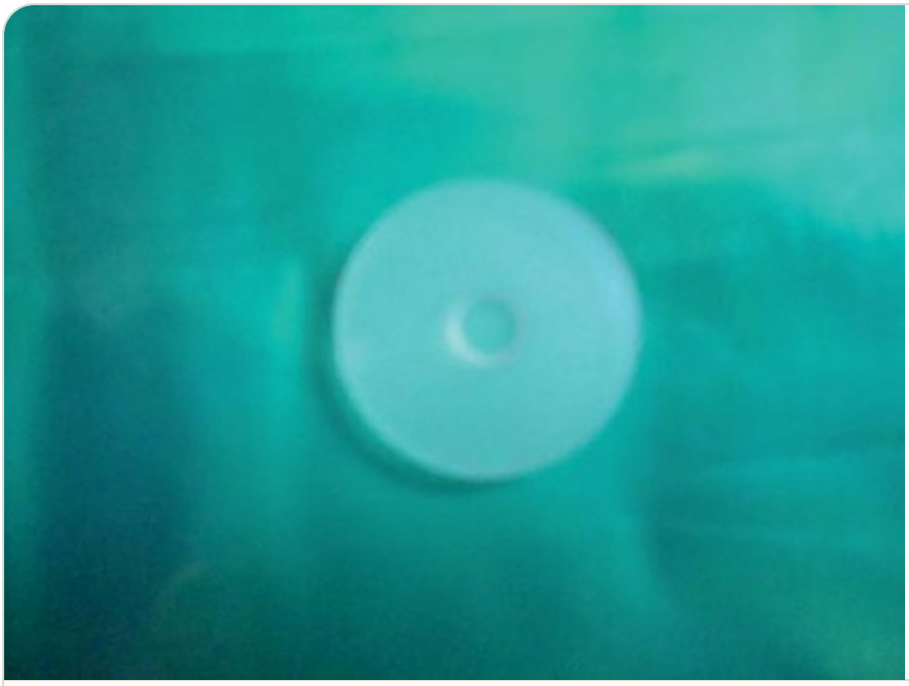


FIGURE 2 A silicone septal button closing a perforation.
Dhingra, fig. 26.12

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Local versus systemic causes** of nasal crusting.
- **The work-up:** ANCA, ACE, urinalysis, chest imaging and biopsy from the edge.
- **Atrophic rhinitis:** primary and secondary, Klebsiella ozaenae, Young's operation.
- **Granulomatosis with polyangiitis** in the upper airway.
- **Cocaine** and levamisole.

5.2 Common pitfalls

COMMON PITFALLS

- Treating vasculitis as chronic rhinosinusitis.
- Biopsying crusts instead of the active edge.
- Repairing a perforation in an active cocaine user.
- Over-resecting turbinates and creating an empty nose.

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CLINICAL · APPROACH TO

Approach to the watering eye

A watering eye is shared territory between the ophthalmologist and the rhinologist. Tears overflow either because too many are produced or because too few drain away, and a surprising proportion of patients referred with "blocked tear ducts" actually have a dry eye producing reflex tearing. The rhinologist's part is to confirm a drainage obstruction, find its level, look in the nose for a cause, and offer endoscopic dacryocystorhinostomy when the obstruction lies at or below the lacrimal sac.

AT A GLANCE

- Hypersecretion versus drainage failure.
- Syringing and probing localise the block.
- Endoscopic DCR for nasolacrimal duct obstruction.
- Bloody tears or a mass at the medial canthus: image and biopsy.

1 CHAPTER ONE Oversupply or poor drainage?

The first question.

1.1 Two mechanisms

Hypersecretion comes from an irritated ocular surface: dry eye with reflex tearing (the cause in up to 40% of referrals), blepharitis, trichiasis, entropion, ectropion, allergy, foreign bodies and, rarely, gustatory tearing ("crocodile tears") after facial nerve regeneration. **Impaired drainage** comes from a failure of the lid pump (facial palsy, lax lower lid), a malpositioned or stenosed punctum, a canalicular block, or an obstruction of the sac or nasolacrimal duct. Nasolacrimal obstruction is common in older adults and in women, and becomes more frequent with age.

1.2 Lacrimal anatomy

Tears drain through the upper and lower **puncta** into the **canaliculi**, which run vertically for about 2 mm and then horizontally for about 8 to 10 mm before joining as the **common canaliculus**, which enters the **lacrimal sac** in the lacrimal fossa.

The **nasolacrimal duct** runs down from the sac for about 12 mm and opens beneath the inferior turbinate, about 1 to 1.5 cm behind its anterior end, under the mucosal fold called the **valve of Hasner**. The lower canaliculus carries most of the flow. In the nose, the sac lies deep to the frontal process of the maxilla, in front of the attachment of the middle turbinate and the uncinate process.

DO NOT MISS

- **A firm mass above the medial canthal tendon, blood-stained tears or bloody reflux on syringing:** lacrimal sac tumour — image and biopsy.
- **A watering eye with nasal blockage, bleeding or numbness:** sinonasal tumour invading the duct.
- **Acute dacryocystitis with orbital cellulitis signs:** urgent treatment.
- **A neonate with a blue swelling at the medial canthus:** dacryocystocele, which can extend into the nose and obstruct the airway.

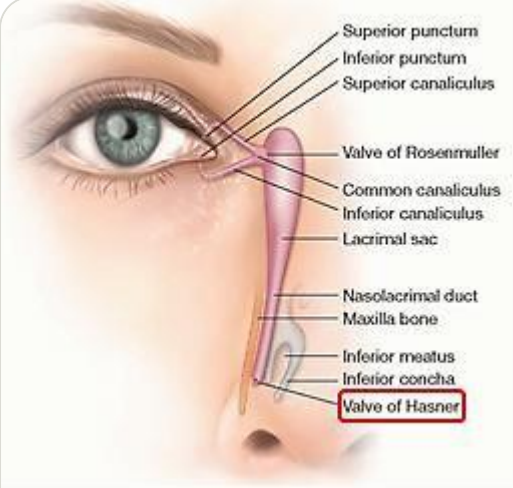


FIGURE 1 Lacrimal drainage: canaliculi, sac, nasolacrimal duct and Hasner's valve.
Anatomy atlas · ENT Steps p. 100

2 CHAPTER TWO Assessment

Find the level.

2.1 History and examination

Ask whether the watering is constant or intermittent, unilateral or bilateral, and associated with stickiness, discharge, swelling at the inner canthus, pain or a red eye. Ask about previous facial trauma, nasal or sinus surgery, radiotherapy, sarcoidosis, granulomatosis with polyangiitis and drugs (topical glaucoma drops, docetaxel).

Examine the lids for position and laxity, the puncta for size and position, and the ocular surface for dryness and inflammation. Press over the lacrimal sac: mucus or pus regurgitating from the puncta indicates a mucocoele or dacryocystitis with a blocked duct. Examine the nose with an endoscope for septal deviation, turbinate or polyp disease, previous surgery, tumours and access to the lacrimal region.

2.2 Tests

The fluorescein dye disappearance test places 2% fluorescein in both eyes; retention after five minutes indicates impaired drainage. **Jones tests** look for dye in the inferior meatus: the primary test (dye recovered under the turbinate) shows a patent system under physiological conditions, and the secondary test, which syringes saline after a failed primary test, separates partial obstruction (dye recovered) from complete obstruction or pump failure. **Lacrimal syringing** with a cannula in the lower canaliculus is the key test: free flow to the nose means a patent system; reflux through the same punctum means a canalicular block; reflux through the opposite punctum means a block in the common canaliculus or below, typically the nasolacrimal duct. **Dacryocystography** and **dacryoscintigraphy** show anatomy and function respectively, and CT or MRI is used after trauma or surgery, for suspected tumour, or with sinus disease.

For personal study. Follow local protocols and senior advice for patients.

3 CHAPTER THREE Causes of obstruction

By level.

LEVELS OF LACRIMAL OBSTRUCTION

LEVEL	CAUSES	TREATMENT
Punctum	Stenosis, eversion with ectropion, inflammation	Punctoplasty, lid surgery
Canaliculi	Viral infection, chemotherapy drugs, trauma, canaliculitis (*Actinomyces*)	Intubation, canaliculo-DCR, Lester Jones tube
Common canaliculus	Membrane, scarring	DCR with membrane removal and intubation
Sac and nasolacrimal duct	Primary acquired obstruction (idiopathic), dacryolithiasis, sinus disease and surgery, trauma, granulomatous disease, tumours	Dacryocystorhinostomy
Lower duct in infants	Imperforate valve of Hasner	Massage, probing

3.1 Primary acquired nasolacrimal duct obstruction

Primary acquired obstruction is idiopathic inflammation and fibrosis of the duct, most common in middle-aged and older women. It causes constant watering and often a mucocoele, and predisposes to acute dacryocystitis.

3.2 Congenital nasolacrimal duct obstruction

About 20% of newborns have a watering eye from an imperforate valve of Hasner. Most resolve by the age of one year with lacrimal sac massage and, if needed, antibiotic drops for infection. Persistent cases are treated by probing, with intubation or balloon dilatation for failures, and endoscopy of the inferior meatus helps in complex cases. A **dacryocystocele** may extend into the nose as an intranasal cyst and needs endoscopic marsupialisation if it causes respiratory distress.

4 CHAPTER FOUR Management

Matching treatment to level.

4.1 Dacryocystorhinostomy

Dacryocystorhinostomy (DCR) creates a permanent opening from the lacrimal sac into the nose, bypassing an obstructed nasolacrimal duct. The **endonasal** approach raises a mucosal flap over the frontal process of the maxilla and the lacrimal bone, removes bone with a drill or punches to expose the whole sac, opens the sac widely and lays its flaps against the nasal mucosa. It avoids a skin scar, preserves the lacrimal pump and allows simultaneous treatment of nasal disease. Silicone stents are used selectively, especially for canalicular disease. Success rates for endonasal DCR are comparable to the external approach, at about 90% for nasolacrimal duct obstruction when the sac is fully exposed.

The **external** approach through a skin incision remains widely used by ophthalmologists. Failure is usually from an opening that is too small or too high, scarring, or synechiae to the middle turbinate or septum.

4.2 Other treatments

Dry eye is treated with lubricants and, if needed, punctal plugs. Canalicular obstruction that cannot be bypassed is treated with a **Lester Jones tube**, a glass bypass from the medial canthus to the nose. Acute dacryocystitis is treated with systemic antibiotics and, if needed, drainage, followed by DCR once the infection settles.

THE WATERING EYE

1 Ocular surface and lids

→ **dry eye, blepharitis, lid malposition**
ophthalmic treatment, lubricants

→ **normal**
step 2

2 Syringe the system

→ **free flow**
functional block or pump failure — Jones tests

→ **reflux same punctum**
canalicular block

→ **reflux opposite punctum or mucopus**
sac or duct obstruction, step 3

3 Sac or duct obstruction

→ endoscopy of the nose

→ **mass, bloody tears or sinus disease**
CT or MRI, biopsy

→ **primary acquired obstruction**
endonasal or external DCR

4 Infant

→ massage and wait to one year

→ **persisting**
probing, then intubation

→ **intranasal cyst with breathing difficulty**
endoscopic marsupialisation

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Hypersecretion versus impaired drainage;** dry eye as a common cause.
- **Lacrimal anatomy:** canaliculi, sac, duct and the valve of Hasner.
- **Syringing and Jones tests:** interpretation.
- **Endonasal DCR:** steps, landmarks, success, causes of failure.
- **Congenital nasolacrimal duct obstruction** and dacryocystocele.

5.2 Common pitfalls

COMMON PITFALLS

- Performing DCR for a dry eye or a lid problem.
- Performing DCR for a canalicular block, where it does not help without additional steps.
- Making the sac opening too small or too high.
- Missing a lacrimal sac tumour when syringing produces blood.

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CLINICAL · APPROACH TO

Approach to a sore throat

A sore throat is one of the commonest reasons anyone sees a doctor, and almost all acute sore throats are viral and self-limiting. The ENT surgeon sees the other end of the spectrum: the patient who cannot swallow saliva, the quinsy, the supraglottitis that will obstruct the airway, the neck space abscess, and the adult with a persistent one-sided sore throat and earache that is a cancer. The approach separates acute from chronic, recognises the airway and the deep collection, uses antibiotics with judgement, and knows when tonsillectomy is justified.

AT A GLANCE

- Centor or FeverPAIN to decide on antibiotics.
- Trismus, drooling, muffled voice: quinsy or deep neck abscess.
- Stridor or toxic patient: epiglottitis — airway first.
- Over 3 weeks, unilateral, otalgia: examine for cancer.

1 CHAPTER ONE Acute or chronic?

Then: is the airway safe?

1.1 The first questions

An **acute** sore throat lasts days to two weeks and is usually infective. A **chronic** sore throat persists beyond three weeks and has a different list: reflux, chronic tonsillitis, tonsilloliths, postnasal drip, smoking, voice misuse, and malignancy. In either case the first judgement is about the airway and the swallow.

RED FLAGS IN SORE THROAT

- **Drooling, inability to swallow saliva, muffled voice, stridor or sitting forward:** supraglottitis or a deep neck abscess — protect the airway, do not lie the patient flat, do not examine the throat with a tongue depressor in a child.
- **Trismus with unilateral swelling and uvular deviation:** quinsy.
- **Neck swelling, torticollis or reduced neck movement, especially in a child:** retropharyngeal or parapharyngeal abscess.

- **Persistent sore throat with fever, rigors and neck tenderness after tonsillitis:** Lemierre's syndrome.
- **Sore throat in a patient taking carbimazole, clozapine or chemotherapy:** neutropenia — urgent full blood count.
- **Persistent unilateral sore throat, referred otalgia, dysphagia, hoarseness, weight loss or a neck lump, especially in a smoker:** cancer until proven otherwise.

CHAPTER TWO

2 The acute sore throat

Assess, score, decide.

2.1 History and examination

Ask about the duration, fever, ability to swallow food and fluids, voice change, breathing, earache and trismus, and about cough and coryza, which point to a viral cause. Examine the oropharynx, the tonsils, the neck nodes and the airway, and, when the history suggests supraglottitis, examine the larynx with a flexible endoscope in a safe setting rather than with a tongue depressor.

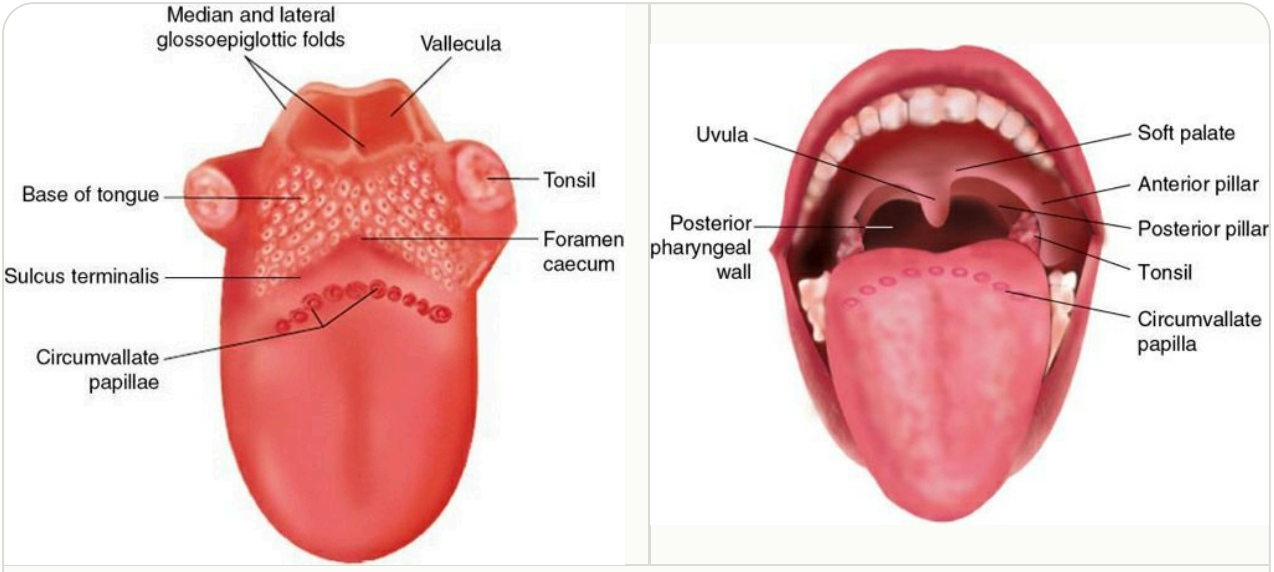


FIGURE 1 First, the base of the tongue and valleculae. Second, the structures seen in the oropharynx.

Dhingra, figs 47.6 and 47.7

2.2 Who needs antibiotics?

Most acute sore throats are viral, and antibiotics shorten symptoms by only about a day in bacterial tonsillitis. Scoring systems identify patients more likely to have streptococcal infection. **FeverPAIN** scores one point each for fever, purulence, attending rapidly (within three days), inflamed tonsils and no cough or coryza: a score of 0 to 1 needs no antibiotic, 2 to 3 a delayed or no antibiotic, and 4 to 5 an immediate antibiotic or a delayed one. The **Centor** criteria (tonsillar exudate, tender anterior cervical nodes, fever and absence of cough) are used similarly, with antibiotics considered at three or four. Phenoxymethylpenicillin for five to ten days is the first choice, with clarithromycin in penicillin allergy. **Amoxicillin is avoided** because it causes a rash in infectious mononucleosis.

Admission is needed for patients who cannot swallow fluids, are dehydrated, have a suspected collection or airway compromise, or are systemically unwell. They need intravenous fluids, analgesia, intravenous benzylpenicillin and often a single dose of dexamethasone.

2.3 The important acute causes

Acute tonsillitis is viral or bacterial (group A streptococcus is the commonest bacterium). **Infectious mononucleosis** from Epstein–Barr virus gives a severe exudative tonsillitis with palatal petechiae, generalised lymphadenopathy and splenomegaly; it is confirmed by a monospot or EBV serology and atypical lymphocytes. Patients avoid contact sports for several weeks because of splenic rupture, and steroids are used for airway compromise. **Peritonsillar abscess (quinsy)** produces trismus, a hot potato voice, unilateral swelling above the tonsil and a deviated uvula; it is drained by needle aspiration or incision. **Supraglottitis** produces severe odynophagia with relatively normal oropharynx, a muffled voice and drooling. **Deep neck space infections** need contrast CT and drainage. **Lemierre's syndrome** is a septic thrombophlebitis of the internal jugular vein, usually with *Fusobacterium necrophorum*, after a sore throat. Other causes include HIV seroconversion, diphtheria in the unvaccinated, gonococcal pharyngitis, and herpangina or hand, foot and mouth disease in children.

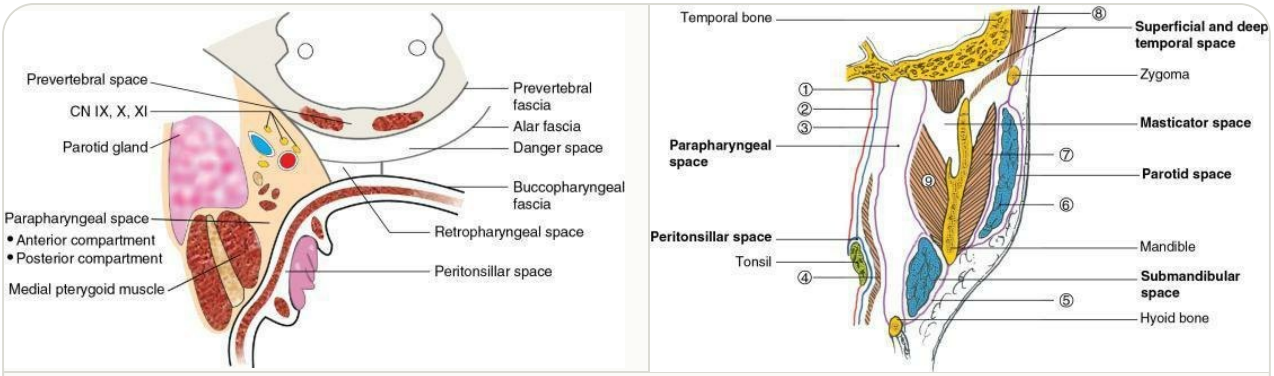


FIGURE 2 First, the spaces around the pharynx where abscesses form. Second, the deep spaces of the head and neck.

Dhingra, figs 52.6 and 52.9

3 CHAPTER THREE The chronic sore throat

A different list.

3.1 Causes

CAUSES OF A CHRONIC SORE THROAT

CAUSE	CLUES
Chronic or recurrent tonsillitis	Repeated episodes, halitosis, enlarged nodes
Tonsilloliths	Halitosis, white stones, foreign body sensation
Laryngopharyngeal reflux	Throat clearing, hoarseness, globus, morning symptoms
Postnasal drip	Nasal symptoms, throat clearing
Smoking, alcohol, dry mouth	History; mouth breathing; drugs
Voice misuse and muscle tension	Pain with voice use
Glossopharyngeal neuralgia and Eagle syndrome	Sharp pain triggered by swallowing; elongated styloid
Malignancy	Unilateral, progressive, otalgia, dysphagia, neck node, smoker

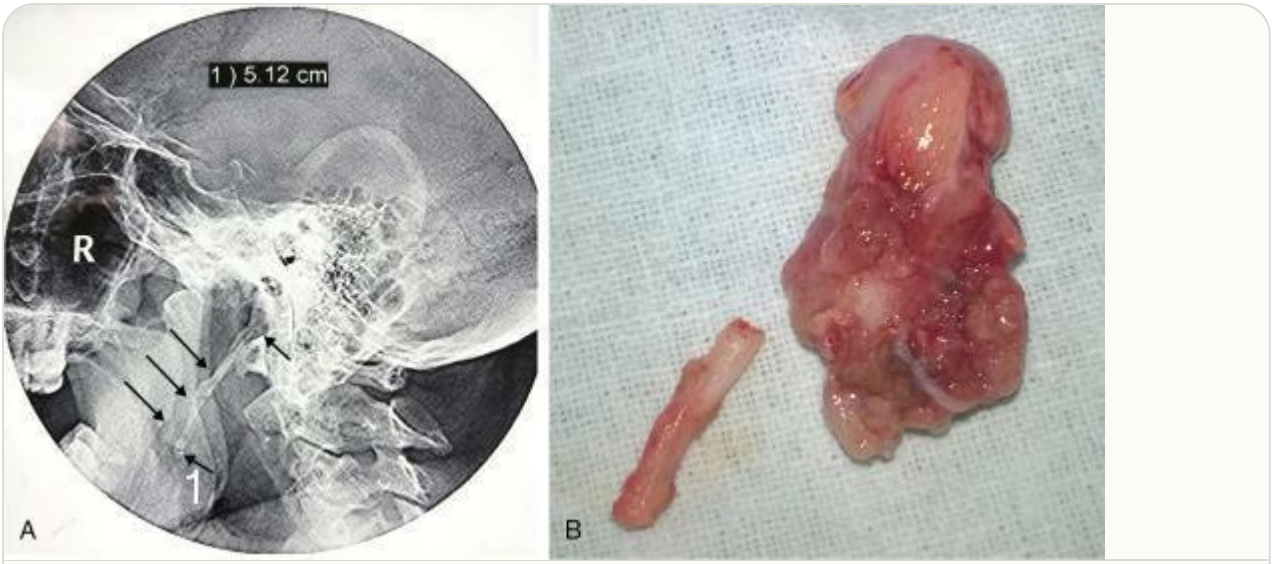


FIGURE 3 An elongated styloid process (Eagle syndrome): (A) on radiograph, (B) excised.
Dhingra, fig. 53.6

3.2 Assessment

Every adult with a sore throat lasting more than three weeks, especially if unilateral, needs a full head and neck examination with flexible endoscopy of the nasopharynx, oropharynx, hypopharynx and larynx, palpation of the tongue base and tonsils, and examination of the neck. Asymmetrical tonsils in an adult, a firm or ulcerated tonsil, or a neck node lead to biopsy or tonsillectomy for histology, and imaging.

4 CHAPTER FOUR Tonsillectomy

When it is justified.

4.1 Indications

In adults and children, the **SIGN** criteria for tonsillectomy in recurrent tonsillitis are sore throats due to acute tonsillitis that are disabling and prevent normal functioning, with **seven or more** well-documented, clinically significant, adequately treated episodes in the preceding year, **five or more** in each of the preceding two years, or **three or more** in each of the preceding three years. Other indications

include two episodes of quinsy, suspected malignancy, and obstructive sleep-disordered breathing in children.

THE SORE THROAT

1. **Airway and swallow**
   - → **drooling, stridor, muffled voice**
     airway team, endoscopy in a safe setting, admit
   - → **trismus with unilateral swelling**
     quinsy — drain
   - → **neck swelling or torticollis**
     contrast CT for deep neck abscess
2. **Acute, airway safe**
   - → **FeverPAIN 0–1**
     no antibiotic
   - → **2–3**
     delayed or none
   - → **4–5**
     penicillin
   - → **unable to swallow fluids**
     admit, IV penicillin, fluids, steroid
   - → **on carbimazole or clozapine**
     urgent blood count
3. **Glandular fever features?**
   - → monospot, avoid amoxicillin, no contact sport, steroids for airway compromise
4. **Chronic, beyond three weeks**
   - → full endoscopic examination
   - → **asymmetric tonsil, unilateral pain or neck node**
     biopsy and imaging

→ **recurrent tonsillitis meeting criteria**
tonsillectomy

→ **reflux or postnasal drip**
treat the cause

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The airway red flags** and how to examine safely.
- **FeverPAIN and Centor;** antibiotic choice and avoiding amoxicillin.
- **Infectious mononucleosis:** diagnosis and advice.
- **Quinsy, deep neck infection and Lemierre's syndrome.**
- **SIGN criteria** for tonsillectomy.
- **The persistent unilateral sore throat** as a cancer symptom.

5.2 Common pitfalls

COMMON PITFALLS

- Examining a child with suspected supraglottitis with a tongue depressor.
- Giving amoxicillin in glandular fever.
- Missing agranulocytosis in a patient on carbimazole.
- Treating a unilateral chronic sore throat with repeated antibiotics in a smoker.

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CLINICAL · APPROACH TO

Approach to a mouth ulcer

Most mouth ulcers are traumatic or aphthous and heal within two weeks. The approach is built around one rule: **a single ulcer that has not healed within three weeks must be biopsied** to exclude squamous cell carcinoma. Beyond that, the pattern of the ulcers, single or multiple, recurrent or persistent, with or without blisters, skin, eye, gut or genital disease, points to the cause, and simple blood tests find the treatable deficiencies behind many recurrent ulcers.

AT A GLANCE

- Single, persistent over 3 weeks, indurated: biopsy.
- Recurrent and healing: aphthous, check blood count, ferritin, B12, folate, coeliac.
- Multiple with skin, eye or genital lesions: systemic disease.
- Remove trauma and review in 2 weeks.

1 CHAPTER ONE Patterns

Single or multiple, acute or recurrent.

1.1 Sorting ulcers

MOUTH ULCERS BY PATTERN

PATTERN	THINK OF	CLUES
Single, persistent over three weeks	Squamous cell carcinoma; traumatic ulcer; tuberculosis; syphilis	Induration, rolled edge, fixed; lateral tongue or floor of mouth; risk factors
Single, with a clear cause	Traumatic ulcer	Sharp tooth, denture, bite; heals within two weeks once the cause is removed
Recurrent, multiple, healing	Recurrent aphthous stomatitis	Onset in youth; minor, major or herpetiform
Acute, multiple, with fever	Viral infection	Primary herpes, hand, foot and mouth disease, herpangina

PATTERN	THINK OF	CLUES
Recurrent with genital ulcers and eye disease	Behçet's disease	Uveitis, skin lesions, pathergy
With gut symptoms or anaemia	Crohn's disease, coeliac disease, haematinic deficiency	Diarrhoea, weight loss, low ferritin, B12 or folate
Blisters, erosions and desquamative gingivitis	Pemphigus vulgaris, mucous membrane pemphigoid, lichen planus	Nikolsky sign, skin or eye involvement, white striae
Target lesions and crusted lips	Erythema multiforme	Recent herpes or drug
Drug-related	Nicorandil, NSAIDs, methotrexate, chemotherapy, bisphosphonates	Timing with the drug
Immunosuppressed	Neutropenia, HIV, CMV, fungal	Systemic disease or treatment

DO NOT MISS

- **A single ulcer over three weeks,** especially indurated, with a raised edge, on the lateral tongue or floor of mouth, or with a neck node: urgent biopsy for squamous cell carcinoma.
- **Leukoplakia or erythroplakia** next to an ulcer.
- **Widespread erosions with skin blisters:** pemphigus vulgaris — urgent dermatology.
- **Ulcers with eye involvement:** pemphigoid or Behçet's disease — ophthalmology.
- **Ulcers with sore throat and fever in a patient on carbimazole or clozapine:** neutropenia.

2 CHAPTER TWO History and examination

The whole patient.

2.1 History

Ask how long each ulcer lasts and how often they come, the age of onset, the number and size, any trigger (trauma, stress, menstruation, foods, stopping smoking), and whether they scar. Ask about genital ulcers, eye symptoms, rashes, joint pain, bowel symptoms, weight loss, fatigue and fevers, and take a full drug history. Ask about smoking, alcohol, betel quid and HIV risk.

2.2 Examination

Note the site, size, number, base and edge of each ulcer, and feel it: induration and fixity suggest malignancy. Look for a sharp tooth or ill-fitting denture, white or red patches, blisters, gingival involvement and lichenoid striae. Examine the neck, skin, eyes and, when relevant, genitalia.

3 CHAPTER THREE Key causes

Recognising each.

3.1 Recurrent aphthous stomatitis

This is the commonest cause of recurrent ulcers, beginning in childhood or adolescence. **Minor** aphthae are under 1 cm, on non-keratinised mucosa, and heal in 10 to 14 days without scarring. **Major** aphthae are larger, deeper and last weeks, healing with scarring. **Herpetiform** ulcers are crops of many pinhead ulcers that coalesce. Triggers include trauma, stress, hormonal changes, sodium lauryl sulphate toothpastes, and deficiencies of iron, vitamin B12 and folate. Patients with recurrent ulcers should have a full blood count, ferritin, B12, folate and coeliac serology.



FIGURE 1 Multiple aphthous ulcers on the uvula and faucial pillars.
Dhingra, fig. 43.1

3.2 Malignancy

Oral squamous cell carcinoma presents as a non-healing ulcer, often painless at first, with induration and a raised everted edge. Risk factors are smoking, alcohol, betel quid and chewing tobacco. Biopsy is taken from the edge, including normal tissue. The tongue and the floor of mouth are the highest-risk sites.



FIGURE 2 First, a traumatic ulcerated mass of the tongue from sharp teeth. Second, a carcinoma to compare.
Dhingra, figs 43.2 and 43.3

3.3 Mucocutaneous disease

Lichen planus produces white lacy striae with erosions, often bilateral on the buccal mucosa; erosive lichen planus carries a small risk of malignant change and needs follow-up. **Pemphigus vulgaris** produces fragile blisters that rupture into painful erosions, often starting in the mouth; **mucous membrane pemphigoid** produces tougher blisters and desquamative gingivitis and can scar the eyes and larynx. Both need biopsy of perilesional tissue for **direct immunofluorescence**, as well as routine histology.

3.4 Systemic disease

Behçet's disease combines recurrent oral ulcers with genital ulcers, uveitis and skin lesions. Crohn's disease, coeliac disease, SLE, reactive arthritis, cyclic neutropenia and the PFAPA syndrome in children (periodic fever, aphthous stomatitis, pharyngitis and adenitis) are other causes.



FIGURE 3 Palatal ulceration in a patient with a white cell count of 2,500 per cubic millimetre during chemotherapy.
Dhingra, fig. 43.4

4 CHAPTER FOUR Investigation and management

Test, biopsy, treat.

4.1 Investigation

Blood tests for recurrent ulcers include a full blood count, ferritin, B12, folate, and coeliac serology; inflammatory markers and an HIV test when indicated. Any persistent single ulcer is biopsied, and perilesional tissue is sent fresh for immunofluorescence when a blistering disease is suspected. Swabs for herpes simplex PCR help in acute vesicular ulcers.

4.2 Treatment

The cause is treated: a sharp tooth smoothed, a denture adjusted, a deficiency replaced, a causative drug stopped with the prescriber. Aphthous ulcers are managed with chlorhexidine or benzydamine mouthwashes, topical corticosteroids (such as betamethasone dissolved as a mouthwash, or beclometasone spray), avoiding triggers including sodium lauryl sulphate, and colchicine or other systemic agents in severe cases under specialist care. Blistering diseases, Behçet's disease and lichen planus are managed with oral medicine, dermatology and rheumatology colleagues.

THE MOUTH ULCER

1. Single and persistent over three weeks?
   - → **yes**
     urgent biopsy — exclude squamous cell carcinoma
   - → **single with obvious trauma**
     remove the cause, review in two weeks, biopsy if not healed
2. Recurrent and multiple?
   - → **bloods**
     full blood count, ferritin, B12, folate, coeliac serology

→ **minor or major aphthae**
topical treatment

→ **genital ulcers or eye disease**
Behçet's disease

3 Blisters, erosions or desquamative gingivitis?

→ biopsy with immunofluorescence

→ pemphigus, pemphigoid, lichen planus — specialist care

4 Acute with fever

→ herpes or other viral infection — supportive care, antiviral if severe

→ **on carbimazole or clozapine**
blood count

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three-week rule** and biopsy from the edge.
- **Recurrent aphthous stomatitis:** types and blood tests.
- **Pemphigus versus pemphigoid:** level of the split, immunofluorescence.
- **Behçet's disease** and other systemic causes.
- **Drug-induced ulceration:** nicorandil.

5.2 Common pitfalls

COMMON PITFALLS

- Giving repeated topical steroids for a single persistent ulcer that is a cancer.
- Biopsying only the necrotic centre of an ulcer.
- Missing iron, B12, folate or coeliac disease behind recurrent ulcers.

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CLINICAL · MOUTH & THROAT

Snoring and obstructive sleep apnoea

Every patient with obstructive sleep apnoea snores, but not every snorer has apnoea. The difference matters: primary snoring is a social nuisance, while apnoea, with its repeated nightly suffocation, raises blood pressure, strains the heart, causes strokes and puts sleepy drivers on the road. The bed partner gives the history, the sleep study gives the diagnosis, positive airway pressure is the first treatment, and surgery is chosen by finding where the airway collapses.

AT A GLANCE

- CPAP first in adults.
- Children usually need adenotonsillectomy.
- Find the site before surgery.
- Plan for post-op airway risk.
- CHAT trial: surgery improved behaviour and sleep but not attention; nearly half the waiting group normalised.

1 CHAPTER ONE Definitions

Words of the sleep study.

1.1 Terms

- **Apnoea:** cessation of airflow for 10 seconds or more during sleep.
- **Hypopnoea:** a reduction in airflow (commonly by 30 to 50%) with a fall in oxygen saturation of 3 to 4% or an arousal.
- **Apnoea–hypopnoea index (AHI):** apnoeas and hypopnoeas per hour of sleep; also called the respiratory disturbance index. Under 5 is normal.
- **Arousal:** a brief awakening on the EEG caused by respiratory effort; the arousal index is their number per hour.
- **Sleep efficiency:** minutes asleep divided by minutes in bed.
- **Multiple sleep latency test:** timed daytime naps to measure sleepiness and detect narcolepsy.

SEVERITY OF OBSTRUCTIVE SLEEP APNOEA (ADULTS)

GRADE	AHI (EVENTS PER HOUR)
Normal	Under 5
Mild	5 to 14
Moderate	15 to 29
Severe	30 or more

1.2 Types of apnoea

- **Obstructive:** the upper airway collapses while breathing effort continues.
- **Central:** the airway is open but the brain does not signal breathing; no effort is made.
- **Mixed:** a central start followed by obstruction.

1.3 Spectrum of sleep-disordered breathing

Primary snoring (no arousals, AHI under 5, no daytime sleepiness), **upper airway resistance** (effort-related arousals and sleepiness without frank apnoeas), and **obstructive sleep apnoea**.

2 CHAPTER TWO Mechanism

Why the throat closes in sleep.

2.1 Snoring

In sleep the pharyngeal muscles relax and the airway narrows; breathing through the narrowing sets the soft palate, uvula, pillars and tongue base vibrating, at up to 90 dB. Snoring affects about a quarter of men and fifteen per cent of women, and increases with age.

2.2 Obstruction

The pharynx has no bony support between the palate and the larynx; its patency depends on muscle tone against the suction of inspiration. Tone falls most in REM sleep and stage 2 sleep, when events are most frequent. Each apnoea ends with an arousal that restores tone, fragmenting sleep.

2.3 Sites

Obstruction may be at more than one level, and may differ from night to night:

- **Level I:** nose and nasopharynx: septal deviation, turbinate hypertrophy, valve collapse, polyps, adenoids, tumours.
- **Level II:** palate and tonsils: a long soft palate and uvula, large tonsils, bulky lateral walls.
- **Level III:** tongue base, hypopharynx and larynx: macroglossia, retrognathia, lingual tonsils, an omega-shaped or floppy epiglottis, tumours.

Obesity and a thick neck (collar over about 42 cm in men) narrow every level; alcohol, sedatives and hypnotics relax the muscles further.

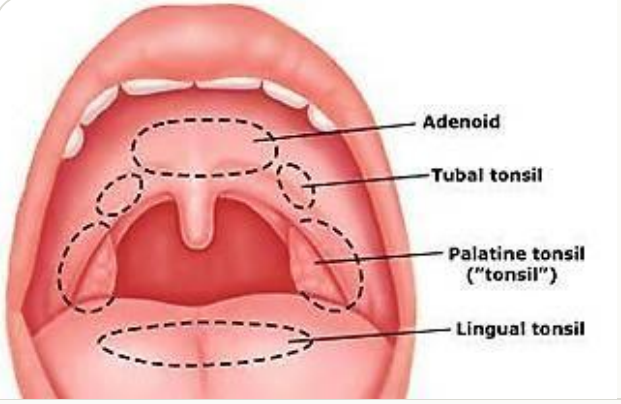


TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	Palatine tonsils
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

FIGURE 1 Waldeyer's ring seen through the mouth, and the adenoids compared with the palatine tonsils.

Anatomy atlas · ENT Steps p. 199, ENT Steps p. 208

2.4 Sleep architecture

An adult sleeps seven to eight hours in three to five cycles of 90 to 120 minutes. Non-REM sleep (75 to 80%) moves from light stages to deep slow-wave sleep; REM

sleep (20 to 25%) brings dreaming, irregular breathing and heart rate, and near-paralysis of skeletal muscle, which is why obstruction is worst then.

3 CHAPTER THREE Consequences

More than noise.

3.1 Personal

Loud snoring disturbs the partner (the snoring spouse syndrome); witnessed choking and gasping, restless unrefreshing sleep, excessive daytime sleepiness, morning headaches, fatigue, poor memory and concentration, irritability and depression, reduced libido, nocturia.

3.2 Medical

- **Hypertension:** apnoea is an independent risk factor; treatment lowers blood pressure, even resistant hypertension.
- **Heart:** arrhythmias, ischaemia, heart failure, pulmonary hypertension and cor pulmonale; untreated severe apnoea increases mortality.
- **Brain:** stroke.
- **Metabolic:** insulin resistance; polycythaemia.
- **Perioperative risk:** difficult airways and sensitivity to opioids and sedatives.

3.3 Social

Road traffic and workplace accidents. Ask every patient about sleepiness while driving; in some jurisdictions the doctor must advise the patient about notifying the licensing authority.

CHILDREN ARE DIFFERENT

In children the commonest cause is adenotonsillar hypertrophy. They show snoring, pauses and gasping, restless sleep with the neck hyperextended, sweating, bedwetting, morning headache, and by day hyperactivity, poor attention and school

performance, and sometimes failure to thrive, rather than sleepiness. An AHI above 1 is abnormal in a child, and adenotonsillectomy is first-line treatment.

4 CHAPTER FOUR Evaluation

Finding the cause and the level.

4.1 History

Take it from the bed partner as well as the patient: snoring, pauses, gasping, choking, restlessness and sweating at night; sleepiness, fatigue, headaches, memory and mood by day. Ask about sleep habits and shift work, alcohol, sedatives and caffeine, sleep position, nasal obstruction, menopause and weight change, and driving.

EPWORTH SLEEPINESS SCALE

Each situation is scored 0 (would never doze), 1 (slight chance), 2 (moderate chance) or 3 (high chance of dozing), for a total out of 24:

- Sitting and reading.
- Watching television.
- Sitting inactive in a public place, such as a theatre or meeting.
- As a passenger in a car for an hour without a break.
- Lying down to rest in the afternoon.
- Sitting and talking to someone.
- Sitting quietly after lunch without alcohol.
- In a car, stopped for a few minutes in traffic.

A total over 10 indicates excessive daytime sleepiness. The STOP-BANG and Berlin questionnaires screen for apnoea.

4.2 Examination

- **General:** body mass index (obese at 30 or more), neck circumference (over 42 cm in

men or 37.5 cm in women), blood pressure, signs of heart failure and hypothyroidism.

- **Nose:** septum, turbinates, valve, polyps; endoscopy for adenoids.
- **Mouth and oropharynx:** tonsil size, palate length and position, uvula, tongue size and scalloping, lateral walls, dental occlusion and retrognathia.
- **Hypopharynx and larynx:** flexible endoscopy of the tongue base, lingual tonsil and epiglottis.

FRIEDMAN GRADING

TONSILS	PALATE POSITION (TONGUE RELAXED IN THE MOUTH)
0: removed	I: tonsils, uvula and pillars all visible
1: hidden within the pillars	II: uvula visible, not the tonsils
2: reach the pillars	III: soft palate visible, not the uvula
3: beyond the pillars	IV: only the hard palate visible
4: meeting in the midline	

FRIEDMAN, NOT MALLAMPATI

For sleep surgery the palate is graded with the tongue relaxed inside the mouth (Friedman), not protruded as in the anaesthetist's Mallampati score. Large tonsils with a favourable palate predict success from palatal surgery.

4.3 Endoscopic tests

- **Müller manoeuvre:** with the nose pinched and mouth closed, the patient inhales forcefully while the endoscopist watches which level collapses; simple but poorly reproducible.
- **Drug-induced sleep endoscopy (DISE):** endoscopy under propofol or midazolam sedation shows collapse during sleep-like conditions; the *VOTE* system records the velum, oropharyngeal lateral walls, tongue base and epiglottis, with the degree and pattern (anteroposterior, lateral or concentric). Used before surgery, after failed surgery and when findings do not match the sleep study.

4.4 Imaging

Lateral cephalometry for craniofacial skeletal anomalies and the posterior airway space; a lateral neck radiograph for adenoids in children.

4.5 Sleep studies

Polysomnography is the gold standard. A full attended study (level 1) records EEG, electro-oculogram and chin and leg EMG (sleep staging), ECG, nasal and oral airflow, thoracic and abdominal effort, oxygen saturation, body position and snoring; it separates obstructive from central apnoea and primary snoring from apnoea. Home studies (level 3: airflow, effort, oximetry, position) are now widely used for uncomplicated adults; oximetry alone is insufficient. A split-night study diagnoses in the first half and titrates CPAP in the second, though events may be under-counted.

5 CHAPTER FIVE Non-surgical treatment

First-line.

5.1 Lifestyle

- Weight loss; bariatric surgery for morbid obesity.
- Avoid alcohol in the evening, sedatives and hypnotics; stop smoking.
- **Positional therapy:** sleeping on the side, with a device (or a ball sewn into the back of a shirt) when apnoea is supine-dependent.
- Treat nasal obstruction and allergy.

5.2 Positive airway pressure

Continuous positive airway pressure (CPAP) is a pneumatic splint that holds the airway open, usually at 5 to 20 cm of water, through a nasal, nasal-pillow or full-face mask. Bilevel (BiPAP) delivers a higher inspiratory and lower expiratory pressure for those who need high pressures; auto-titrating devices adjust continuously. It is the most

effective non-surgical treatment and the standard for moderate to severe apnoea, but only about half of patients use it adequately (at least four hours a night, five nights a week) in the long term, because of discomfort, claustrophobia, dryness, leaks and inconvenience. Coaching and follow-up improve adherence.

5.3 Oral appliances

A *mandibular advancement device* holds the jaw forward and draws the tongue and soft tissues away from the posterior wall; a *tongue-retaining device* holds the tongue forward. Effective for snoring and mild to moderate apnoea, and useful in retrognathia; side effects are jaw pain, dental movement and dryness.

6 CHAPTER SIX Surgery

Matched to the level.

6.1 Indications

Failure of or intolerance to CPAP, favourable anatomy, patient preference, and serious cardiovascular disease. Surgery is tailored to the level(s) of obstruction; multilevel procedures may be combined or staged.

6.2 Nasal surgery

Septoplasty, turbinate reduction, polypectomy, valve surgery and adenoidectomy. Nasal surgery seldom cures apnoea, because the nose is not a collapsing segment, but it improves sleep quality and CPAP tolerance and can lower the pressure needed.

6.3 Palate and oropharynx

- **Tonsillectomy (with adenoidectomy):** the main treatment in children; in adults with grade 3 or 4 tonsils, isolated tonsillectomy can be very effective.

- **Uvulopalatopharyngoplasty (UPPP):** introduced by Fujita in 1981; tonsillectomy with trimming of the uvula and free edge of the palate and suturing of the pillars to widen the oropharynx. Overall success is about 50%, rising above 80% in well-selected patients (Friedman tonsils 3 to 4, palate I to II). Complications: pain, bleeding, velopharyngeal insufficiency, nasopharyngeal stenosis and voice change.
- **Uvulopalatal flap:** the uvula is folded back onto the denuded palate, sparing the muscle.
- **Expansion sphincter pharyngoplasty:** the palatopharyngeus is rerouted superolaterally to widen the lateral dimension.
- **Uvulopalatoplasty (UPP, laser or radiofrequency assisted) and palatal stiffening (cautery-assisted or implants):** useful for snoring, much less for apnoea.

Dhingra notes that palatal surgery relieves snoring in about 80% but apnoea in only about half, with relapse when another level takes over.

6.4 Tongue base and hypopharynx

Radiofrequency reduction of the tongue base in repeated sessions, lingual tonsillectomy, midline glossectomy (coblation, laser, robotic), tongue suspension, hyoid suspension (to the mandible or thyroid cartilage), genioglossus advancement, and supraglottoplasty for epiglottic collapse.

6.5 Skeletal surgery

Maxillomandibular advancement (Le Fort I osteotomy of the maxilla and sagittal split of the mandible, fixed forwards with plates) enlarges the whole skeletal framework of the airway and is the most effective operation apart from tracheostomy, especially in retrognathic patients and after failed soft tissue surgery; it changes facial appearance and needs orthodontic care.

6.6 Hypoglossal nerve stimulation

An implanted stimulator fires the hypoglossal nerve with inspiration, protruding the tongue; for moderate to severe apnoea in patients who cannot use CPAP, with a suitable BMI and without concentric palatal collapse on DISE.

6.7 Tracheostomy

The only operation that reliably cures obstructive apnoea, by bypassing every level; reserved for life-threatening disease, severe cardiopulmonary complications and failure of everything else, because of its personal and social cost.

OPERATIVE GUIDES IN THEATRE 5 operations

Operations for sleep-disordered breathing.

Tonsillectomy
Tonsillectomy

Uvulopalatopharyngoplasty and Related Modifications
UPPP

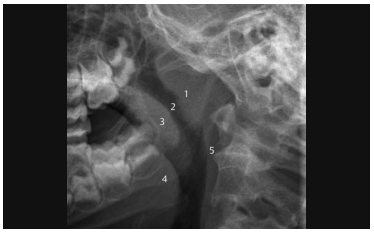
Drug-Induced Sedated Endoscopy
DISE

Hypoglossal Nerve Stimulation Therapy
Hypoglossal nerve stimulation

Maxillomandibular Advancement for Obstructive Sleep Apnea
Maxillomandibular advancement

IMAGING IN THE APP 1 case

Adenoid hypertrophy on imaging.



Adenoid Hypertrophy
Infection & inflammation · 1 image

6.8 Perioperative care

Patients with apnoea are sensitive to opioids and sedatives and may have difficult airways. Monitor oxygen saturation overnight after surgery in severe apnoea, with cardiovascular disease, after tongue base surgery, and in young children or those with comorbidities; restart CPAP after surgery when appropriate.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

Dhingra

Calls tracheostomy the gold standard of surgical treatment, lists uvulopalatoplasty as the commonest procedure and quotes 80% success for snoring but 50% for apnoea.

Lee

Places CPAP as the gold standard of therapy, emphasises patient selection by Friedman stage and DISE, reporting UPPP success above 85% in favourable anatomy, and names maxillomandibular advancement the most effective operation short of tracheostomy.

They agree that surgery must be matched to the level of collapse.

SLEEP-DISORDERED BREATHING IN CHILDREN AND SELECTION FOR SURGERY (OXFORD TEXTBOOK)

About 10 to 15% of children snore, and 1 to 4% have obstructive sleep apnoea, more in obese and Black children. It usually begins between two and six years, when the tonsils and adenoids are largest relative to the airway; tonsillar hypertrophy is grade 3 or more on the Brodsky scale. Prematurity is a risk factor, and sleep apnoea can cause failure to thrive, poor behaviour and attention, and enuresis. Most children can be treated on a careful history and examination, ideally with a parental video; overnight **oximetry** helps triage (a normal trace does not exclude apnoea), and **polysomnography** is kept for uncertain cases and high-risk children: those under two, with obesity, Down syndrome, craniofacial or neuromuscular disease, or cardiac complications.

The **Childhood Adenotonsillectomy Trial (CHAT, 2013)** randomised children aged five to nine with non-severe apnoea to early adenotonsillectomy or watchful waiting for seven months. Surgery did not improve attention and executive function, but it improved symptoms, behaviour, quality of life and sleep study results; even so, about 45% of the waiting group normalised spontaneously, less often in obese and Black children and those with more severe disease. Watchful waiting is therefore reasonable for snoring and mild disease, with intranasal steroids and **montelukast** helping in the short term, especially with rhinitis or residual symptoms after surgery. Weight loss is recommended for obese children. Children with persistent apnoea after adenotonsillectomy, or without large tonsils, need drug-induced sleep endoscopy or MRI to find other levels of obstruction.

In adults, **transoral robotic** tongue base surgery requires a BMI below about 35 (ideally near 30) and a mouth opening of at least 2.5 cm; up to 30% of adults with apnoea are not overweight, and nasal surgery helps by easing CPAP use and reducing negative airway pressure.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definitions:** apnoea, hypopnoea, AHI and severity grades.
- **Obstructive versus central apnoea.**
- **Consequences** of untreated apnoea.
- **Evaluation:** Epworth score, Friedman staging, Müller manoeuvre, DISE, polysomnography.
- **CPAP** and its adherence problem.
- **Surgery by level,** UPPP selection, and maxillomandibular advancement.
- **Children:** adenotonsillectomy first.

7.2 Common pitfalls

COMMON PITFALLS

- Operating on the palate when the tongue base is the main site of collapse.
- Using oximetry alone to diagnose or exclude apnoea.
- Giving opioids freely after surgery to a patient with severe apnoea.
- Promising that nasal surgery will cure apnoea.

Written for this app from Oxford Textbook of Otolaryngology ch. 51, 75, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 16, Dhingra ch. 55 and K. J. Lee ch. 2. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · APPROACH TO

Approach to the hoarse voice

Hoarseness, or dysphonia, is any change in the quality of the voice: rough, breathy, strained, weak or tiring. Most of it is inflammatory, functional or due to benign lesions, but persistent hoarseness is the commonest presenting symptom of laryngeal cancer, and a hoarse voice with a paralysed vocal fold may be the first sign of a tumour in the chest or neck. The rule is simple: any adult with hoarseness lasting more than three weeks should have the larynx examined, and the examination should be good enough to see the vocal folds vibrate.

AT A GLANCE

- Over 3 weeks: laryngoscopy.
- Smoker, pain, otalgia, dysphagia, neck node: urgent suspected-cancer pathway.
- Breathy voice and weak cough: palsy — image skull base to arch.
- Normal larynx: think muscle tension dysphonia, but look again if it does not improve.

1 CHAPTER ONE The voice and its failure

How the folds make sound.

1.1 Physiology in brief

The vocal fold has a layered structure. The **cover** (epithelium and the jelly-like superficial lamina propria, Reinke's space) slides over the **transition** (the vocal ligament) and the **body** (the vocalis muscle). Air from the lungs sets the cover vibrating as a *mucosal wave*. The voice fails when the folds cannot close (paralysis, atrophy, a gap), when the cover is stiff or heavy (scar, sulcus, Reinke's oedema, cancer), when a lesion interrupts closure (nodules, polyps, cysts), or when the muscles around the larynx work against it (muscle tension dysphonia, dystonia).

RED FLAGS IN HOARSENESS

- **Hoarseness over three weeks in a smoker or drinker,** or any persistent hoarseness over 45: urgent laryngoscopy on a suspected cancer pathway.
- **Stridor or breathlessness:** airway assessment first.

- **Hoarseness with dysphagia, odynophagia, referred otalgia, haemoptysis, weight loss or a neck lump.**
- **A breathy voice and weak cough after thyroid, chest, neck or cardiac surgery, or with no obvious cause:** vocal fold paralysis — image the vagus and recurrent laryngeal nerve course.
- **Hoarseness in a child with breathing difficulty:** recurrent respiratory papillomatosis.

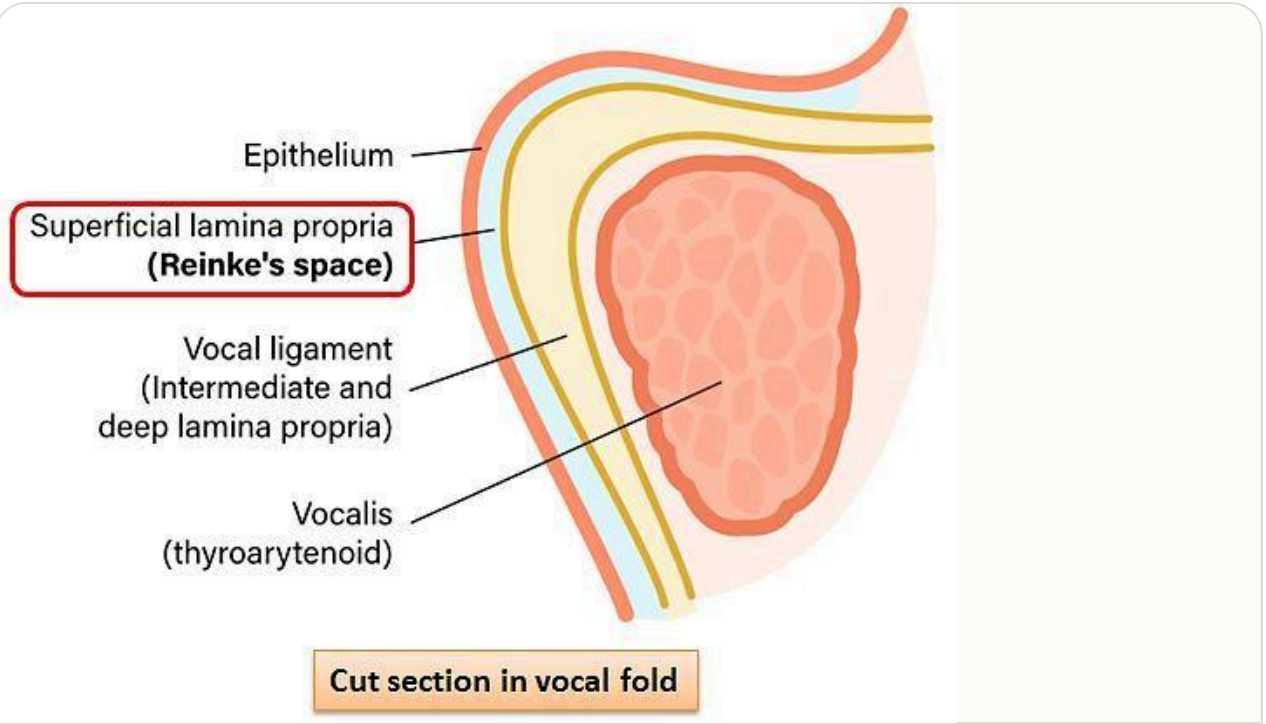


FIGURE 1 Layers of the vocal fold: the cover (epithelium and Reinke's space) over the vocal ligament and muscle.

Anatomy atlas · ENT Steps p. 246

2 CHAPTER TWO History

Onset, use and risk.

2.1 The questions

Ask how the voice changed and when: sudden onset after shouting suggests a haemorrhage or polyp; onset after a cold suggests laryngitis; onset after an operation or intubation suggests nerve injury or arytenoid dislocation; gradual

progression suggests a lesion or cancer. Ask whether it fluctuates, worsens with use or improves with rest. Ask about voice demands (teachers, singers, call-centre workers), smoking and alcohol, reflux symptoms, inhaled steroids, thyroid disease, neurological symptoms (tremor, weakness, swallowing problems), breathing, swallowing, pain and weight. The **Voice Handicap Index** (VHI-10) measures the effect on the patient's life.

3 CHAPTER THREE Examination

See the folds move.

3.1 Laryngeal examination

Listen to the voice: breathy (a gap), rough (irregular vibration), strained (muscle tension or dystonia), tremulous or with voice breaks. Examine the neck for lymph nodes, the thyroid and laryngeal tenderness. Examine the larynx with a **flexible nasendoscope**, ideally a distal-chip scope, looking at the mucosa, any lesion, the movement of each fold and arytenoid, and the pattern of closure; ask the patient to sniff, say "ee" and cough. **Videostroboscopy** shows the mucosal wave and is needed to distinguish nodules from cysts, to detect scar and sulcus, and to judge early cancer. Perceptual rating (GRBAS) and acoustic analysis document the voice.



FIGURE 2 Flexible laryngoscopy in clinic.
Dhingra, fig. 62.6

4 CHAPTER FOUR Causes

By mechanism.

CAUSES OF HOARSENESS

GROUP	CAUSES	CLUES
Inflammatory	Acute laryngitis, chronic laryngitis, laryngopharyngeal reflux, inhaled steroid laryngitis	Infection, smoking, reflux, inhalers
Benign lesions	Nodules, polyps, cysts, Reinke's oedema, sulcus, granuloma	Voice use, smoking (Reinke's), intubation or reflux (granuloma)
Premalignant and malignant	Leukoplakia, dysplasia, squamous cell carcinoma	Smoker, progressive, white or irregular lesion
Papillomatosis	Recurrent respiratory papillomatosis (HPV 6 and 11)	Children and young adults, recurrence

GROUP	CAUSES	CLUES
Neurological	Vocal fold paralysis, laryngeal dystonia, essential tremor, Parkinson's disease, motor neurone disease, myasthenia	Breathy voice, strangled voice breaks, tremor, other signs
Functional	Muscle tension dysphonia, functional aphonia	Normal folds, strained voice, normal cough
Other	Hypothyroidism, ageing (presbyphonia), arthritis of the cricoarytenoid joint, trauma, amyloid	Context

4.1 Some key causes

Vocal fold nodules are bilateral, symmetrical swellings at the junction of the anterior and middle thirds, caused by voice overuse; voice therapy is the treatment. **Polyps** are usually unilateral, often after a haemorrhage, and are removed by microlaryngoscopy if therapy does not help. **Reinke's oedema** is diffuse swelling of the superficial lamina propria in smokers, giving a low-pitched voice, especially in women; stopping smoking is essential, and surgery reduces the swelling while preserving the cover. **Leukoplakia** carries a risk of malignant change that depends on the grade of dysplasia and is excised for histology.

Vocal fold paralysis is more often left-sided, because the left recurrent laryngeal nerve has a longer course around the aortic arch. Causes are surgery (thyroid, cardiothoracic, anterior cervical spine), malignancy (lung, oesophagus, thyroid, skull base), and idiopathic. Unexplained paralysis is investigated with CT from the skull base to the aortic arch. **Muscle tension dysphonia** is the commonest functional voice disorder, with a strained voice, laryngeal tension and normal or compensatory supraglottic squeezing; it responds to voice therapy. **Laryngeal (spasmodic) dystonia** produces strangled voice breaks, and is treated with botulinum toxin.

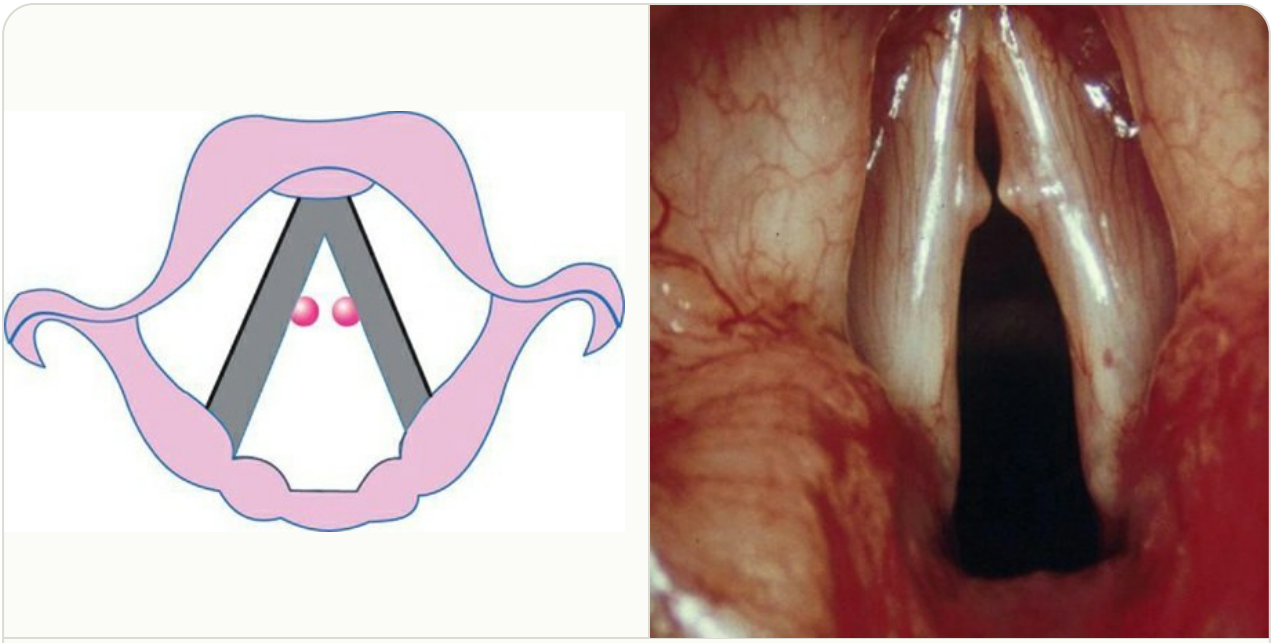


FIGURE 3 First, the site of vocal nodules at the junction of the anterior and middle thirds. Second, nodules on both folds.
Dhingra, figs 61.1 and 61.2

5 CHAPTER FIVE Management

Principles.

5.1 For all patients

Voice hygiene and **voice therapy** with a speech and language therapist are central to most benign and functional voice problems. Smoking cessation, hydration, treatment of reflux when its features are present, and review of inhaled steroid technique are also important.

5.2 Surgery

Phonosurgery under general anaesthesia uses cold steel or lasers to remove lesions while preserving the superficial lamina propria. Suspicious lesions are biopsied or excised for histology. Vocal fold paralysis that does not recover is treated with **injection medialisation** (temporary or long-lasting), **medialisation thyroplasty**, arytenoid adduction or laryngeal reinnervation. Cancer is staged and treated on the head and neck pathway.

THE HOARSE VOICE

1 Red flags or over three weeks?

- **yes**
  laryngoscopy, urgently if smoker, over 45, stridor or other head and neck symptoms
- **recent cold, improving**
  wait and review

2 Laryngoscopy

- **lesion suspicious for cancer or dysplasia**
  microlaryngoscopy and biopsy, staging
- **benign lesion**
  voice therapy, phonosurgery if needed
- **inflammation**
  smoking, reflux and inhaler review

3 Mobility

- **fold paralysed**
  CT skull base to aortic arch, injection or thyroplasty if no recovery
- **normal folds, strained voice**
  muscle tension dysphonia — voice therapy
- **voice breaks**
  dystonia — botulinum toxin

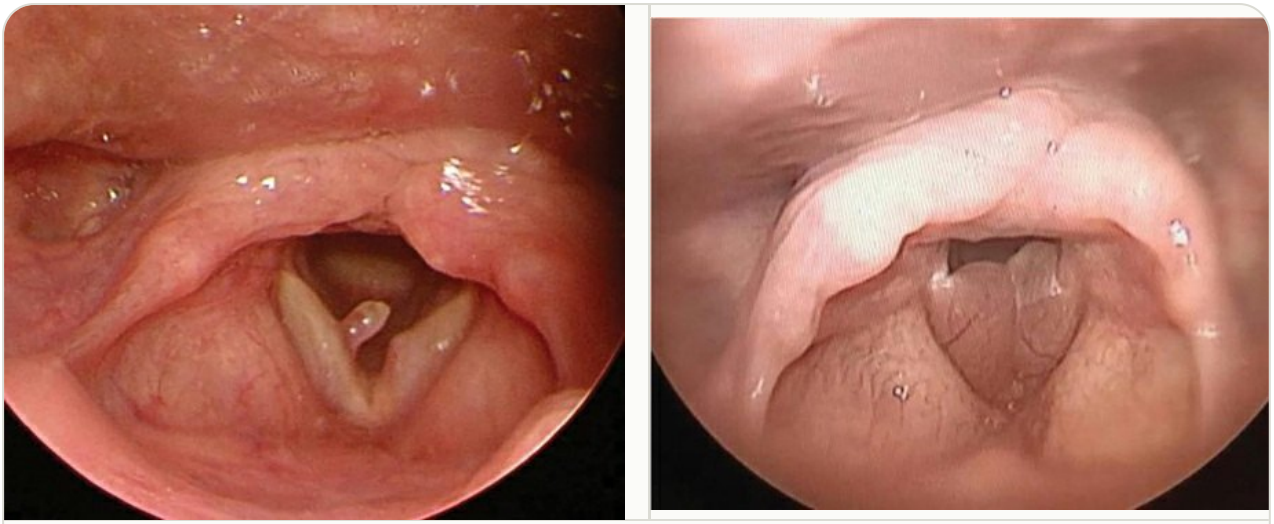


FIGURE 4 First, a pedunculated vocal fold polyp. Second, bilateral Reinke's oedema in a smoker.
Dhingra, figs 61.4 and 61.5

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three-week rule** and urgent referral for persistent hoarseness.
- **The body–cover model** and the mucosal wave; stroboscopy.
- **Nodules, polyps, cysts and Reinke's oedema:** features and treatment.
- **Vocal fold paralysis:** causes, investigation and treatment options.
- **Functional dysphonia** and the role of voice therapy.

6.2 Common pitfalls

COMMON PITFALLS

- Prescribing antibiotics or a proton pump inhibitor for months without examining the larynx.
- Missing a lung cancer in a patient with left vocal fold paralysis.

- Operating on nodules before voice therapy.
- Stripping the vocal fold and leaving it scarred.

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CLINICAL · APPROACH TO

Approach to difficulty swallowing

Difficulty swallowing is never a trivial symptom. It can mean a cancer of the pharynx or oesophagus, a neurological disease, a pharyngeal pouch or a motility disorder, and it carries the risks of aspiration pneumonia, malnutrition and dehydration. The approach rests on three questions: is the problem in the mouth and pharynx (*oropharyngeal*) or in the oesophagus; is it structural or neuromuscular; and are there features of malignancy that demand urgent endoscopy?

AT A GLANCE

- Coughing and nasal regurgitation: oropharyngeal (neurological).
- Solids then liquids, weight loss: cancer.
- Liquids and solids from the start: achalasia.
- Regurgitation of old food, gurgling: pharyngeal pouch.

1 CHAPTER ONE Where is the problem?

Oropharyngeal or oesophageal.

1.1 Two patterns

Oropharyngeal dysphagia is difficulty starting a swallow: food sticks in the throat at once, the patient coughs or chokes on swallowing, food or fluid comes back through the nose, the voice is wet, and repeated swallows are needed. It is usually neuromuscular (stroke, Parkinson's disease, motor neurone disease, myasthenia, myopathy) or caused by a pharyngeal pouch, a cricopharyngeal bar, head and neck cancer or its treatment.

Oesophageal dysphagia is a sensation of food sticking after swallowing, usually felt behind the sternum. Difficulty with **solids** that progresses to liquids suggests a mechanical obstruction: cancer if rapid with weight loss, a peptic stricture if slow with reflux, a ring or web if intermittent. Difficulty with **solids and liquids from the start** suggests a motility disorder such as achalasia. Patients localise the level poorly: a lower oesophageal obstruction can be felt in the neck.

RED FLAGS IN DYSPHAGIA

- **Progressive dysphagia, weight loss, odynophagia, haematemesis or anaemia:** upper GI endoscopy urgently, at any age.
- **Dysphagia with hoarseness, otalgia, a neck lump or a smoking and drinking history:** head and neck cancer.
- **Total dysphagia with inability to swallow saliva:** food bolus obstruction — emergency.
- **Coughing on swallowing, chest infections and weight loss:** aspiration; assess safety of oral intake.
- **New neurological signs** (ptosis, fatigable weakness, fasciculation, dysarthria): neurological cause.

CHAPTER TWO

2 History and examination

Detail matters.

2.1 History

Ask what is difficult (solids, liquids, tablets, saliva), where food sticks, and how long the problem has lasted and whether it is getting worse. Ask about coughing and choking, nasal regurgitation, a wet voice, regurgitation of undigested food (often hours later, with halitosis and gurgling in a pouch), chest infections, heartburn, weight loss, pain, voice change and neurological symptoms. Review medications that cause oesophagitis (doxycycline, bisphosphonates, potassium, NSAIDs) or dry the mouth. The **EAT-10** questionnaire screens for and measures dysphagia.

2.2 Examination

Examine the mouth, dentition, tongue movement and palate, the cranial nerves, the neck (nodes, thyroid, a pouch that gurgles), and the larynx and pharynx with a flexible endoscope, looking for pooling of saliva in the piriform fossae (a sign of obstruction or poor clearance), lesions and vocal fold movement. Observe a trial swallow of water when safe, and look for general signs of neurological disease.

3 CHAPTER THREE Investigation

Choose by the question.

SWALLOWING INVESTIGATIONS

TEST	WHAT IT SHOWS	USE
Flexible endoscopy in clinic	Mucosa, pooling, cord movement	First test in ENT
FEES (fibreoptic endoscopic evaluation of swallowing)	Pharyngeal phase, residue, penetration and aspiration	Oropharyngeal dysphagia, at the bedside
Videofluoroscopy	All phases of the swallow, aspiration, cricopharyngeal opening	Oropharyngeal dysphagia
Barium swallow	Pouch, webs, rings, strictures, motility	Structural and pouch assessment
Upper GI endoscopy (OGD)	Oesophageal mucosa, biopsy, dilatation	Oesophageal dysphagia, red flags
High-resolution manometry	Oesophageal motility (Chicago classification)	Achalasia, motility disorders
CT neck and chest	Malignancy and staging	Suspected cancer

In eosinophilic oesophagitis, biopsies are taken even when the oesophagus looks normal.

4 CHAPTER FOUR Causes

Structural, neuromuscular, oesophageal.

4.1 Pharyngeal pouch

A **pharyngeal pouch (Zenker's diverticulum)** is a pulsion diverticulum through **Killian's dehiscence**, between the thyropharyngeus and cricopharyngeus parts of the inferior constrictor, with failure of cricopharyngeal relaxation. It affects older adults and causes dysphagia, regurgitation of undigested food, halitosis, gurgling, aspiration and weight loss. A barium swallow confirms it. Symptomatic pouches are

treated by **endoscopic stapling or laser diverticulotomy**, which divides the wall between pouch and oesophagus together with the cricopharyngeus, or by open excision or inversion with cricopharyngeal myotomy for small or difficult pouches.

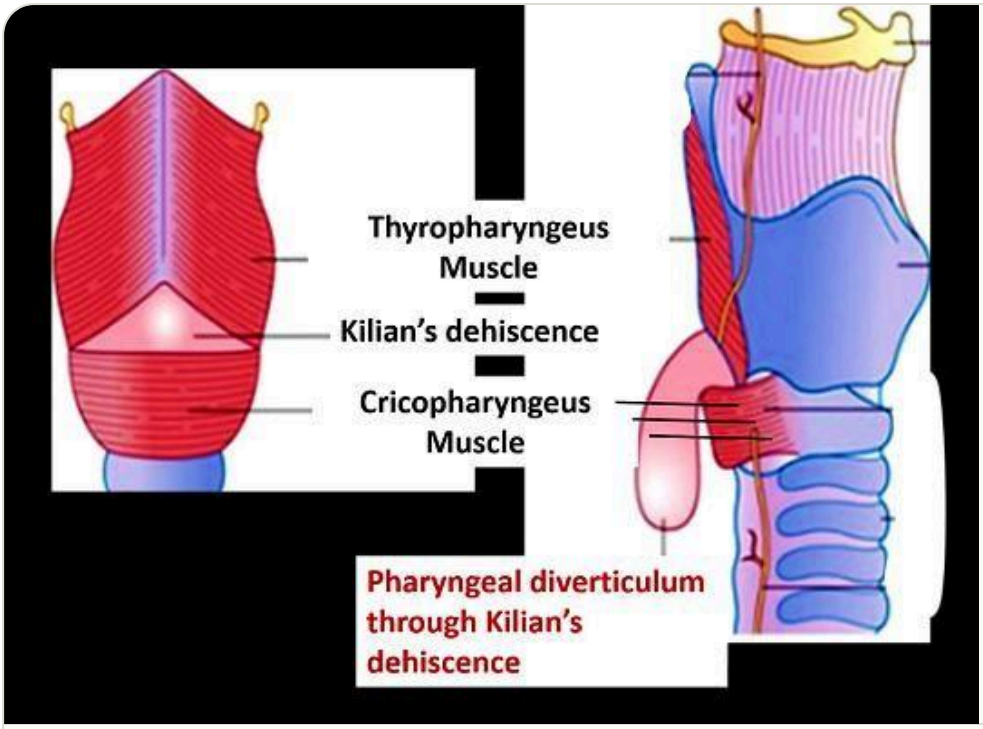


FIGURE 1 Killian's dehiscence and a pharyngeal pouch.
Anatomy atlas · ENT Steps p. 198

4.2 Other structural causes

Cricopharyngeal dysfunction without a pouch may respond to dilatation, botulinum toxin or myotomy. **Post-cricoid webs** with iron-deficiency anaemia form the Plummer–Vinson (Paterson–Brown-Kelly) syndrome, a risk factor for post-cricoid cancer. **Cervical osteophytes**, an enlarged thyroid and head and neck cancer, or the fibrosis that follows its treatment by radiotherapy, also narrow the pharynx.

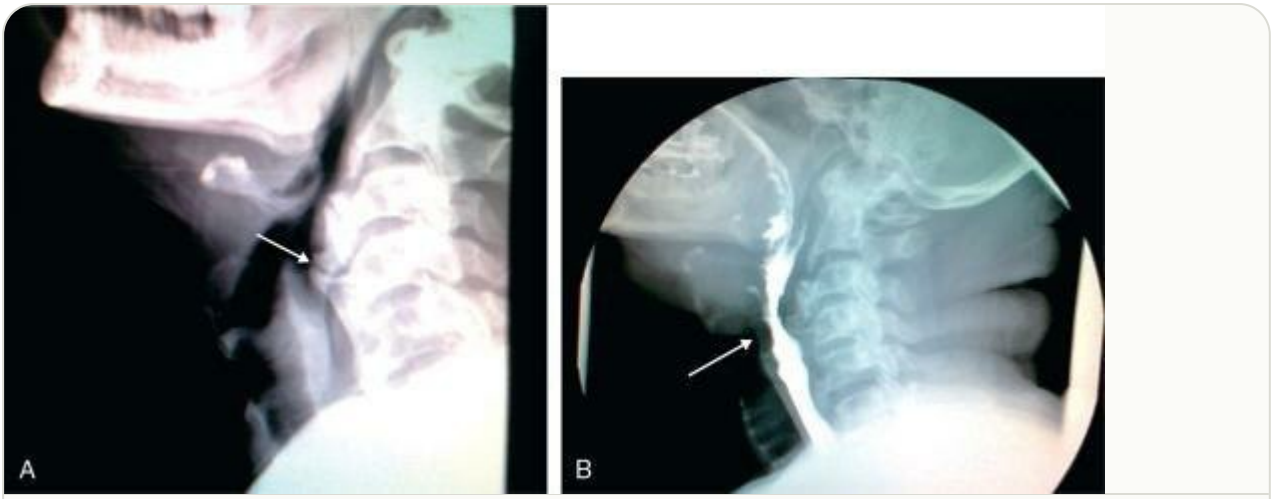


FIGURE 2 Large cervical osteophytes pressing on the oesophagus on barium swallow.
Dhingra, fig. 69.1

4.3 Neuromuscular causes

Stroke is the commonest cause of oropharyngeal dysphagia; most patients improve in the first weeks. Progressive neurological diseases (Parkinson's disease, motor neurone disease, multiple sclerosis, myasthenia gravis, inclusion body myositis, oculopharyngeal muscular dystrophy) need neurological diagnosis and long-term swallow management.

4.4 Oesophageal causes

Oesophageal cancer is the cause not to miss. **Peptic strictures** follow long-standing reflux. **Schatzki rings** cause intermittent food bolus obstruction. **Achalasia** is a failure of lower oesophageal sphincter relaxation with absent peristalsis, treated with pneumatic dilatation, Heller myotomy or peroral endoscopic myotomy. **Eosinophilic oesophagitis** typically affects young atopic men, presents with food bolus obstruction, and responds to proton pump inhibitors, topical steroids and dietary elimination.

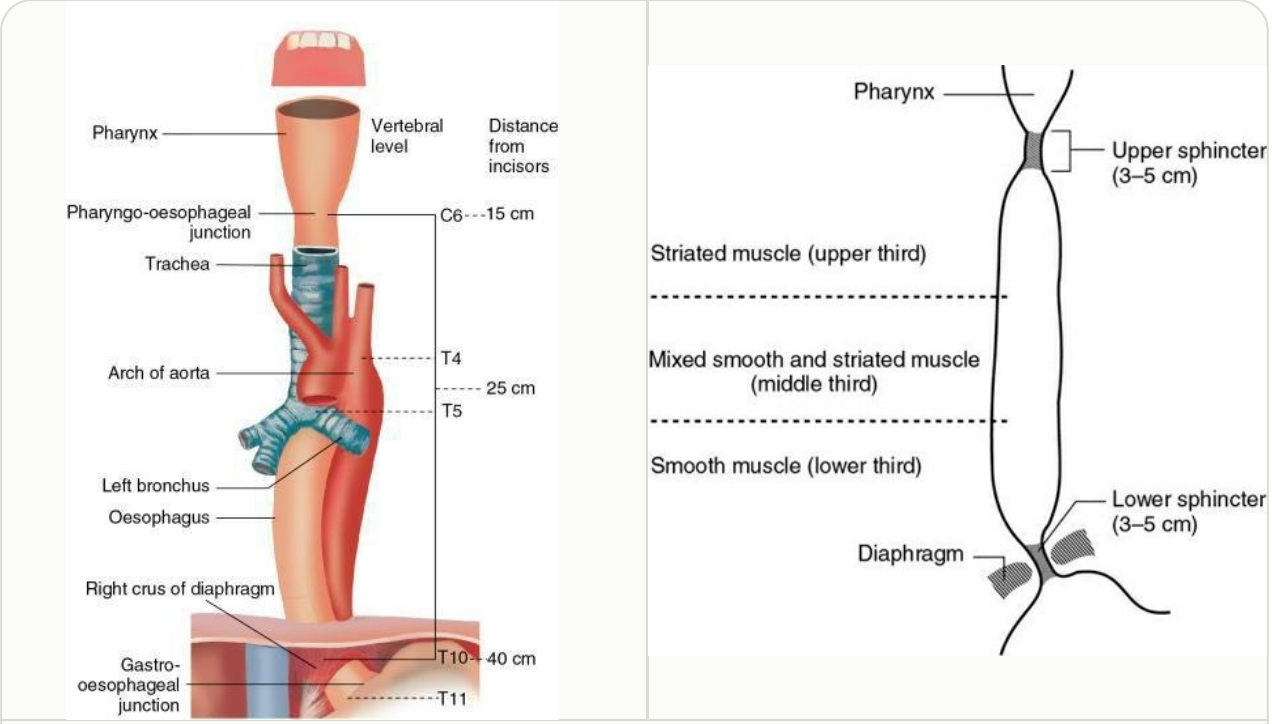


FIGURE 3 The oesophagus and its normal constrictions, with distances from the upper incisors.
Dhingra, figs 67.1 and 67.2

5 CHAPTER FIVE Management

Safety, cause, function.

5.1 Principles

Every patient needs an assessment of whether oral intake is safe, by a **speech and language therapist**, who can advise on texture modification, postures (chin tuck, head turn), exercises and compensatory manoeuvres. Nutrition and hydration are protected, by nasogastric or gastrostomy feeding when necessary. The cause is then treated. A **food bolus obstruction** in the oesophagus needs endoscopic removal if it does not pass, and every patient needs subsequent endoscopy to exclude a stricture, eosinophilic oesophagitis or cancer.

DIFFICULTY SWALLOWING

1 Red flags?

→ **progressive, weight loss, odynophagia, anaemia**

urgent OGD

→ **hoarseness, otalgia, neck lump**
head and neck examination and imaging

→ **cannot swallow saliva**
food bolus — urgent endoscopy

2 Oropharyngeal or oesophageal?

→ **coughing, nasal regurgitation, wet voice**
oropharyngeal, step 3

→ **food sticks after swallowing**
oesophageal — OGD, then barium or manometry

3 Oropharyngeal

→ flexible endoscopy, FEES or videofluoroscopy

→ **regurgitation and gurgling**
barium swallow for pouch — endoscopic stapling

→ **neurological signs**
neurology, speech therapy

4 Unsafe swallow

→ speech therapy, texture changes, postures

→ non-oral feeding if needed

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Oropharyngeal versus oesophageal dysphagia;** solids versus liquids.
- **Red flags** and urgent endoscopy.
- **FEES versus videofluoroscopy.**
- **Pharyngeal pouch:** Killian's dehiscence, presentation, endoscopic stapling and its limits.
- **Plummer–Vinson syndrome, achalasia and eosinophilic oesophagitis.**

6.2 Common pitfalls

COMMON PITFALLS

- Attributing progressive dysphagia to reflux without endoscopy.
- Performing a rigid oesophagoscopy blind in a patient with a pouch, risking perforation.
- Forgetting a swallow safety assessment before restarting oral intake.

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CLINICAL · APPROACH TO

Approach to a lump in the throat

Globus pharyngeus is the persistent feeling of a lump, tightness or something stuck in the throat, with no difficulty swallowing food. It is common, accounts for a large share of new referrals to ENT clinics, and is benign; but patients are often frightened that it is cancer, and a minority of cancers present with throat symptoms that can be mistaken for globus. The approach is to recognise the typical pattern, look for the features that make it atypical, examine the throat thoroughly, and then explain and treat with confidence rather than investigate endlessly.

AT A GLANCE

- Globus: midline, comes and goes, eating relieves it.
- Red flags: true dysphagia, pain, otalgia, lateralised, voice change, weight loss, neck lump.
- Flexible nasendoscopy in clinic; no routine barium.
- Reassurance, reflux measures, speech therapy.

1 CHAPTER ONE Typical globus

The classic history.

1.1 The pattern

Typical globus is felt in the midline, usually around the level of the cricoid or the thyroid notch. It is present **between meals** and when swallowing saliva, and often **improves when eating or drinking**. It is not painful and does not cause true dysphagia, odynophagia, weight loss or regurgitation. It fluctuates, worsens with stress, tiredness and throat clearing, and has often been present for months. It is most common in middle age and slightly more common in women.

1.2 What lies behind it

The cause is usually multifactorial. Suggested contributors include **laryngopharyngeal reflux**, raised cricopharyngeal tone, postnasal drip, throat clearing, muscle tension around the larynx, stress and anxiety, and heightened

sensitivity of the pharynx. Thyroid swelling, cervical osteophytes, a long uvula and tongue-base lymphoid hypertrophy are occasionally relevant.

FEATURES THAT ARE NOT GLOBUS

- **Pain, odynophagia or referred otalgia.**
- **True dysphagia,** food sticking, regurgitation or weight loss.
- **Unilateral or lateralised symptoms,** or a sensation that is progressive.
- **Hoarseness,** haemoptysis or a neck lump.
- **A smoker or heavy drinker over 50.**

Any of these features calls for investigation for malignancy and other causes, not a diagnosis of globus.

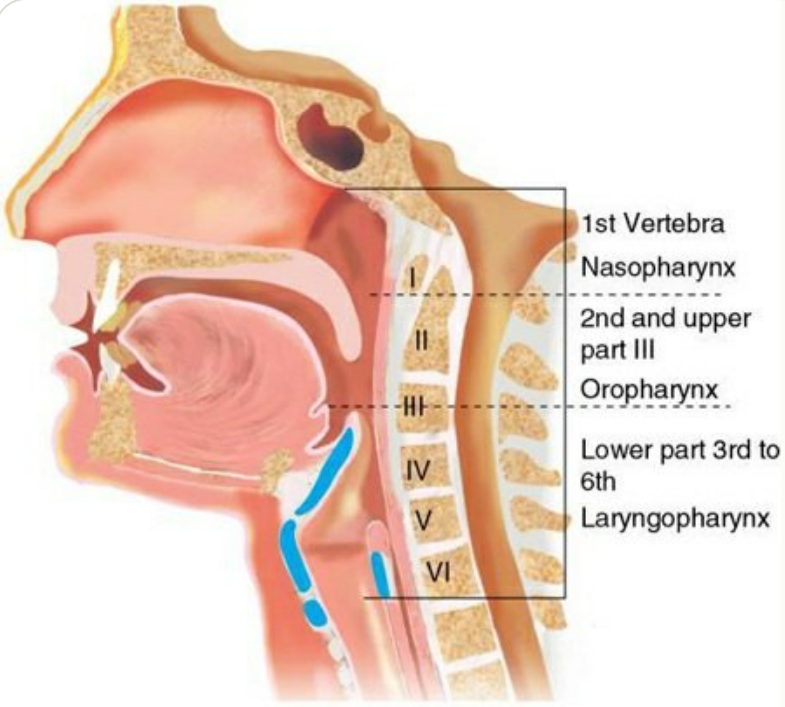


FIGURE 1 Divisions of the pharynx and their vertebral levels.
Dhingra, fig. 47.3

2 CHAPTER TWO Assessment

Examine, then stop.

2.1 Examination

Examine the mouth and oropharynx, palpate the tongue base and tonsils, examine the neck and thyroid, and perform a **flexible endoscopy** of the nasal cavity, nasopharynx, oropharynx, hypopharynx and larynx, including pooling in the piriform fossae and the postcricoid area. Look for signs associated with reflux (interarytenoid thickening, oedema), but recognise that these are common and non-specific.

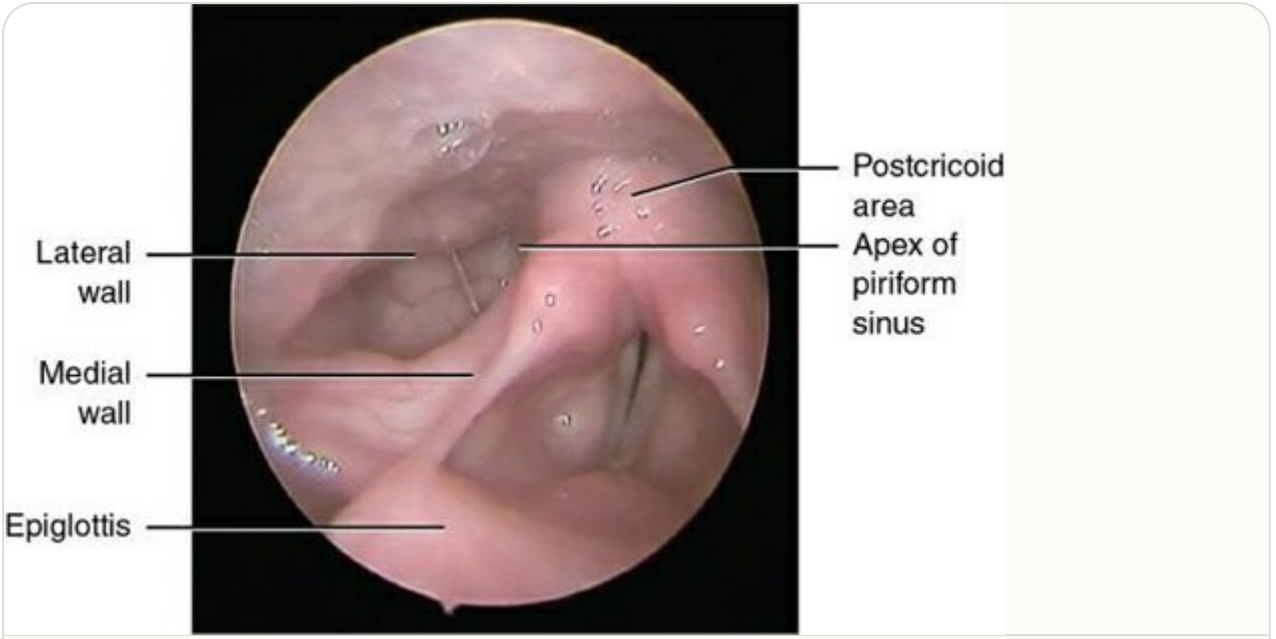


FIGURE 2 The piriform fossa, where saliva pools in obstruction and where hypopharyngeal cancer hides.
Dhingra, fig. 47.8

2.2 Investigation

In typical globus with a normal examination, **no further investigation is needed**. A routine barium swallow adds little and should not be requested just to reassure. Investigations are reserved for atypical features: an upper GI endoscopy or barium swallow for dysphagia, an ultrasound for a thyroid swelling, and imaging or an examination under anaesthesia when malignancy is suspected.

3 CHAPTER THREE Management

Explanation is treatment.

3.1 For most patients

A clear explanation that the sensation is common, real, benign and not a cancer, after a full examination, is the most effective treatment, and many patients improve with reassurance alone. Reduce throat clearing (sipping water or swallowing instead), stop smoking and keep well hydrated.

Speech and language therapy helps patients with throat clearing, muscle tension and voice symptoms. A **trial of reflux treatment** (diet and lifestyle changes, alginates after meals and at bedtime, and a proton pump inhibitor for patients with typical reflux symptoms) is reasonable, though its effect in isolated globus is unproven. For persistent, troublesome symptoms, low-dose amitriptyline and help with anxiety may be tried.

3.2 Review and discharge

Patients with typical globus and a normal examination can be discharged with advice to return if new features appear: pain, dysphagia, weight loss, hoarseness or a lump.

A LUMP IN THE THROAT

1. **Typical?**
   - → midline, between meals, eased by eating, no pain or dysphagia: step 2
   - → pain, dysphagia, weight loss, hoarseness, unilateral or neck lump: investigate for malignancy — endoscopy and imaging
2. **Full examination**
   - → flexible endoscopy, neck and thyroid
   - → **abnormal**

investigate the finding

→ **normal**
step 3

3 Treat

→ explanation and reassurance

→ reduce throat clearing, hydrate, stop smoking

→ speech therapy

→ trial of reflux treatment if reflux features

→ discharge with safety-net advice

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The typical features** of globus and why eating eases it.
- **The red flags** that exclude a diagnosis of globus.
- **A thorough examination,** including flexible endoscopy.
- **Why routine barium swallow is not needed.**
- **Explanation, speech therapy and a reflux trial.**

4.2 Common pitfalls

COMMON PITFALLS

- Labelling a unilateral or painful throat symptom as globus.

- Investigating typical globus repeatedly, reinforcing the patient's fear.
- Prescribing proton pump inhibitors indefinitely without review.

Written for this app from Oxford Textbook of Otolaryngology ch. 48, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 63. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · APPROACH TO

Approach to stridor and the threatened airway

Stridor is a sign, not a diagnosis: the sound of air forced through a narrowed airway. Its timing tells you where the narrowing is, its history tells you what it is, and the work of breathing tells you how long you have. Judge the severity first, keep the patient calm and upright, and never let a full examination or a scan come before a safe airway.

AT A GLANCE

- Noise timing tells the level.
- Keep calm, sit up, call early.
- Use adrenaline and steroids as a bridge.
- Have a surgical airway ready.

1 CHAPTER ONE Definition

Noise and where it comes from.

1.1 Stridor

A harsh, noisy breathing sound produced by turbulent airflow through a narrowed larynx or trachea. *Stertor* is the lower-pitched snoring of the nose, nasopharynx or oropharynx; *wheeze* is the musical expiratory sound of the smaller airways.

1.2 Phase and site

- **Inspiratory:** supraglottis, pharynx and extrathoracic airway collapse on inspiration (laryngomalacia, epiglottitis, retropharyngeal abscess, supraglottic tumours).
- **Biphasic:** a fixed narrowing at the glottis, subglottis or cervical trachea (subglottic stenosis, bilateral cord palsy, papillomas, croup).
- **Expiratory:** the intrathoracic trachea and bronchi (tracheomalacia, bronchial foreign body, tracheal stenosis, compression by masses or vessels).

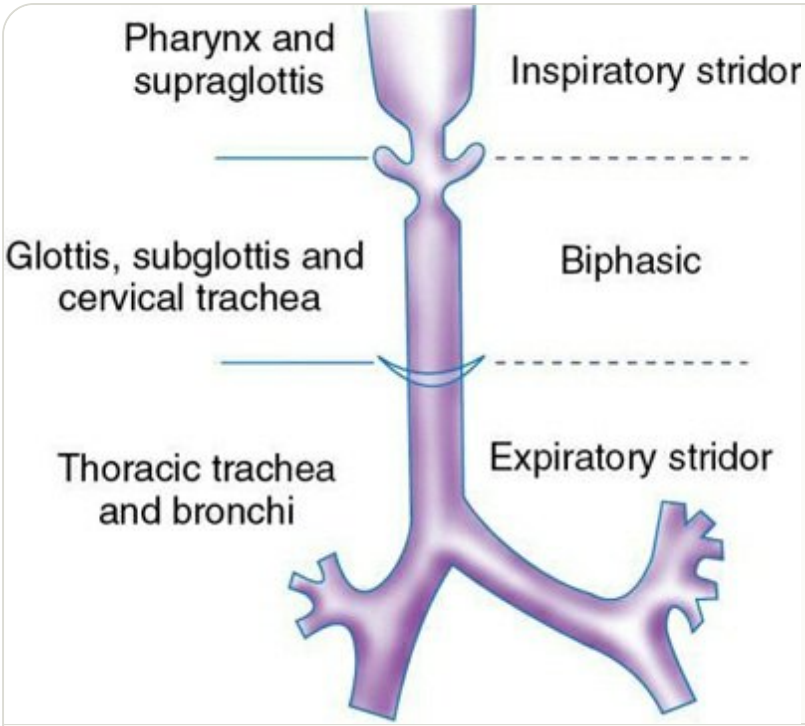


FIGURE 1 Types of stridor and their site of origin.
Dhingra, fig. 59.3

1.3 Associated sounds

- Snoring or snorting: nose and nasopharynx.
- Gurgling and a muffled voice: pharynx and supraglottis.
- Hoarse voice or weak cry: the vocal folds. The cry is normal in laryngomalacia and subglottic stenosis.
- Barking cough: subglottis (croup).
- Expiratory wheeze: bronchi.

2 CHAPTER TWO Causes

From the nose to the bronchi.

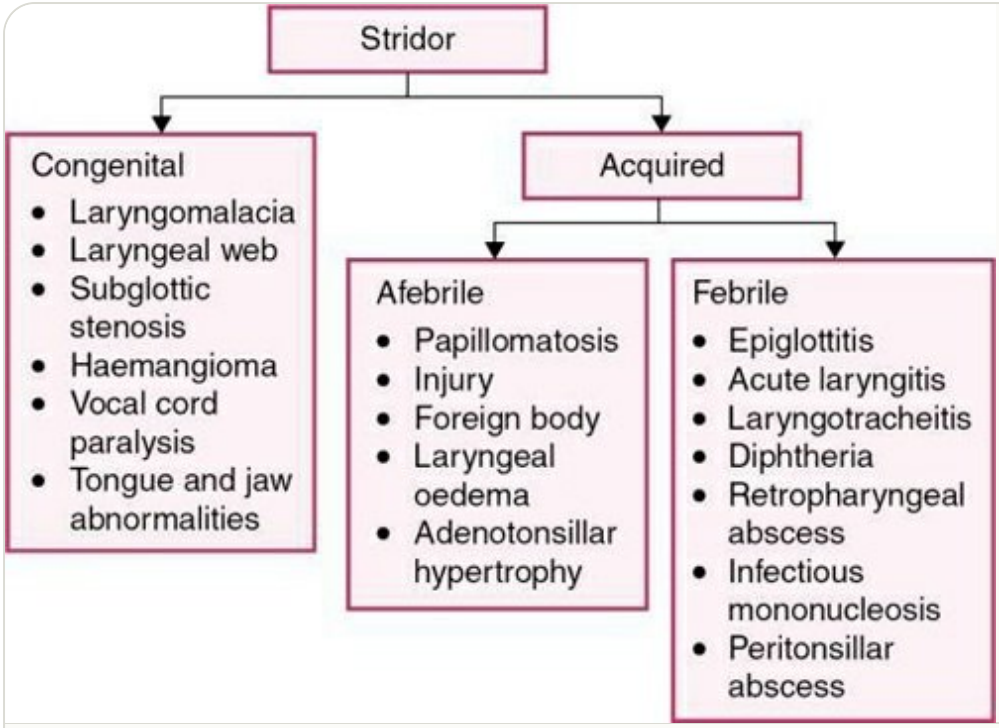


FIGURE 2 Causes of stridor in infants and children: congenital, and acquired with or without fever.
Dhingra, fig. 59.5

CAUSES OF STRIDOR BY SITE

SITE	CONGENITAL	ACQUIRED
Nose	Choanal atresia	Severe nasal obstruction in neonates
Tongue and jaw	Macroglossia, lingual thyroid, micrognathia (Pierre Robin)	Haemangioma, lymphangioma, Ludwig's angina
Pharynx	Dermoid, cysts	Adenotonsillar hypertrophy, retropharyngeal abscess, tumours
Larynx	Laryngomalacia, web, cysts, vocal fold paralysis, subglottic stenosis, haemangioma, cleft	Epiglottitis, croup, diphtheria, oedema, foreign body, trauma, papillomas, carcinoma, bilateral palsy, post-intubation stenosis, tetany, laryngospasm

SITE	CONGENITAL	ACQUIRED
Trachea and bronchi	Tracheomalacia, stenosis, atresia	Bacterial tracheitis, foreign body, stenosis after intubation or tracheostomy, tumours
Outside the airway	Vascular rings, oesophageal atresia and fistula, congenital goitre, cystic hygroma	Oesophageal foreign body, goitre, mediastinal and neck masses

2.1 In adults

Laryngeal and tracheal tumours, bilateral vocal fold paralysis (after thyroidectomy), post-intubation and tracheostomy stenosis, laryngeal trauma, angioedema, epiglottitis, deep neck infection, goitre and foreign bodies.

3 CHAPTER THREE Assessment

How bad, and what.

3.1 Severity

- **Mild:** stridor only when active or crying, no recession.
- **Moderate:** stridor at rest, suprasternal, intercostal, subcostal and sternal recession, tracheal tug, nasal flaring, tachypnoea, tachycardia.
- **Severe:** marked recession, agitation and restlessness (hypoxia), drooling, inability to speak or feed.
- **Impending arrest:** fatigue, falling respiratory rate and quieter stridor (less air moving, not improvement), pallor or cyanosis, drowsiness.



FIGURE 3 A child with bilateral abductor paralysis: recession of the suprasternal notch, sternum, intercostal spaces and epigastrium on inspiration.

Dhingra, fig. 59.4

A QUIETER CHILD IS NOT ALWAYS A BETTER CHILD

Diminishing stridor with tiring, drowsiness or a falling oxygen saturation means respiratory failure. Oximetry can stay normal until late; watch the work of breathing and the level of consciousness.

3.2 History

- **Onset:** at birth or soon after (congenital), or later (acquired).
- **Speed:** sudden (foreign body, oedema, epiglottitis) or gradual and progressive (laryngomalacia, haemangioma, papillomas, stenosis, tumours).
- **Duration and course:** short (infection, foreign body) or long-standing.
- **Fever and toxicity:** epiglottitis, tracheitis, retropharyngeal abscess, diphtheria.
- **Feeding:** choking, cyanosis and aspiration (cleft, paralysis, fistula, vascular ring, oesophageal foreign body).
- **Cyanotic spells** and apnoeas: need for airway support.
- **Witnessed choking** on a toy, nut or food.
- **Previous intubation,** surgery (thyroid, cardiac) or trauma.

- **Voice and cry** quality.

3.3 Examination

Observe from a distance first: posture, colour, recession, drooling, voice. Note the phase of stridor, the effect of position (prone improves laryngomalacia, micrognathia and macroglossia), and auscultate over the nose, open mouth, neck and chest to localise. Examine the nose, tongue, jaw and pharynx; in adults, and in calm older children, flexible nasendoscopy shows the larynx.

FLEXIBLE ENDOSCOPY IN THE CLINIC

Under topical anaesthesia, the awake child can be examined for laryngomalacia, vocal fold movement, cysts, papillomas and webs, which an anaesthetised child cannot show: dynamic problems need a breathing patient.

3.4 Investigations

Chosen by the history, and only when the airway allows:

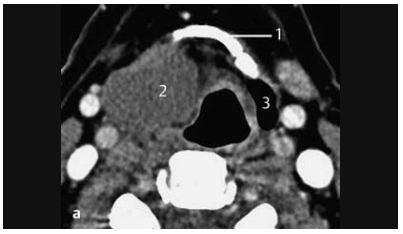
- Lateral and anteroposterior soft tissue radiographs of the neck (steeple sign of croup, thumb sign of epiglottitis, prevertebral swelling, radio-opaque foreign bodies).
- Chest radiographs, with inspiratory and expiratory films or fluoroscopy for radiolucent bronchial foreign bodies (air trapping, mediastinal shift).
- CT or MRI for subglottic stenosis, haemangioma, masses and vascular rings; angiography for vascular anomalies.
- Contrast swallow for clefts, fistulae, rings and aspiration.
- Arterial blood gases in severe obstruction.

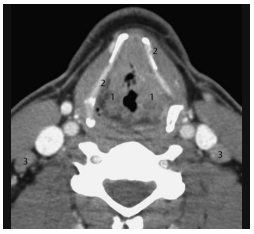
3.5 Microlaryngoscopy and bronchoscopy

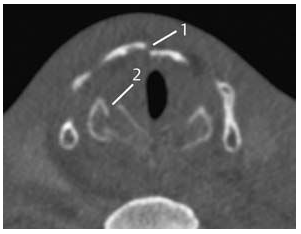
The definitive assessment of the paediatric airway, in theatre with full resuscitation facilities and an experienced anaesthetist. Inhalational induction maintains spontaneous breathing; a quick laryngoscopy is followed by bronchoscopy of the subglottis, trachea and bronchi, collection of secretions and removal of crusts or foreign bodies; then the child is intubated if needed. Look at the whole airway: a second lesion is common.

IMAGING IN THE APP 3 cases

Laryngocele, supraglottic tumour and laryngeal trauma on imaging.







Laryngocele
Infection & inflammation · 10 images

Supraglottic Tumor
Tumours · 9 images

Laryngeal Trauma
Trauma · 10 images

3.6 The common paediatric causes

Laryngomalacia is the commonest cause of stridor in infants, accounting for well over half of congenital cases. Inspiratory stridor begins in the first weeks, is worse with feeding, crying and lying supine, and usually resolves by 18 to 24 months. Short aryepiglottic folds, an omega-shaped epiglottis and prolapsing arytenoid mucosa are seen on awake flexible laryngoscopy. Most infants need only reassurance and treatment of reflux; poor weight gain, apnoeas, cyanosis or pectus excavatum are indications for **supraglottoplasty**.

Vocal fold paralysis is the second commonest congenital cause; bilateral paralysis is associated with Arnold–Chiari malformation and hydrocephalus. **Subglottic stenosis** is congenital or, more often, acquired after intubation, and is graded by the **Cotton–Myer** system by the percentage of lumen obstructed: grade I up to 50%, grade II 51 to 70%, grade III 71 to 99%, grade IV no lumen. **Subglottic haemangioma** typically presents at a few weeks of age with biphasic stridor that worsens over the first months, often with a cutaneous haemangioma in a beard distribution, and responds to **propranolol**. **Croup** (viral laryngotracheobronchitis) produces a barking cough and stridor in children aged six months to three years and is treated with a single dose of oral dexamethasone, 0.15 mg/kg, with nebulised adrenaline for severe distress. Recurrent or atypical croup, or croup under six months, warrants airway endoscopy.

4 CHAPTER FOUR Management

Airway first.

4.1 Immediate

1. Keep the patient calm and upright (in a parent's arms for a child); give humidified oxygen without causing distress.
2. Call senior anaesthetic and ENT help early.
3. Temporising measures as appropriate: nebulised adrenaline, steroids (dexamethasone), heliox (a helium–oxygen mixture flows more smoothly through narrow airways).
4. Treat the cause where possible: antibiotics, removal of a foreign body.

4.2 Securing the airway

- **Intubation:** in theatre, by the most experienced anaesthetist, with an ENT surgeon scrubbed and ready.
- **Tracheostomy:** when intubation is impossible or unsafe, for prolonged obstruction, laryngeal trauma, and for long-term airway management (bilateral paralysis, stenosis, severe congenital anomalies).
- **Cricothyroidotomy:** the emergency surgical airway in adults and older children when the patient can neither be intubated nor oxygenated; needle cricothyroidotomy with jet ventilation in young children.

4.3 The adult threatened airway

The Difficult Airway Society guidelines end with **front-of-neck access** when the patient "can't intubate, can't oxygenate": the recommended technique is a **scalpel–bougie–tube** cricothyroidotomy through the cricothyroid membrane, identified beforehand by palpation or ultrasound. In a patient with a known obstructing laryngeal tumour, the safest plan is often an **awake tracheostomy under local anaesthesia**, or awake fibreoptic intubation, agreed between surgeon and anaesthetist before any sedation. **High-flow humidified nasal oxygen** extends the

time before desaturation during induction and airway surgery. Heliox and steroids buy time but do not treat the cause.

5 CHAPTER FIVE Pathway

Assess, stabilise, secure.

THE CHILD OR ADULT WITH STRIDOR

1 How severe?

- → **silent chest, cyanosis, exhaustion or falling consciousness**
  emergency airway now — theatre, senior anaesthetist and ENT
- → **severe recession or agitation**
  oxygen, nebulised adrenaline, dexamethasone, senior help
- → **mild**
  step 2

2 Phase and age

- → **inspiratory in an infant, worse feeding and supine**
  laryngomalacia — awake flexible laryngoscopy
- → **biphasic, worsening in first months**
  subglottic haemangioma or stenosis
- → **barking cough, toddler**
  croup — dexamethasone
- → **sudden onset while eating or playing**
  foreign body — bronchoscopy

3 Febrile and toxic?

- → **drooling, sitting forward**
  supraglottitis — airway in theatre
- → **high fever, poor response to croup treatment**
  bacterial tracheitis

4 Adult

- → **tumour**
  awake tracheostomy or planned intubation, then biopsy
- → **post-thyroidectomy bilateral palsy**
  reintubate or tracheostomy
- → **angioedema**
  adrenaline, steroids, antihistamine
- → stop ACE inhibitor

OPERATIVE GUIDES IN THEATRE 6 operations

Tracheostomy, laryngomalacia and subglottic stenosis surgery, airway foreign bodies and acquired stenosis.

Open Tracheostomy
Tracheostomy
Percutaneous Tracheostomy
Percutaneous
Laryngomalacia
Laryngomalacia
Subglottic Stenosis
Subglottic stenosis
Tracheoscopy, Bronchoscopy, and Airway Foreign Bodies
Airway foreign body
Acquired Laryngeal Stenosis
Acquired stenosis

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Laryngomalacia:** features, natural history and indications for supraglottoplasty.
- **Cotton–Myer grading** and subglottic haemangioma with propranolol.
- **Front-of-neck access** and the awake tracheostomy.
- **Definition** of stridor, and stertor and wheeze.
- **Inspiratory, biphasic, expiratory** and their sites.
- **Causes** in children by site, congenital and acquired.
- **History points** and severity signs.
- **Management:** airway first, then diagnosis by endoscopy.

6.2 Common pitfalls

COMMON PITFALLS

- Sending a child with stridor for radiographs unaccompanied.
- Mistaking quieter stridor in a tiring child for improvement.
- Stopping at the first lesion found and missing a second.

Written for this app from Oxford Textbook of Otolaryngology ch. 52, 74, Oxford Specialist Handbook ch. 3, 23, Oxford Handbook of ENT ch. 3, 25, Dhingra ch. 59, 64 and K. J. Lee ch. 50, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · APPROACH TO

Approach to a lump in the neck

A lump in the neck is one of the commonest reasons to see an ENT surgeon, and one of the most dangerous to mismanage. In a child it is almost always a reactive node or a congenital cyst. In an adult over forty who smokes, a persistent lateral lump is a metastatic squamous carcinoma until proved otherwise. The rule is simple: a full head and neck examination with endoscopy, imaging, and fine-needle aspiration come before any open biopsy. A node cut out casually seeds the neck and spoils the cure.

AT A GLANCE

- Adult neck lump is cancer until shown otherwise.
- Scope, FNA, image.
- No open biopsy first.
- A cystic node in middle age may be HPV-related.

1 CHAPTER ONE Approach

Where, how old, how long.

1.1 The age rule

In children and young adults, inflammatory and congenital masses dominate, and lymphoma and thyroid cancer are the malignancies to remember. In adults over about forty, neoplasms dominate, and most are metastases, usually from squamous carcinoma of the upper aerodigestive tract. The old "rule of 80" puts it memorably: about 80% of non-thyroid neck masses in adults are neoplastic, 80% of those are malignant, and 80% of the malignant ones are metastatic, mostly from a primary above the clavicles.

1.2 The time rule

Duration is almost as informative as age. A mass present for days is usually inflammatory, one present for months neoplastic, and one present for years

congenital or benign.

1.3 Site

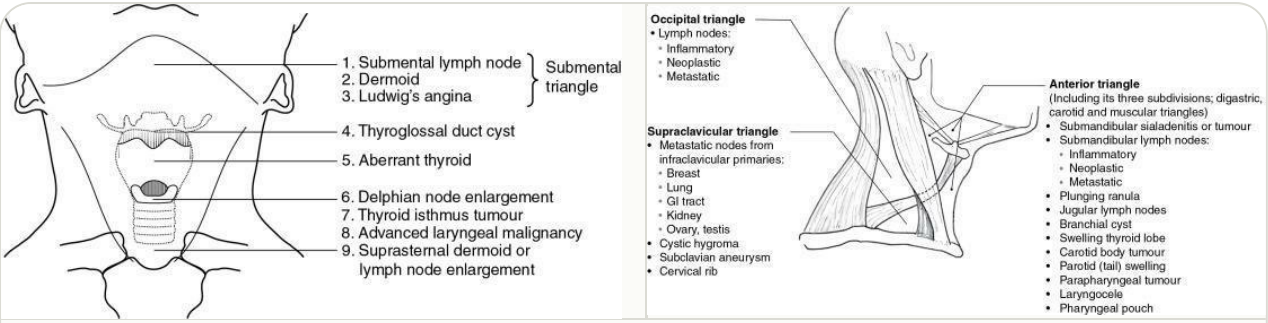


FIGURE 1 First, swellings in the midline of the neck. Second, lateral swellings grouped by the triangles of the neck.

Dhingra, figs 77.1 and 77.2

NECK MASSES BY SITE

SITE	COMMON CAUSES
Midline, above the hyoid	Submental nodes, sublingual dermoid, thyroglossal cyst (suprahyoid), plunging ranula
Midline, below the hyoid	Thyroglossal cyst (commonest at the hyoid), thyroid isthmus, Delphian and pretracheal nodes, laryngocele, suprasternal dermoid, thymic cyst
Submandibular triangle	Submandibular nodes and gland (sialadenitis, stone, tumour), plunging ranula
Carotid triangle	Jugulodigastric and deep cervical nodes, branchial cyst, carotid body tumour, vagal schwannoma, laryngocele, carotid aneurysm
Parotid region and angle of the jaw	Parotid tumour or node, parapharyngeal mass, sebaceous cyst
Posterior triangle	Accessory chain nodes (nasopharynx, lymphoma, tuberculosis), cystic hygroma, lipoma, neural tumours, cervical rib
Supraclavicular	Virchow node (stomach and other infraclavicular primaries), lung and breast metastases, lymphoma, tuberculosis
Within sternomastoid	Sternomastoid tumour of infancy

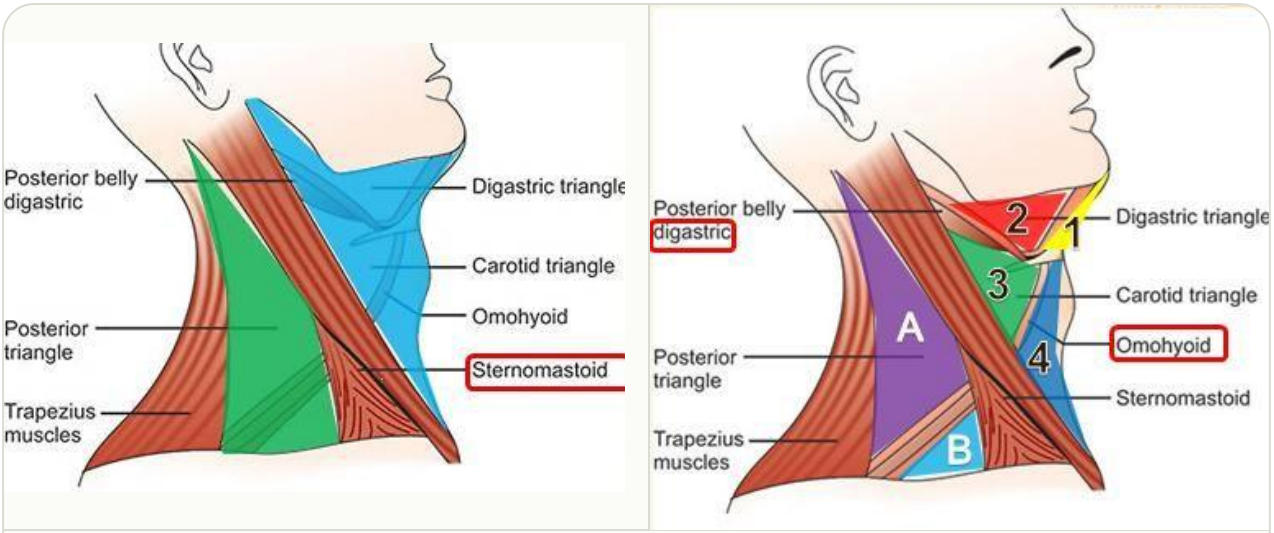


FIGURE 2 The triangles of the neck and their subdivisions.
Anatomy atlas · ENT Steps p. 270

2 CHAPTER TWO Lymph nodes

Levels and drainage.

2.1 The levels

NECK NODE LEVELS

LEVEL	BOUNDARIES	DRAINS
IA (submental)	Between the anterior bellies of digastric, above the hyoid	Lower lip, chin, floor of mouth, tongue tip
IB (submandibular)	Digastric bellies and the mandible	Oral cavity, anterior nasal cavity, face, submandibular gland
II (upper jugular)	Skull base to the hyoid, around the internal jugular vein. IIA in front of the accessory nerve, IIB behind it	Oral cavity, oropharynx, nasopharynx, larynx, parotid
III (middle jugular)	Hyoid to the cricoid	Oropharynx, hypopharynx, larynx
IV (lower jugular)	Cricoid to the clavicle	Hypopharynx, larynx, thyroid, cervical oesophagus

LEVEL	BOUNDARIES	DRAINS
V (posterior triangle)	Behind sternomastoid, in front of trapezius. VA above the cricoid, VB below	Nasopharynx, oropharynx, scalp and skin, thyroid
VI (central)	Hyoid to the sternal notch, between the carotids	Thyroid, larynx (subglottis), hypopharynx, cervical oesophagus
VII (superior mediastinal)	Below the sternal notch	Thyroid, oesophagus

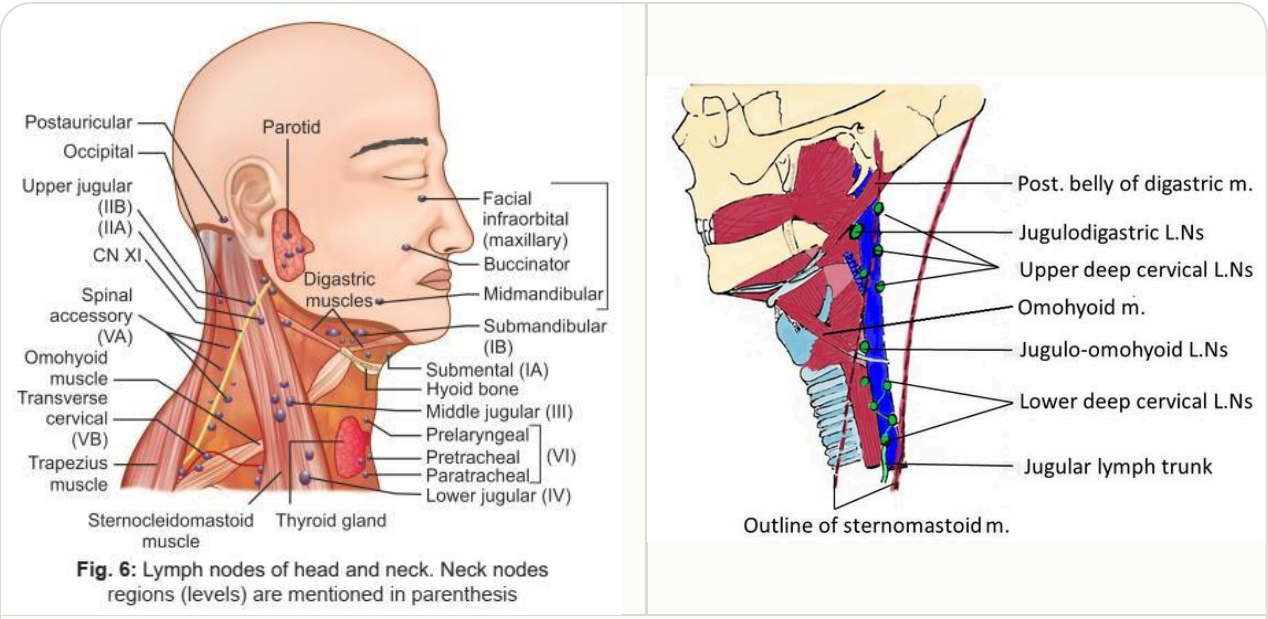


FIGURE 3 The lymph node levels of the neck, and the deep cervical chain.

Anatomy atlas · ENT Steps p. 285, Doss p. 198

READING THE NODE BACK TO THE PRIMARY

- **Jugulodigastric node (level II):** tonsil, tongue base, nasopharynx, supraglottis.
- **Posterior triangle nodes:** suggest the nasopharynx.
- **Level VI:** suggests the thyroid.
- **Left supraclavicular node (Virchow, *Troisier sign*):** suggests an abdominal or thoracic primary.
- **Cystic level II node in an adult:** a metastasis from *HPV-positive oropharyngeal carcinoma* (tonsil or tongue base) or from papillary thyroid carcinoma, not a "branchial cyst", until proved otherwise.

3 CHAPTER THREE Assessment

History, examination, tests.

3.1 History

Ask how long the lump has been present, whether it is growing or painful, and whether it changes with meals or infections. Then ask about the upper aerodigestive tract: hoarseness, dysphagia, odynophagia, referred otalgia, nasal obstruction or bleeding, a unilateral serous otitis, and weight loss. Record the risk factors (tobacco, alcohol and betel quid; sexual history for HPV; exposure to tuberculosis; HIV risk; contact with cats for *Bartonella* or toxoplasmosis; travel; and previous skin cancer or radiation) and any "B" symptoms of fever, night sweats and weight loss that suggest lymphoma or tuberculosis.

3.2 Examination

Palpate the neck from behind, level by level, noting the size, consistency, fixity, pulsation and transillumination of the lump and whether it moves with swallowing (thyroid, thyroglossal cyst) or with tongue protrusion (thyroglossal cyst). Then examine the whole head and neck: scalp, skin, ears, nose and oral cavity, palpate the tonsils and tongue base, and look at the nasopharynx, larynx and hypopharynx with a flexible nasendoscope. Finish with the axillae, groins, liver and spleen.

3.3 Investigation

1. **Ultrasound with FNA** is the first test for most adult masses. Ultrasound shows whether the mass is cystic or solid, the node architecture and its vascularity; FNA gives cytology, with p16/HPV, EBV, thyroglobulin, flow cytometry for lymphoma, and culture and acid-fast stains as needed. A non-diagnostic aspirate is repeated or replaced by a core biopsy.
2. **Contrast CT or MRI of the neck** shows the extent and may reveal a primary.
3. **PET-CT** is used for an unknown primary, done *before* panendoscopy and biopsies, and for staging and distant disease.
4. **Chest imaging.**

5. **Blood tests:** count, ESR, serology for EBV, HIV, toxoplasma and *Bartonella*, and tests for tuberculosis.
6. **Panendoscopy under anaesthesia** with directed biopsies, tonsillectomy and tongue-base mucosectomy to find an occult primary.

DO NOT EXCISE AN ADULT NECK NODE FIRST

An open biopsy of a node that turns out to be metastatic squamous carcinoma breaches the fascial planes and contaminates the field. It raises rates of local recurrence and complications. If open biopsy is unavoidable after negative FNA and endoscopy (for example, to type a suspected lymphoma), plan it so that the incision can be included in a neck dissection, with frozen section, and be ready to proceed to dissection.

OPERATIVE GUIDES IN THEATRE 2 operations

Fine-needle aspiration, and management of the unknown primary.

Office-Based Procedures: Fine-Needle Aspiration Biopsy of a Mass in the Head and Neck
FNA

Management of the Unknown Primary Cancer
Unknown primary

4 CHAPTER FOUR Inflammatory causes

The commonest masses.

4.1 Reactive lymphadenitis

Reactive nodes are tender, mobile and often bilateral, and accompany a sore throat, a dental infection or a skin infection. They settle within two to six weeks. A node that persists beyond that, grows, is hard, or lies in the supraclavicular fossa needs investigation.

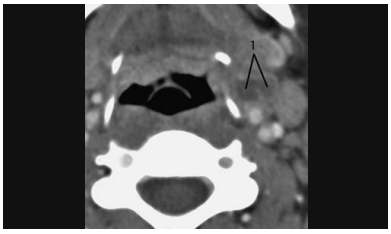
4.2 Acute suppurative lymphadenitis

Suppurative lymphadenitis, caused by *Staphylococcus aureus* and group A streptococcus, mainly affects young children. The node is hot and tender and may

form an abscess. It is treated with antibiotics, and an abscess is drained by aspiration or incision. A neck abscess in an adult should raise the suspicion of a necrotic metastasis or tuberculosis.

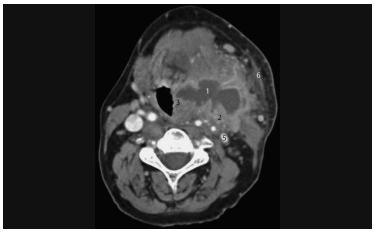
IMAGING IN THE APP 2 cases

Cervical lymphadenitis and neck abscess.



Cervical Lymphadenitis

Infection & inflammation · 9 images



Abscess of the Neck

Infection & inflammation · 1 image

4.3 Tuberculous lymphadenitis (scrofula)

Tuberculous lymphadenitis is very common in endemic areas and more frequent with HIV. It affects any age, usually the deep cervical and posterior triangle nodes, and passes through recognisable stages: discrete nodes, then matted nodes with periadenitis, then caseation and a *cold abscess*, then a "collar-stud" abscess through the deep fascia, and finally sinus formation.

The diagnosis rests on FNA or biopsy showing caseating granulomas, acid-fast bacilli on Ziehl–Neelsen staining, culture and GeneXpert with rifampicin resistance testing, supported by a chest radiograph, a tuberculin test or interferon-gamma release assay, and an HIV test. Treatment is standard antituberculous therapy: isoniazid, rifampicin, pyrazinamide and ethambutol for two months, then isoniazid and rifampicin for four. Nodes may enlarge temporarily during treatment, a *paradoxical reaction*. Surgery is reserved for drug failure or residual masses, and abscesses are aspirated rather than incised.

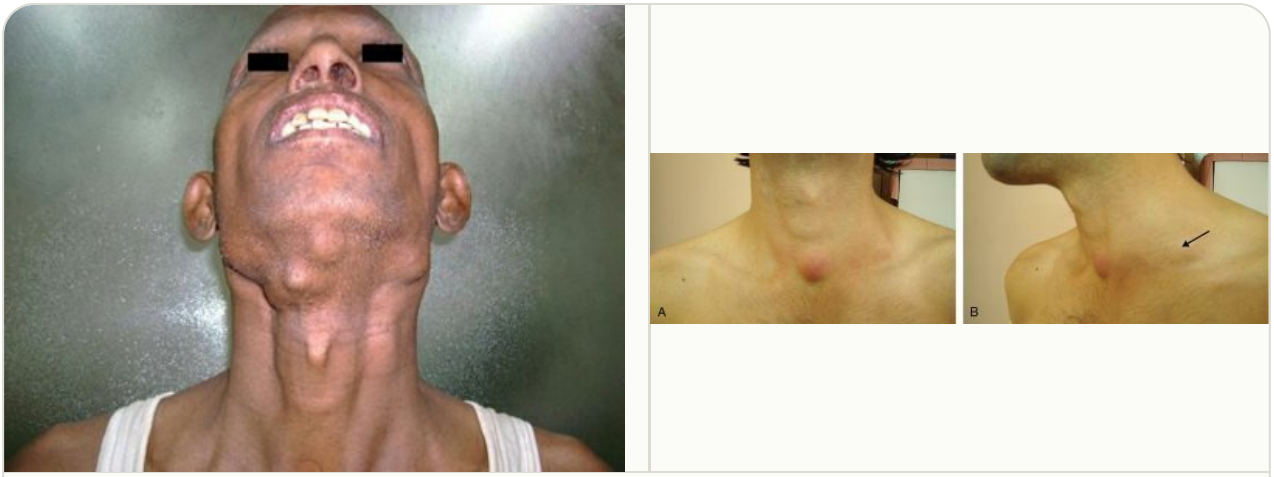


FIGURE 4 First, multiple tuberculous nodes in the neck. Second, (A) a caseating suprasternal node and (B) supraclavicular tuberculous nodes.

Dhingra, figs 77.8 and 77.9

4.4 Atypical mycobacteria

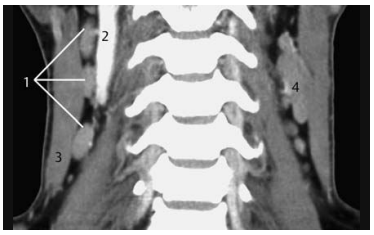
Atypical mycobacterial infection affects children aged one to five as a painless node, usually submandibular or preauricular, with violaceous, thin overlying skin and a tendency to form sinuses. The chest film is normal and the tuberculin reaction weak. Complete excision is the treatment, or curettage where the facial nerve is at risk, with clarithromycin as an alternative; incision and drainage leaves a chronic sinus.

4.5 Other infections

Cat-scratch disease from *Bartonella henselae* causes a tender, often suppurating node after a cat scratch; it is diagnosed by serology and treated with azithromycin when severe. **Toxoplasmosis** causes posterior cervical nodes in young adults and is diagnosed by serology. **Infectious mononucleosis** from EBV gives generalised nodes, exudative tonsillitis and splenomegaly. **HIV** causes persistent generalised lymphadenopathy, and its nodes may harbour tuberculosis, lymphoma or Kaposi sarcoma. **Actinomycosis** produces a woody swelling with sinuses near the jaw.

IMAGING IN THE APP 1 case

Cervical lymphadenopathy in HIV.



Lymphadenopathy, HIV

Infection & inflammation · 1 image

4.6 Non-infective inflammatory causes

Sarcoidosis produces non-caseating granulomas. **Kikuchi–Fujimoto disease**, histiocytic necrotising lymphadenitis, affects young women with tender nodes and fever and settles on its own. **Kimura disease** affects young Asian men with a preauricular mass, eosinophilia and a high IgE. **Castleman disease** is unicentric or multicentric, and **Rosai–Dorfman disease** causes massive painless nodes.

5 CHAPTER FIVE Neoplastic causes

Metastases and lymphoma.

5.1 Metastatic squamous carcinoma

The typical patient has a hard, painless, enlarging node, often in level II or III. He may be an older smoker and drinker, or a younger non-smoker with HPV-positive oropharyngeal cancer, whose nodes are often cystic. A node fixed to deep structures or to the skin means extranodal spread. The primary is sought in the tonsil, tongue base, nasopharynx, pyriform sinus, larynx and oral cavity, by examination, endoscopy, imaging and panendoscopy. Supraclavicular nodes may come from below the clavicle: from the lung, breast, oesophagus, stomach, pancreas, kidney, testis or ovary.

5.2 Carcinoma of unknown primary

When a squamous metastasis has no primary on examination, endoscopy and imaging, the search proceeds in a set order:

1. PET-CT.
2. Panendoscopy with biopsies.
3. Ipsilateral (or bilateral) tonsillectomy and transoral tongue-base mucosectomy.
4. p16 and EBV staining of the node, which point to the oropharynx or nasopharynx.

Treatment is neck dissection with radiotherapy to the neck and the likely mucosal sites, or primary chemoradiotherapy. The outcome is better than many expect, especially with HPV-positive disease.

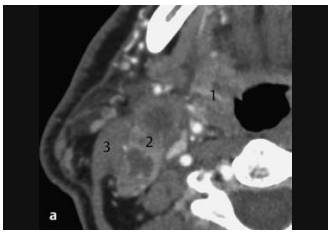
5.3 Lymphoma

Hodgkin lymphoma affects young adults, with rubbery, painless, discrete nodes, often low in the neck or supraclavicular, sometimes with a mediastinal mass, B symptoms and pain after alcohol; the diagnostic cell is the Reed–Sternberg cell. **Non-Hodgkin lymphoma** affects older adults and involves both nodes and extranodal sites, including Waldeyer's ring (tonsil, nasopharynx, tongue base), the thyroid, the salivary glands and the sinonasal tract, where NK/T-cell lymphoma occurs.

FNA with flow cytometry may suggest lymphoma, but a *core or excision biopsy* is usually needed to classify it, and fresh tissue must be sent. Treatment follows staging, with chemotherapy with or without radiotherapy.

IMAGING IN THE APP 1 case

Non-Hodgkin lymphoma in the neck.



Lymphoma, Non-Hodgkin

Tumours · 13 images

5.4 Other neoplasms

Thyroid carcinoma, including cystic nodal metastases of papillary carcinoma, salivary tumours, paragangliomas and schwannomas, lipomas and, rarely, sarcomas complete the neoplastic list.

6 CHAPTER SIX Other masses

Not nodes.

6.1 Swellings that are not nodes

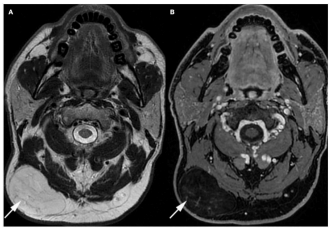
A **laryngocele** is an air-filled dilatation of the laryngeal saccule. An *external* laryngocele pierces the thyrohyoid membrane to form a lateral neck swelling that enlarges on a Valsalva manoeuvre, coughing or blowing, as in trumpeters and glassblowers; infection turns it into a *laryngopyocele*, and a ventricular or supraglottic carcinoma obstructing the saccule must always be excluded. A **pharyngeal pouch** forms a soft swelling low in the left neck that gurgles, with regurgitation. A **plunging ranula** is a soft, transilluminant submandibular swelling from the sublingual gland, cured by removing that gland.

A **cervical rib** is a hard supraclavicular lump with thoracic outlet symptoms: arm paraesthesia from pressure on the lower trunk of the brachial plexus, and coldness, claudication, aneurysm or emboli from compression of the subclavian artery. It is excised if it causes symptoms. A **lipoma** is soft and lobulated with a slipping edge, commonly in the posterior triangle and nape, and **sebaceous and epidermoid**

cysts lie in the skin with a punctum. Finally, a **carotid aneurysm**, or an ectatic, tortuous carotid in an elderly hypertensive patient, can present as a pulsatile lump.

IMAGING IN THE APP 1 case

Lipoma of the neck.



Lipoma of the neck

Tumours · 1 image

7 CHAPTER SEVEN Children

A different list.

7.1 Neck lumps in children

Most neck lumps in children are reactive lymph nodes, which are common, small, mobile and settle within weeks. Congenital lesions follow: **thyroglossal duct cysts** in the midline, moving with tongue protrusion; **branchial cysts, sinuses and fistulae**, most from the second arch along the anterior border of sternocleidomastoid; **dermoid cysts**; **lymphatic malformations** (cystic hygroma), which are soft, transilluminate and may respond to sclerotherapy; and **haemangiomas**. Infection gives acute suppurative lymphadenitis and abscess, and **non-tuberculous mycobacterial** infection gives a violaceous, painless node in a toddler, often in the submandibular region, which is best treated by excision when feasible.

Malignancy is uncommon but must be considered. Features that call for urgent assessment are a node larger than 2 cm or growing, a hard, fixed or matted node, a supraclavicular node, nodes persisting beyond six weeks, and systemic symptoms

(fever, night sweats, weight loss, bruising, pallor, hepatosplenomegaly). Lymphoma is the commonest head and neck malignancy in children, followed by rhabdomyosarcoma, neuroblastoma and thyroid cancer. Ultrasound is the first test, avoiding radiation; a chest X-ray excludes a mediastinal mass before any anaesthetic, and an excision biopsy is often needed for diagnosis.

8 CHAPTER EIGHT Pathway

Lump to diagnosis.

THE ADULT NECK LUMP

1. **Clinical picture**
   - → **acute, tender, with infection** — antibiotics and review in two to three weeks
   - → **midline moving with swallowing** — thyroid pathway
   - → **any red flag or persisting over three weeks** — step 2
2. **Full head and neck examination**
   - → flexible endoscopy, skin, oral cavity, tonsils and tongue base
   - → **obvious primary** — biopsy it and stage
3. **Ultrasound with FNA or core**
   - → **squamous carcinoma** — CT or MRI, PET-CT, panendoscopy, tonsillectomy and tongue-base mucosectomy if no primary
   - → **lymphoma suspected** — core biopsy or planned excision for flow cytometry

→ **thyroid**
thyroid pathway

→ **tuberculosis**
culture, treat

4 Still no diagnosis

→ repeat core biopsy

→ open biopsy planned for possible neck dissection, with frozen section

9 CHAPTER NINE Neck dissection

The principle.

9.1 Types and indications

Neck dissection removes the lymph nodes and fat of the neck en bloc. A **radical** dissection removes levels I to V with the accessory nerve, internal jugular vein and sternomastoid; a **modified radical** dissection removes the same levels while preserving one or more of those structures; a **selective** dissection removes only the levels at risk, such as the supraomohyoid dissection (I to III) for the oral cavity and the lateral dissection (II to IV) for the larynx and pharynx; and an **extended** dissection removes additional nodes or structures. A **therapeutic** dissection treats clinically positive nodes, while an **elective** dissection treats a clinically negative neck when the risk of occult metastases exceeds about 15 to 20%.

OPERATIVE GUIDES IN THEATRE 3 operations

Selective, modified radical and radical neck dissection.

Selective Neck Dissection
Selective neck dissection
Modified Radical Neck Dissection
Modified radical
Radical Neck Dissection
Radical

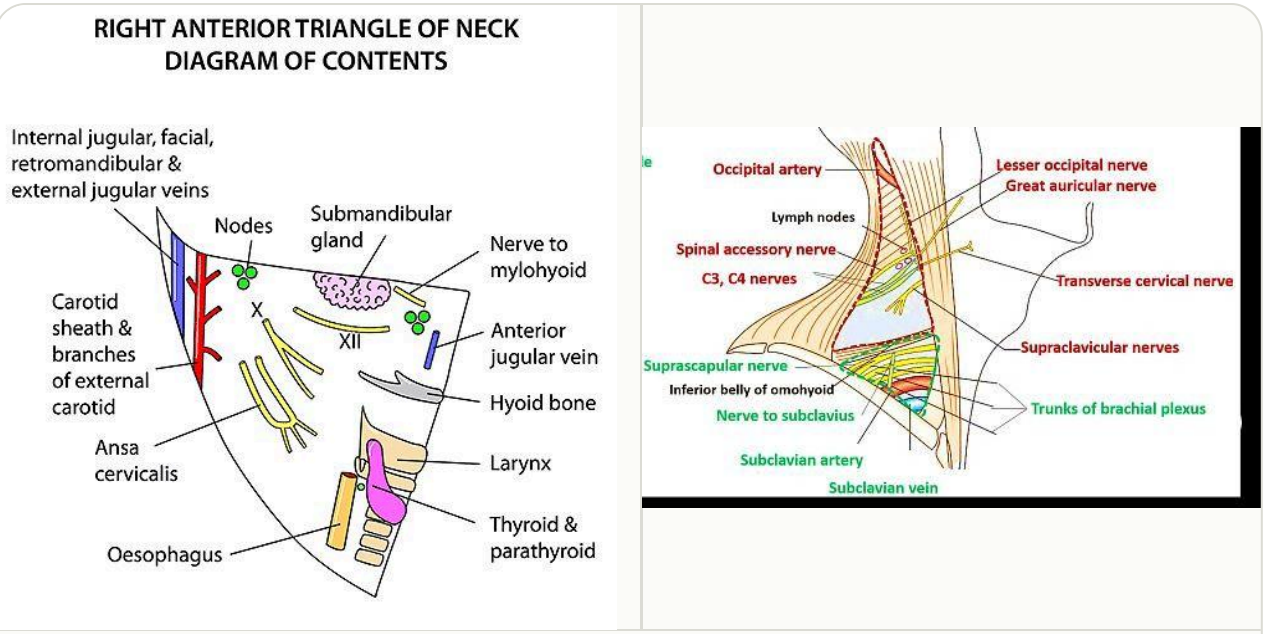


FIGURE 5 Contents of the anterior and posterior triangles.

Anatomy atlas · ENT Steps p. 271, ENT Steps p. 272

10 CHAPTER TEN For the examination

What the examiner wants to hear.

10.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Neck lumps in children** and the features that call for biopsy.
- **A systematic approach:** age, duration and site.
- **Node levels and drainage;** Virchow node.
- **The work-up of an adult neck mass:** full examination and endoscopy, then ultrasound and FNA, then imaging, then panendoscopy. Why open biopsy is avoided.
- **Unknown primary:** PET-CT before panendoscopy, tonsillectomy and tongue-base mucosectomy, p16 and EBV.
- **Tuberculous lymphadenitis:** stages, diagnosis and treatment.
- **Lymphoma:** needs tissue for typing.

For personal study. Follow local protocols and senior advice for patients.

10.2 Common pitfalls

COMMON PITFALLS

- Calling a cystic level II mass in an adult a branchial cyst.
- Excision biopsy of a node before examining the upper aerodigestive tract.
- Incising a tuberculous or atypical mycobacterial abscess, which leaves a chronic sinus.
- Missing the nasopharynx in a patient with posterior triangle nodes and a unilateral effusion.

Written for this app from Oxford Textbook of Otolaryngology ch. 50, 67, Oxford Specialist Handbook ch. 3, 12, Oxford Handbook of ENT ch. 3, 14, Dhingra ch. 77 and K. J. Lee ch. 11, 38, 40. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · APPROACH TO

Approach to a swollen salivary gland

A swollen salivary gland is either inflamed, obstructed, infiltrated or growing a tumour. The history usually separates these: swelling that comes with meals and goes again is obstruction; acute painful swelling with fever is infection; bilateral painless swelling is systemic; and a persistent discrete lump is a neoplasm until a needle says otherwise. Ultrasound with a fine-needle or core biopsy is the workhorse of investigation, and facial weakness with a parotid lump is malignancy until proven otherwise.

AT A GLANCE

- Meal-related swelling: stone or stricture.
- Bilateral diffuse: mumps, Sjögren, sialosis, HIV, sarcoid.
- Discrete lump: ultrasound and FNA.
- Children: parotid lumps are more often malignant than in adults.

1 CHAPTER ONE The pattern

Time course and side.

1.1 Sorting the swelling

SALIVARY SWELLING BY PATTERN

PATTERN	THINK OF	CLUES
Recurrent, swelling at meals	Stone or stricture	Submandibular more than parotid; pain with eating, settles in hours
Acute, painful, red, fever	Acute bacterial sialadenitis	Elderly, dehydrated, postoperative; pus from the duct
Acute, bilateral, with malaise in a child or young adult	Mumps or other viral parotitis	Contacts; unvaccinated; orchitis
Recurrent parotid swelling in a child	Juvenile recurrent parotitis	Episodes settle by puberty; sialectasis on ultrasound

PATTERN	THINK OF	CLUES
Bilateral, painless, persistent	Sialosis, Sjögren's syndrome, IgG4-related disease, sarcoidosis, HIV	Dry eyes and mouth; alcohol, diabetes; cysts
Discrete persistent lump	Neoplasm	Painless, slow (benign); pain, fixity, palsy, nodes (malignant)

RED FLAGS FOR MALIGNANCY

- **Facial nerve weakness** with a parotid lump.
- **Pain, rapid growth, fixity to skin or deep tissues, skin ulceration or trismus.**
- **Cervical lymphadenopathy.**
- **A lump in the submandibular, sublingual or minor salivary glands,** where the proportion of malignant tumours is higher.
- **A parapharyngeal mass** pushing the tonsil medially (deep lobe tumour).
- **Sjögren's syndrome with a new persistent swelling:** lymphoma.

2 CHAPTER TWO History and examination

Which gland, and why?

2.1 History

Ask how long the swelling has been present, whether it fluctuates or is constant, whether it is related to eating, whether it is painful, and whether it is one gland or several. Ask about a bad taste or pus in the mouth, dry eyes and dry mouth, joint pains, fever, weight loss and night sweats, alcohol intake, diabetes, medications (anticholinergics, diuretics), radiotherapy, HIV risk, and previous skin cancers of the face and scalp, which metastasise to parotid nodes.

2.2 Examination

Inspect and palpate the gland, bimanually for the submandibular gland and the floor of mouth, feeling for a stone along Wharton's duct. Look at the duct openings: the

parotid (Stensen's) duct opposite the upper second molar and the submandibular duct beside the frenulum; massage the gland to see whether clear saliva, cloudy saliva or pus comes out. Test every branch of the facial nerve, examine the oropharynx for medial displacement of the tonsil, check the neck for nodes, and inspect the skin of the face and scalp.

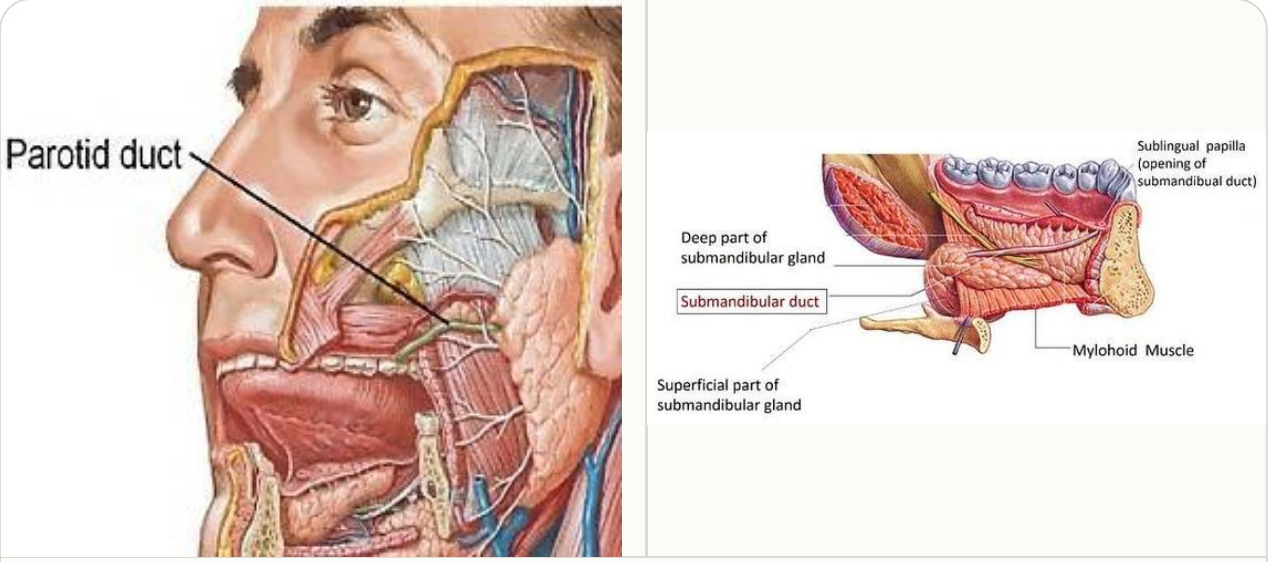


FIGURE 1 First, the parotid gland and Stensen's duct. Second, the sublingual and submandibular glands with their ducts.

Anatomy atlas · ENT Steps p. 181, ENT Steps p. 191

3 CHAPTER THREE Investigation

Ultrasound first.

3.1 Imaging and tissue

Ultrasound is the first-line investigation for almost all salivary swellings. It shows stones, duct dilatation, sialectasis, diffuse inflammatory change, cysts and solid masses, and guides **fine-needle aspiration cytology** or **core biopsy**. Cytology is reported using the Milan system, which grades the risk of malignancy; core biopsy gives more tissue when cytology is inconclusive or lymphoma is suspected. **MRI** defines the extent of a tumour, deep lobe involvement and perineural spread; **CT** shows stones and bony invasion; and **sialography** or **MR sialography** shows ducts and strictures. **Sialendoscopy** is both diagnostic and therapeutic for obstruction.

Blood tests include inflammatory markers, autoantibodies (anti-Ro, anti-La) for Sjögren's syndrome, serum IgG4, ACE and HIV when indicated.

4 CHAPTER FOUR Causes

Obstruction, infection, systemic, neoplasm.

4.1 Stones and strictures

About 80 to 90% of salivary stones occur in the submandibular gland, because its saliva is more alkaline and mucous and its duct runs uphill. Patients have **meal-time syndrome**: pain and swelling with eating that settle over hours. Stones near the duct opening are removed transorally; stones up to about 4 to 5 mm can be retrieved with a sialendoscope basket, and larger ones fragmented by laser or extracorporeal lithotripsy or removed by a combined endoscopic and open approach. Gland excision is reserved for failure. Parotid problems more often come from strictures, which are dilated endoscopically.

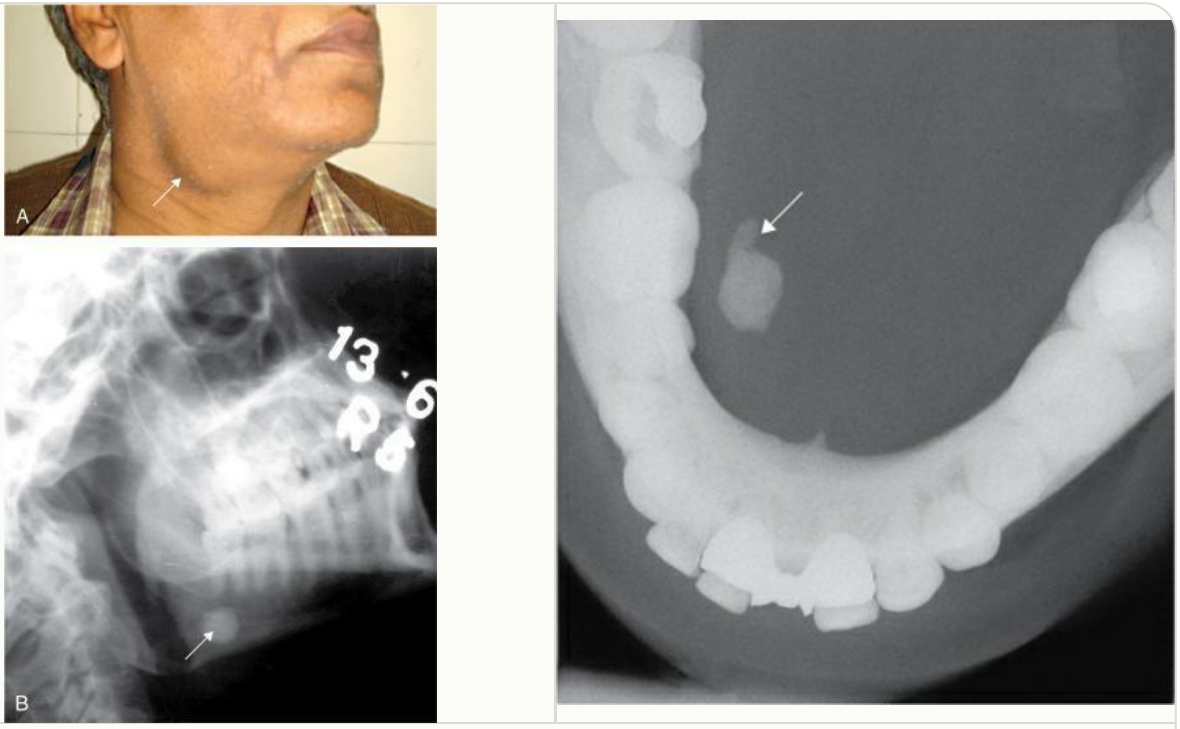


FIGURE 2 Submandibular sialadenitis, and a radio-opaque stone in the submandibular duct.
Dhingra, figs 45.2 and 45.3

4.2 Infection

Acute bacterial sialadenitis, usually *Staphylococcus aureus*, occurs in dehydrated, elderly or postoperative patients with reduced salivary flow; treatment is rehydration, sialogogues, massage, oral hygiene and antibiotics, with drainage of any abscess. **Mumps** causes bilateral painful parotitis with malaise and is managed supportively. **Juvenile recurrent parotitis** causes repeated parotid swellings in children and usually resolves by puberty.

4.3 Systemic and infiltrative causes

Sjögren's syndrome causes dry eyes, dry mouth and parotid swelling, and carries a markedly raised risk of MALT lymphoma. **IgG4-related disease** causes firm bilateral submandibular swellings (Küttner tumour) that respond to steroids. **Sarcoidosis** can cause parotid swelling; with uveitis, fever and facial palsy it forms Heerfordt's syndrome. **HIV** causes lymphoepithelial cysts. **Sialosis** is a painless bilateral enlargement associated with alcohol, diabetes, malnutrition and drugs.

4.4 Tumours

Most salivary tumours arise in the parotid, and about 80% of parotid tumours are benign, most commonly **pleomorphic adenoma**, followed by **Warthin's tumour** (often bilateral, in older male smokers). The smaller the gland, the more likely a tumour is malignant: about half of submandibular tumours and most sublingual and minor salivary gland tumours are malignant. Malignant tumours include mucoepidermoid carcinoma (the commonest), adenoid cystic carcinoma (with perineural spread), acinic cell carcinoma, salivary duct carcinoma and carcinoma ex pleomorphic adenoma, as well as metastases from cutaneous squamous cell carcinoma and melanoma in parotid nodes. Benign parotid tumours are removed by superficial or partial parotidectomy, or extracapsular dissection, preserving the facial nerve; malignant tumours are managed by the head and neck multidisciplinary team.

THE SWOLLEN SALIVARY GLAND

1 Pattern

→ **meal-related**
stone or stricture — ultrasound, sialendoscopy

→ **acute painful with pus**
bacterial sialadenitis — fluids, antibiotics, drain abscess

→ **bilateral painless**
bloods for Sjögren's, IgG4, sarcoid, HIV

→ **discrete lump**
step 2

2 Discrete lump

→ ultrasound with fine-needle cytology or core biopsy

→ MRI if malignant, deep or fixed

3 Red flags?

→ **facial weakness, pain, fixity, nodes**
treat as malignant — MDT

→ **benign cytology**
parotidectomy or extracapsular dissection, or observation in selected patients

4 Sjögren's with persistent swelling

→ biopsy to exclude lymphoma



FIGURE 3 Squamous cell carcinoma of the parotid: (A) a parotid swelling and (B) facial palsy, the red flag.

Dhingra, fig. 46.1

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Pattern recognition:** meal-time, acute, bilateral and discrete swellings.
- **Why stones form in the submandibular gland.**
- **Ultrasound and fine-needle cytology** as first-line tests; the Milan system.
- **The proportion of malignancy** by gland.
- **Facial palsy with a parotid lump** means malignancy.
- **Sjögren's syndrome and lymphoma risk.**

5.2 Common pitfalls

COMMON PITFALLS

- Performing an open biopsy of a parotid lump, which risks seeding a pleomorphic adenoma and the facial nerve.

- Missing a cutaneous primary in a patient with a parotid node.
- Removing a submandibular gland for a stone that could be removed endoscopically.

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